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An integrated and practical database solution for storage and efficient processing of large genomic datasets

Master's thesis
in COMPUTER SCIENCE

Warsaw, 2026-07-28

Abstract

Genomic workflows commonly connect specialized programs through files. This arrangement separates sequences from relational metadata, repeats parsing and indexing work, and prevents the database planner from representing the cost of alignment. This thesis presents BioSeqDB, a PostgreSQL extension that stores compact biological sequences and runs a minimap2-derived seeding, chaining, and extension pipeline during database scans.

BioSeqDB provides alphabet-generic packed sequence views, scalar and single instruction, multiple data (SIMD) minimizer extraction, and Vulkan implementations for selected sketching and chaining operations. Bounded CPU dynamic-programming kernels and GPU alignment routes perform final extension. Optional SneakySnake and GateKeeper filters reject unsuitable candidate regions before that stage. The database layer adds persistent minimizer indexes, bounded ingestion, external-provider relation definitions, coordinated caching, and generated PostgreSQL boundary declarations.

Alignment is not evaluated as an independent scalar function for every row. Planner support introduces custom paths whose execution nodes batch sequences, execute the explicitly selected implementation when the input and hardware pass admission, and place alignment results into planned expressions. The same operation can therefore participate in joins, filtering, metadata projection, and persistent storage. The thesis describes how the mapper's data and control flow become a database execution primitive while retaining explicit resource bounds and support for heterogeneous processors.

Keywords

sequence alignment, bioinformatics, minimap2, PostgreSQL, databases, genomic data management, SIMD, GPU computing

Thesis domain (Socrates-Erasmus subject area codes)

11.3 Informatics and Computer Science

Subject classification

Applied computing, life and medical sciences, bioinformatics

Tytuł pracy w języku polskim

Zintegrowane i praktyczne rozwiązanie bazodanowe do przechowywania i wydajnego przetwarzania dużych zbiorów danych genomicznych

Contents

Introduction . . . 7

1. Biological foundations of sequence analysis . . . 11
   1.1. Biological context . . . 11
      1.1.1. DNA, ribonucleic acid (RNA), and protein alphabets . . . 12
   1.2. Alignment operation . . . 12
   1.3. Technical considerations . . . 13
      1.3.1. FASTA . . . 13
      1.3.2. FASTQ . . . 14
      1.3.3. CIGAR . . . 14

2. Sequence alignment and search strategies . . . 15
   2.1. Alignment foundation and specification of a problem . . . 15
   2.2. Dynamic programming definition . . . 16
   2.3. Seed and extend . . . 18
   2.4. Limitations of existing tools . . . 20

3. The BioSeqDB alignment pipeline . . . 23
   3.1. Alignment as a database execution pipeline . . . 23
   3.2. The logical alignment chain . . . 24
      3.2.1. Target preparation and minimizers . . . 25
      3.2.2. Seeds, chains, and candidate regions . . . 25
      3.2.3. Reduction, extension, and output . . . 25
   3.3. Compile-time algorithm selection and runtime routing . . . 26
   3.4. Batching, memory, and target reuse . . . 28
   3.5. Relation to minimap2 and GPU mappers . . . 29
   3.6. Final invariants and current boundaries . . . 30

4. Minimizer extraction . . . 33
   4.1. From every base to a sparse, comparable sketch . . . 33
      4.1.1. Canonical k-mers and minimizer windows . . . 33
      4.1.2. From minimizers to candidate anchors . . . 34
      4.1.3. Minimizers, spaced seeds, and linked seeds . . . 35
      4.1.4. Fuzzy seed hashing with BLEND . . . 36
   4.2. The minimap2 event contract . . . 36
   4.3. Data-independent sliding minima . . . 38
      4.3.1. Rolling ntHash . . . 39
      4.3.2. Chunked vector extraction . . . 40
   4.4. BioSeqDB minimizer backends . . . 41

4.4.1. Scalar reference . . . 41
4.4.2. ExactSimd continuation . . . 41
4.4.3. PNT1 as a separate ntHash scheme . . . 42
4.4.4. Vulkan tile extraction . . . 43
4.4.5. Comparison and selection . . . 44
4.4.6. Limitations . . . 45
4.5. Conclusion . . . 45

5. Chaining seed anchors . . . 47
5.1. From local seed hits to candidate mappings . . . 47
5.2. Anchor geometry and ordering . . . 47
5.3. Scalar chaining recurrence . . . 48
5.3.1. Relation to minimap2 . . . 50
5.4. Endpoint selection and backtracking . . . 50
5.5. Range minimum query chaining . . . 51
5.6. SIMD chaining on the CPU . . . 52
5.6.1. Exact ordered decisions . . . 52
5.6.2. Bounded batch approximation . . . 53
5.7. Memory locality in GPU chaining . . . 53
5.7.1. Vulkan score generation . . . 53
5.7.2. Completion and result projection . . . 55
5.8. Execution policy . . . 55
5.9. Summary . . . 56

6. Final alignment and extension . . . 57
6.1. From a seed chain to a base-level alignment . . . 57
6.2. Affine-gap dynamic programming . . . 57
6.2.1. Banding and stopping criteria . . . 58
6.3. Alignment around a chain . . . 59
6.4. KSW2 with portable SIMD . . . 60
6.4.1. Difference representation . . . 60
6.4.2. Endpoint and traceback rules . . . 60
6.5. Adaptive block alignment . . . 61
6.5.1. Block movement and growth . . . 61
6.5.2. SIMD recurrence within a block . . . 61
6.5.3. Exact diagonal certificate . . . 62
6.6. Traceback, CIGAR, and edit counting . . . 62
6.7. Accelerator boundary and gridded traceback . . . 63
6.8. Runtime selection and design limits . . . 64

7. Alignment prefiltering . . . 65
7.1. Why filter after chaining but before alignment? . . . 65
7.2. SneakySnake chip-maze filtering . . . 66
7.3. GateKeeper shifted-mask filtering . . . 68
7.3.1. The exact rejection guard . . . 70
7.4. Dispatch and query-pipeline integration . . . 71
7.5. Limitations . . . 72
7.6. Conclusion . . . 72

8. Sequence representations and lazy evaluation . . . 73
8.1. One logical sequence, several physical forms . . . 73
8.2. Alphabet-generic compact ranges . . . 74
8.2.1. Logical and physical traits . . . 74
8.2.2. Ranks, symbols, and orientation . . . 75
8.3. Packing and resident PostgreSQL values . . . 76
8.3.1. Packing geometry . . . 76
8.3.2. A flat trailing-payload representation . . . 76
8.4. A lazy view over flat and fragmented storage . . . 77
8.4.1. The reader-facing facade . . . 77
8.4.2. Ownership and failure handling . . . 78
8.5. Persistent slices and lazy composition . . . 78
8.5.1. Immutable piece trees . . . 78
8.5.2. Bound and unbound values . . . 79
8.5.3. SQL views and transforms . . . 79
8.6. Access planning and materialization . . . 80
8.6.1. Consumer requirements . . . 80
8.6.2. Why eager conversion is avoided . . . 81
8.7. Contract for sequence consumers . . . 81

9. PostgreSQL API and custom scans . . . 83
9.1. Alignment as a relational operation . . . 83
9.2. SQL surface and ownership contract . . . 84
9.3. Planner support, custom paths, and custom plans . . . 84
9.4. Typed placeholder injection . . . 85
9.5. Bounded batch execution . . . 86
9.6. Join execution and rescans . . . 87
9.7. User-visible plan information . . . 87
9.8. Conclusion . . . 88

10. Physical sequence storage in PostgreSQL . . . 89
10.1. Storage as part of query execution . . . 89
10.2. Columnar relation layout . . . 89
10.2.1. Logical byte space and reservations . . . 89
10.2.2. Stripes, chunk groups, and streams . . . 90
10.3. Fragmented and packed sequence payloads . . . 92
10.3.1. Locator and fragment geometry . . . 92
10.3.2. Buckets, shards, and the canonical registry . . . 92
10.3.3. Insertion, concurrency, and cleanup . . . 93
10.4. Projection, pruning, and lazy reconstruction . . . 93
10.5. Coordination of repeated query requests . . . 94
10.6. Resulting storage contract . . . 95

11. Persistent sequence indexes . . . 97
11.1. Persistent indexes as derived database state . . . 97
11.2. Logical organization of minimizer indexes . . . 97
11.2.1. Minimizers per fragment . . . 98
11.2.2. Minimizers per sequence . . . 98
11.3. Bounded construction and publication . . . 99

11.4. Compatibility and integrity . . . 101
11.5. Selective cold loading and memory accounting . . . 101
11.6. Runtime index selection . . . 102
11.7. Differences from minimap2 `.mmi` . . . 103
11.8. Operational limits and conclusions . . . 104

12. Bounded sequence ingestion and integrity limits . . . 105
12.1. Bounded sequence ingestion and integrity limits . . . 105
12.2. Layered input pipeline . . . 105
12.3. Incremental parsing and direct compact encoding . . . 106
12.4. Bounded delivery and backpressure . . . 107
12.5. Header-to-column mapping . . . 107
12.6. SQL publication and error behavior . . . 108
12.7. Integrity limits . . . 109
12.8. Conclusion . . . 110

13. External sequence providers . . . 111
13.1. Remote databases as relations . . . 111
13.2. Provider and execution boundaries . . . 112

14. Source-driven SQL code generation . . . 113
14.1. One source for the PostgreSQL boundary . . . 113
14.2. Generated artifacts and safety . . . 114

15. Conclusions . . . 115

References . . . 117

Introduction

Sequencing instruments turn biological material into collections of strings. The useful interpretation of those strings rarely exists in isolation. A read has an instrument and sample provenance. A reference interval belongs to an assembly and annotation release. An alignment has a score, a strand, a position, an edit path, and a relationship to experimental metadata. Conventional workflows keep the strings in FASTA and FASTQ sequence files, Sequence Alignment/Map (SAM) and Binary Alignment/Map (BAM) alignment files, or mapper-specific index files while keeping much of the surrounding information in tables or ad-hoc scripts. Each tool sees only the portion of the workflow represented by its command-line arguments.

Relational database systems solve a different part of the problem. PostgreSQL provides durable storage, transactions, typed values, indexing, joins, extensible operators, and a cost-based planner. It does not, by itself, provide the representations or execution kernels required for billions of packed symbols and computationally expensive sequence searches. A naive user-defined function is not sufficient. Evaluating one alignment function independently for every row prevents batching, hides accelerator opportunities, and gives the planner little information about a predicate that can dominate the query.

BioSeqDB investigates the boundary between these two systems. It is implemented as a PostgreSQL extension whose non-database core is written in C++26. Biological symbols are represented by generic compact ranges rather than by a deoxyribonucleic acid (DNA)-only byte string. The alignment path is derived from minimap2: representative k-mers are extracted as minimizers, matching minimizers become anchors, compatible anchors are chained, and CPU or GPU alignment routes extend the selected regions. The port is not a line-for-line translation. The implementation introduces template-selected alphabets and layouts, Highway SIMD kernels, task-based parallel execution, Vulkan compute paths, persistent PostgreSQL-backed indexes, and prefilters for candidate regions. These changes must preserve the semantics that make minimap2 useful while operating on data that may be lazy, fragmented, cached, or obtained from an external provider.

The database boundary is equally important. An SQL expression such as `mmsearch(a, b)` has a conventional function definition so that PostgreSQL can type and plan it. The ordinary function body is not an alignment fallback: it fails closed if no BioSeqDB custom executor owns the expression. Its planner-support function assigns the unowned expression a high startup cost, while structural planner hooks can construct an owning custom path with explicit nonnegative costs. The resulting custom scan batches input, invokes the alignment pipeline, and makes its results available to planned expressions through injected PostgreSQL nodes. This is what permits alignment to participate in filters, projections, and joins while retaining the execution structure required by central processing unit (CPU) vector units and graphics processing units (GPUs).

The central research question is therefore not whether an existing aligner can be called from SQL. It is whether the aligner's data and control flow can become a database execution

primitive without losing algorithmic fidelity, resource bounds, or the ability to exploit modern processors. The investigation is organized around four supporting questions:

RQ1. Which parts of seeding, chaining, and extension can be expressed over a common compact sequence abstraction without eager conversion?

RQ2. Which stages admit SIMD or GPU implementations, and which input and resource constraints govern their selection?

RQ3. How can PostgreSQL plan and execute a set-valued, batch-oriented search while presenting an ordinary typed SQL interface?

RQ4. How should sequences, indexes, ingestion, external data, caches, and generated boundary code fit around the kernels so that the resulting system remains inspectable and reliable?

This thesis answers these questions through implementation and analysis of algorithmic and execution contracts. Historical designs appear only when they explain a constraint or a rejected optimization. They are not presented as simultaneous production architectures. Comparisons with minimap2, SIMD-Minimizers, the G3SA GPU seed-chain-extend system, Block Aligner, SneakySnake, GateKeeper, and Citus Columnar distinguish inherited mechanisms from BioSeqDB-specific changes.

The main contributions developed in the following chapters are:

C1. An alphabet-generic sequence representation that supports packed, fragmented, lazily composed, and PostgreSQL-backed data through a common range contract.

C2. A staged minimap2-derived mapper with scalar, SIMD, and Vulkan components and explicit runtime dispatch.

C3. Optional GPU-capable SneakySnake and GateKeeper candidate filters integrated without changing the meaning of an accepted alignment.

C4. Persistent sequence and minimizer-index storage derived from a columnar PostgreSQL access method, with bounded loading, caching, and selective cold-index reads.

C5. A planner and custom-scan interface that connects SQL expressions and joins to batch-oriented alignment.

C6. Structural generation and validation of SQL installation definitions and PostgreSQL function-manager boundary declarations from annotated C++ sources.

The document first establishes the biological and algorithmic context, then states the shortcomings that motivate an integrated system. The algorithm chapters follow data through minimizer extraction and chaining. They then describe the expensive downstream alignment stage before stepping back to the prefilter that guards that stage. This expository inversion makes the filter's safety and cost contract concrete without changing the runtime order, which remains chaining, prefiltering, and alignment. It then moves outward through sequence abstractions, PostgreSQL planning, physical storage, persistent indexes, ingestion, external providers, and generated SQL boundary code.

- Chapter 1 introduces biological sequences and sequencing data. It explains variation and the computational questions that motivate database-resident sequence analysis.

- Chapter 2 develops exact dynamic programming and seed-and-extend mapping. It also presents graph-based approaches and the limits of existing tools.
- Chapter 3 presents the complete processing path with its stage contracts and CPU and accelerator routes.
- Chapter 4 explains minimizer sketching and rolling hashing. It compares the SIMD and Vulkan implementations with minimap2 and SIMD-Minimizers.
- Chapter 5 derives the chaining objective and compares its scalar RMQ SIMD and GPU forms. It then states the selection policy.
- Chapter 6 describes affine-gap alignment with KSW2 and block-based SIMD extension. It also covers traceback and accelerator integration.
- Chapter 7 describes SneakySnake and GateKeeper. It explains their filtering guarantees and GPU mappings before showing how BioSeqDB applies them.
- Chapter 8 develops compact sequence ranges and bit-level views. It then presents fragmented and lazy sequences with PostgreSQL-backed adapters and alphabet dispatch.
- Chapter 9 follows an mmsearch expression through planning and batched execution. It explains joins and per-row result substitution.
- Chapter 10 explains the columnar-derived storage layout and fragment metadata. It also covers caching and request coordination.
- Chapter 11 describes index construction and storage with compatibility checks and selective cold loading. It also explains updates and differences from minimap2 indexes.
- Chapter 12 presents the bounded input/output pipeline for FASTA and FASTQ data. It covers compression and header parsing with typed PostgreSQL writes and failure handling.
- Chapter 13 explains generated API clients and the provider abstraction that exposes remote sequence databases as PostgreSQL relations.
- Chapter 14 explains structural source analysis and checked annotations. It then presents dependency ordering and generation of the PostgreSQL boundary.
- Chapter 15 answers the research questions and summarizes the final system. It identifies the principal limitations and possible extensions.

Chapter 1

Biological foundations of sequence analysis

1.1. Biological context

"Significant scrutiny and attention have been given to genome-sequencing costs and how they are calculated since the beginning of the field of genomics in the late 1980s". Institute, The Cost of Sequencing a Human Genome NHGRI (National Human Genome Research Institute) has carefully tracked costs per genome and expose this data publically.

In the last 15 years, there has been seen a massive development of sequencing technologies. Companies like Applied Biosystems, Illumina, Roche, and Pacific Biosciences have developed a plethora of sequencing platforms, e.g. SOLiD, GS FLX, and PacBio. These tools are able to generate an enormous amount of data in a matter of days. As an example, Illumina NextSeq generates up to 360 Gbp, split into reads of length up to 150 bp in merely 2 days (data taken from Illumina website).

Although those platforms vary in detail, the general idea behind them is as follows:

1. DNA is fragmented through physical and chemical methods. Adapters are appended to the ends of the fragments to form what is called a sequencing library.
2. Fragments of unwanted size or self-ligated adapters are removed from the library.
3. The library is immobilized on a chemical bead so that it can be steadily observed.
4. Amplification is performed, e.g. through PCR emulsion
5. A combination of optical devices and fluorophores is used to make reads.

With new advancements in sequencing technologies, the cost of sequencing a genome has been decreasing significantly to the point where most popular open databases store more than 10s of petabytes of data. To make this data usesable researchers often need to perform search operations:

- For bioinvestigative resarch: does this genomic sequence already contains known genes? Does it contain a known mutation? Is it a known pathogen?
- For evolutionary biology: is this genomic sequence a known evolutionary lineage? What is the genetic distance between this sequence and a known sequence?
- For legal research: is this DNA sample from known individual? Does it represent known kinship to someone else?

This search operation is using an operation called "sequence alignment" which is the core concept of this thesis.

1.1.1. DNA, ribonucleic acid (RNA), and protein alphabets

DNA is a polymer containing one of four bases: adenine, cytosine, guanine and thymine. Their conventional one-letter symbols form the alphabet

$$\Sigma_{\mathrm{DNA4}} = \{\mathtt{A}, \mathtt{C}, \mathtt{G}, \mathtt{T}\}.$$

Four symbols are sufficient for an unambiguous DNA string, but experimental data introduces ambiguity. The symbol `N` conventionally means that the nucleotide at a position is unknown or was not resolved. Other International Union of Pure and Applied Chemistry (IUPAC) codes describe sets of possible bases, such as a purine at a position.

RNA uses adenine, cytosine, guanine and uracil, so a basic RNA alphabet replaces `T` with `U`. Many RNA molecules are transcribed from DNA. Crick, "Central Dogma of Molecular Biology"

A protein sequence is written over the twenty common amino-acid symbols. It's alphabet is notably larger, but in most cases we can translate it to a DNA sequence and use the DNA alphabet directly.

Sequence class	Core symbols	Computational consequences
DNA4	`A,C,G,T`	
RNA4	`A,C,G,U`	
DNA5	`A,C,G,T,N`	
RNA5	`A,C,G,U,N`	
PROT20	Twenty common amino-acid symbols	

Table 1.1: Biological alphabets and the distinctions that a sequence-processing system must not erase. The table describes biological data in general. It does not assert that every alphabet or ambiguity convention is exposed by every BioSeqDB SQL path.

1.2. Alignment operation

Due to nature of biological sequences, search must be able to match fragments that are non-exact, thus alignment operation is similar to popular fuzzy-search operations on normal text. The alignment can be viewed as a processes that optimises gaps (insertions and deletions) in a way to optimise similarity metric. High similarity score between parts of sequences can indicate that they share same evolutionary origin or have a common purpose. By comparing newly found sequences to already researched ones, we can deduce lots of relevant biological data. Juliusz K and W, "Master thesis: An integrated and practical database solution for storage and efficient processing of large genomic datasets"

We will consider finite nonemptystrings over the DNA4 alphabet. A gap is a symbol indicated by $-$. An alignment is a process of adding gaps to the sequences so that they are of the same length and at the same time, there is no index where a gap occurs in both sequences.

As an example, let us present an alignment of $GATCAGATA$ and $CATTCAATCA$. One of the possible alignments is:

```
GAT-CAGAT-A
CATTCA-ATCA
```

There are 4 cases that can occur during alignments:

1. Insertion - a gap has been added to the first sequence
2. Deletion - a gap has been added to the second sequence
3. Mismatch - letters at a given index are different
4. Match - letters at a given index are the same

1.3. Technical considerations

A typical workflow for sequence analysis includes 3 data formats - FASTA, FASTQ, and SAM/BAM. Reference genome, stored in FASTA, is aligned with millions of reads in the FASTQ file and the result of alignment is saved as SAM/BAM file. The reason for the reference genome being in FASTA and reads in FASTQ is that we assume that the reference genome is correct, but reads produced by sequencers can contain various errors. Moreover, we're more likely to score alignments with high-quality bases higher than the ones with low-quality.

1.3.1. FASTA

FASTA is a text-based file format for storing nucleotide sequences. It is one of the most widely used formats in bioinformatics. The current standard specification is given by NCBI. Below is a description of the usually encountered version.

It consists of a series of entries separated by a new line. Each entry starts with a single-line header. The header starts with '>' and contains various metadata about the following sequence, like the description of the sequence, library accession number, and submission date. After the header comes the sequence itself represented with standard IUB/IUPAC acid codes, split into lines of length usually not exceeding 80 characters. Aside from A, C, T, G commonly encountered is code N, which means any of A/C/T/G. Lowercase letters are treated the same as uppercase. The usual extension is .fasta and .fa. An example FASTA file:

```
>PartOfHumanGenome|HG_9357/2022|2022-03-01
ACTTTGAGCTAGCATCGATTGGATCAGATGCAT
ACTCATCGGCAGTCGCATCAGCTTGGACTACGT
>PartOfMouseGenome|MG_5417/2019|2019-08-14
ACTATAGGAGCAGANNNNNNNNCTAGAGACATC
TCTGCGTNTNTTCCGGACGTACGTAGCATAGTC
```

The advantages of this format are simplicity and human readability. The downside is that it is not optimal in terms of size. To adhere to this issue, compression software is used, like gzip.

1.3.2. FASTQ

FASTQ is an extension of the FASTA format. It enhances FASTA by adding quality scores of sequences. It is a standard output format of sequencers.

Each entry consists of 4 lines. The first line starts with '@' and contains metadata about the sequence. The second line contains the sequence itself. The third line starts with '+' and may contain some additional metadata. The fourth line contains quality scores, represented by ASCII characters with codes from 33 to 126, one character per nucleotide. The file extension used is .fq and .fastq. Again, gzip can be used to reduce file size. An example FASTQ file:

```
@HumanGenomeSequencerRead
ACTACTAGACGG
+Genome20200301
*!]^#[$*@*!(
```

The quality score is usually a phread score, meaning Q satisfies $Q = -10 \log_{10} p$, where p is the probability of a corresponding nucleotide being incorrect. ASCII codes from 33 to 126 are mapped from 0 to 93.

1.3.3. CIGAR

CIGAR is an acronym for Concise Idiosyncratic Gapped Alignment Report and has been originally defined by the Exonerate alignment program. The CIGAR format is designed to contain the minimal information necessary for the reconstruction of an alignment. This format does not include nucleotides. Sweden, The Cost of Sequencing a Human Genome

- M - match or mismatch
- D - deletion
- I - insertion
- S - soft clipping. Can occur only at the beginning or at the end of the alignment. Indicates that nucleotides are clipped from the alignment.
- H - hard clipping. The same as soft clipping, except that the clipped part is not present in the SEQ field of the SAM file.
- N - skipped region. Used for representing introns (part of the gene that is non-coded) in the alignment.
- = - match
- X - mismatch

Chapter 2

Sequence alignment and search strategies

2.1. Alignment foundation and specification of a problem

Alignment can be done in different ways, where the core algorithms stays the same, but some of the conbditions change. It's important to understand what are the capabilities of standard dynamic programming algorithms and how they can be used to solve different problems.

Formally: Let $X = x_1 \dots x_n$ and $Y = y_1 \dots y_m$ be the sequences we align. An alignment is a sequence of columns

$$A = ((a_1, b_1), \dots, (a_L, b_L)), \qquad (a_r, b_r) \in (\Sigma \cup \{-\})^2 \setminus \{(-,-)\},$$

where first and second elements reciver contents of X and Y after gaps are removed:

1. An element of shape $a_r, -$ signifies an insertion in X
2. An element of shape $-, b_r$ signifies a deletion in Y
3. An element of shape m_r, m_r signifies a match
4. An element of shape a_r, b_r where $a_r \neq b_r$ signifies a substitution

This alignment is an optimized sequence correspondence under a scoring model. For substitution score $s(a, b)$ and gap cost $g(\ell)$, the score is

$$S(A) = \sum_{\text{symbol pairs } r} s(a_r, b_r) - \sum_{\text{gap runs } u} g(|u|).$$

The scoring model may be formulated in different ways, but one of the simplest and most-common conventions is to use an affine gap cost model:

$$g(\ell) = q + \ell e, \qquad \ell \geq 1, \tag{2.1}$$

where q is the opening cost and e is the extension cost.

A gap of size one costs then $q + e$. Some scientific papers mentioning $q + (\ell - 1)e$ use a different parameter convention. Affine scoring charges one opening cost per gap run and can be evaluated with a fixed number of states per matrix cell Gotoh, "An improved algorithm for matching biological sequences."

This scoring model isn't sufficient to define entire problem:

1. Global alignment consumes both complete sequences Needleman and Wunsch, "A general method applicable to the search for similarities in the amino acid sequence of two proteins"

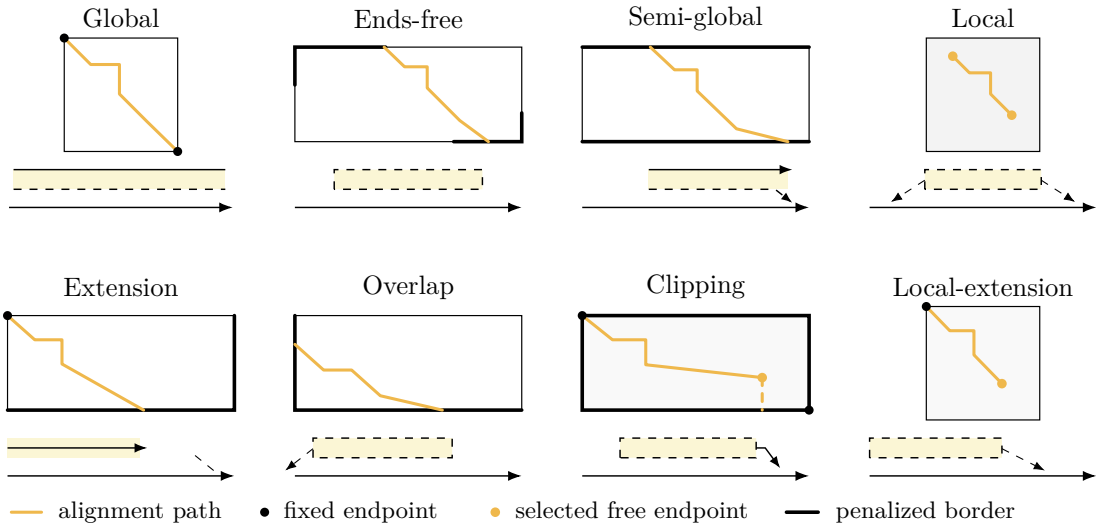


Figure 2.1: Eight alignment boundary modes. Yellow lines depcict possible tracebacks. Black dots mark fixed endpoints (for example for global alignment we always align entire sequence so endpoints are 0 and len(Y)-1), yellow dots mark selected free endpoints, and bold border segments mark penalized or anchored boundaries (usually the border is the length of sequence, but under certain conditions we might have limits on the alignment length and coordinates). In each lower schematic, the solid arrow is the reference and the yellow dashed track is the query. These panels showcase how different libraries/algorithms handle different goals across number of scientific resarch

2. Local alignment may restart at zero and selects the best pair of substrings Smith and Waterman, "Identification of common molecular subsequences"
3. Semi-global and overlap goals consume selected ends while allowing specified prefix/-postfix to remain unaligned
4. Extension begins from a fixed anchor and chooses an endpoint under a stopping rule
5. Clipping will just finish on the rightmost gap and keep postfix unaligned
6. Local-extension fixes the start but permits an interior endpoint

Figure 2.1 shows schematic examples of these modes.

Various practical implementations also use special stopping criteria, which usualy can be upper bound on the $S(A)$ score.

2.2. Dynamic programming definition

Let $H_{i,j}$ be the best score after consuming i symbols of X and j symbols of Y. For a simpler linear gap cost model d, we can define the algorithm's state matrix as:

$$H_{i,j} = \max\{H_{i-1,j-1} + s(x_i, y_j), H_{i-1,j} - d, H_{i,j-1} - d\}. \tag{2.2}$$

This is the intuitive definition based on 3 possible options (replacement or two variants of gap). Free-end alignment (like local) change the score values for the cells from which traceback may begin.

Algorithm 1 Affine-gap alignment with an explicit boundary contract

Require: sequences $X[1 \dots n]$, $Y[1 \dots m]$, scores s, q, e, and a boundary contract B
Ensure: score, endpoints, and an optional CIGAR string
1: initialize H, I, and D according to B
2: for $i \leftarrow 1$ to n do
3: for $j \leftarrow 1$ to m do
4: update $M_{i,j}$, $I_{i,j}$, $D_{i,j}$, and $H_{i,j}$ with Equation 2.3
5: store the predecessor of each winning state when traceback is requested
6: end for
7: end for
8: select the endpoint permitted by B
9: follow predecessors to an allowed start and run-length encode the operations

For affine scoring model we need to hold main state and one state for each gap orientation. Let $I_{i,j}$ and $D_{i,j}$ end in gaps in X and Y, respectively, and let $M_{i,j}$ end in a symbol pair. Under Equation 2.1, one formulation is

$$\begin{aligned} M_{i,j} &= H_{i-1,j-1} + s(x_i, y_j), \\ I_{i,j} &= \max\{H_{i,j-1} - q - e,\ I_{i,j-1} - e\}, \\ D_{i,j} &= \max\{H_{i-1,j} - q - e,\ D_{i-1,j} - e\}, \\ H_{i,j} &= \max\{M_{i,j}, I_{i,j}, D_{i,j}\}. \end{aligned} \tag{2.3}$$

Using H to open a gap permits a transition from any previous state. Local alignment also includes zero in the final maximum.

The algorithm state matrix can be used to retrieve the edit path as CIGAR string. This is the standard compact representation of gapped alignments described in chapter 1SAM/BAM Format Specification Working Group, Sequence Alignment/Map Format Specification

Affine model is some scenarios might not be enough as it doesn't distinguish between short and long gaps. The two-piece affine model can support such feature without complicating the scoring function too much:

$$g_2(\ell) = \min(q_1 + \ell e_1,\ q_2 + \ell e_2)$$

This model permits different cost slopes for short and long gaps. In our software we use single affine model for the fastest alignment modes (adaptive block), while KSW2 variants support dual affine model.

For large datasets evaluating entire large $O(nm)$ matrix can make significant practical difference:

- We can evaluate alignments only close to diagonal. Being close to diagonal means that there aren't many large gaps between matching fragments. Designing algorithms around diagonals can make us skip evaluation of many far cells that do not contribute to the result.
- In case of features that do not require traceback (CIGAR construction requires traceback, but score or coordinates of best match do not) we can just store relevant slice in memory and avoid allocating large memory region.
- We can introduce X-drop stop condition: stop when score falls below best score seen so far by more than X

- We can introduce Z-drop stop condition: the same as X-drop, but if the distance from diagonal is too high then stop as well (this means that we stop when there are too many gaps cummulatively)

Smart evaluation startegy can combine all these approaches. KSW2 and Block aligner for example use all of them. Block aligner Liu and Steinegger, "Block Aligner: an adaptive SIMD-accelerated aligner for sequences and position-specific scoring matrices" evaluates only the most promising rectangular region in the matrix while KSW2 Suzuki and Kasahara, "Introducing difference recurrence relations for faster semi-global alignment of long sequences" focuses on diagonal-centered evaluation.

2.3. Seed and extend

Applying local dynamic programming between a query of length m and a reference of length N costs $O(Nm)$. Most reference positions don't contribute to the result. Long-read mappers therefore organize search into three stages Li, "Minimap2: pairwise alignment for nucleotide sequences."

First step is to use divide and conquer approach and split sequence into representatives known as seeds:

$$\begin{aligned} \text{Seed}(x_1, x_2, ..., x_n) = (s_1, i_1, j_1), (s_2, i_2, j_2), ..., (s_k, i_k, j_k) \\ \text{where} 1 \leq i_a, j_a \leq n \forall a \in 1, 2, ..., k \end{aligned} \tag{2.4}$$

Seed (s_a, i_a, j_a) is some represenative of the sequence substring $x_a = x_{i_a} x_{i_a+1} ... x_{j_a}$. This can be implemented in different ways, this definition is the most abtract one. If a seed occurs in both query and reference, then we can consulde that two represented substrings are very similar. Using some concrete seeding function we can obtain intersection for query and reference:

$$S_{X \cap Y} = (s, i_x, j_x, i_y, j_y) \text{where} (s, i_x, j_x) \in \text{Seed}(X) \text{and} (s, i_y, j_y) \in \text{Seed}(Y) \tag{2.5}$$

1. Seeding Generate $Seed(X)$ and $Seed(Y)$ and obtain $S_{X \cap Y}$
2. Chaining As matching seeds will cover only parts of both sequences that match without errors, we still need to fill-in the gaps between them. We connect seeds from $S_{X \cap Y}$ into chains using similar dynamic programming approach as in usual aligner. This gives us some approximate alignments.
3. Extension Perform usual alignment algorithm on the approximate fragments to obtain final results.

Minimizers are one of many possible seeding model definitions. Given parameters w and k, a window W of length $l = w + k - 1$ contains w consecutive k-mers. The minimizer of the window is the smallest k-mer (subsequence of length k) in the window. For random minimizers, k-mers are ordered by a pseudo-random order, usually given by comparing hashes of the k-mers. In case of ties, the leftmost smallest k-mer is chosen. Roberts et al., "Reducing storage requirements for biological sequence comparison" Overlapping windows often retain the same representative, which is emitted once.

Canonical minimizers: DNA is double-stranded and majority of sequencing technologies do not distinguish these two strands. Genomic sequences thus need to satisy additional constraint: sequence and its reverse complement ($A \leftrightarrow T, C \leftrightarrow G$) should be handled in the same way by the software (they should be indistinguisable) In our case this means that for minimizer of a window W at position p we need to have minimizer $\overline{W^c}$ at position $w - 1 - p = |W| - k - p$.

Groot Koerkamp and Martayan, "SimdMinimizers: Computing Random Minimizers, fast" In most of the software due to simplicity truly cannonical minimizers aren't used. Instead we take minimum of hashes of k-mer and its reverse complement: $m^{\text{cannonical}} = \min(m, \overline{m^c})$

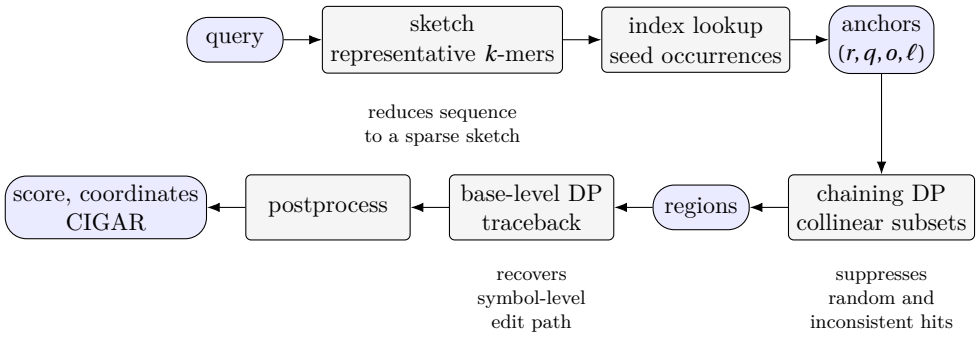


Figure 2.2: The seed–chain–extend decomposition

In case of Minimap2 and BioseqDB $S_{X\cap Y}$ is represented as tuple (r, q, o, ℓ) which holds reference coordinate r, query coordinate q, orientation o and span ℓ (length).

After sorting anchors by ending reference coordinate, with implementation-specific tie keys, a chaining score can be written

$$f(i) = \max\left(w_i,\ \max_{j<i}\{f(j) + \alpha(j,i) - \beta(j,i)\}\right), \tag{2.6}$$

where α rewards newly covered matching bases and β penalizes large gaps and diagonal changes. A representative minimap2 reward is

$$\alpha(j,i) = \min\big(\min(y_i - y_j,\ x_i - x_j),\ w_i\big).$$

Testing every $j < i$ is $O(A^2)$ for A anchors. Minimap2 scans a bounded recent window and stops after a configured number of unproductive indices, making the practical cost much smaller Li, "Minimap2: pairwise alignment for nucleotide sequences."

Algorithm 2 Conceptual bounded predecessor chaining

1. Input. Anchors $a_1, \ldots, a_A$ in canonical coordinate order.
2. Endpoint loop. For each $i = 1, \ldots, A$, perform the following steps:
   (a) Initialize $f[i] \leftarrow \text{span}(a_i)$ and $p[i] \leftarrow \perp$.
   (b) For each candidate $j < i$ in the active distance and iteration window, perform the following steps:
      i. If a_j and a_i have valid order, strand, gap, and bandwidth, compute
      $$v \leftarrow f[j] + \text{reward}(a_j, a_i) - \text{gapPenalty}(a_j, a_i).$$
      ii. If the candidate is valid and $v > f[i]$, set $f[i] \leftarrow v$ and $p[i] \leftarrow j$.
      iii. Optionally stop the predecessor loop after the configured skip heuristic fires.
3. Backtracking. Backtrack from high-scoring endpoints through p.
4. Result. Return the chains that remain after filtering by score, seed count, and overlap.

2.4. Limitations of existing tools

Most of the current most popular tools implement variations of the seed and extend approach described in the previous section. However the algorithm outlined here is heavily underspecified to be directly implemented as a computer program:

- $S_{X \cap Y}$ can be implemented as interection of two hash maps where $Seed(X)$ is realized as map from seed to its start/end: $S \to \natural^2$ where seed itself is represented by it's hash $S = \mathbb{Z}$. This opens few important questions: what hashing strategy to choose? Which map implementation to choose?
- As $Seed(X)$ can be computed before alignment starts we can store it in memory and persist it to disk to speed up processing in the future. Which binary format we want to choose?
- Anchors as tuples will be implemented as some concrete binary structure and authors need to specify their representation (do we store them as AOS or SOA? What are widths of elements and their types?). Each decision will imapct performance as there can be millions of anchors per long sequence and additional bytes sum up to MBs of additional memory to be processed (and slowdowns).
- Finally, what is the most effective way to parallelize this computation (the most straightforward approach is to split input sequences X and Y into chunks)

Most of the current state of the art tools choose to:

1. File-based operation
2. Suboptimal or non-extensible internal data structures
3. Optimise only selected parts of the algorithm like hash function properties or one specific stage.

This leads to different consequences for the user.

File-based operation: Operating directly on the disk with simplistic wire format means that seed index is usually one giant binary blob not prepared for highly-concurrent access and interface doesn't offer modern tools like buffer pooling or lazy loading. The format of indexes are largely undocumented by authors. Missing of lazy loading feature is the most hurting to the user. If index is large it will be loaded entirely into memory and can overwhelm SWAP space. This degrades performance by several orders of magnitude and can be avoided by splitting sequences into smaller chunks. User doesn't know what is the exact relationship between sequence length and required memory space, so this process takes time and involves guessing. Additionally index implementation sometimes offer complex parameters to adjust its partitioning (like in case of MMseqs2), but entire responsibility is still shifted to the user and that leads to user frustration andrecurring OOM issues or severly degraded performance. Youngblut, MMseqs2 Github issue 338 Zhan, MMseqs2 Github issue 517 Software not only treats indexes as files, but it operates on FASTA files directly. This means the user cannot easily sort/split/extract or filter records without using external tools.

Suboptimal or non-extensible internal data structures: Minimap2 uses 31 bits to encode anchor's position meaning that practically entire human genome (>3Gbps) cannot be aligned at once and require splitting via other tools before any operation is performed. LexiMap (not using seed and extend approach) limits maximum genome size to 268Mbp Shen, LexicMap FAQs.

Additionally software is implemented without proper abstractions - minimap2 code is non-easily extensible C code, while G3SA doesn't offer extension points for new types of indexes or sequences interfaces.

Selective optimisations: Optimisations are usually fixed for one single platform (AVX512 or SSE) and either get severe performance penalties (scalar fallbacks) or user cannot use them at all. SIMD support is quite popular even though it supports one or two selected platforms, but GPU support exists usually only for NVIDIA graphics cards. This leads to a difficult situation for the user as they need to choose between several versions of the same tool (Saurabh Kalikar, mm2-fast: Accelerating minimap2 for long-read sequencing applications on modern CPUs, Sadasivan H, mm2-gpu: Accelerating Minimap2 for Accurate Long Read Alignment on GPUs), while better, more modern software like G3SA exists purely on NVIDIA hardware g3sa_pape Modern hardware like MacBook Pro M3 with arm64 architecture won't be able to utilise any integrated GPU acceleration. While minimap2 offers compatibility layer to port SSE code to NEON, it does not vectorize all operations and code ported from SSE doesn't account for lane widths and instruction level parallelism available in M2-M4 chips. Feintzeig, SIMD + Parallelism: Ray Tracer Optimization Part 3

I wanted to address these limitations by implementing BioseqDB, Postgres extension that will provide the following features:

- Tight integration with Postgres meaning that user can operate on sequences (loaded FASTA files) directly using SQL rather than resorting to external toolkits.
- Support for indexes and storage structures designed with parallelism and lazy-loading from the start, meaning that software can be run on very RAM-constrained device and still perform well. At the same time design will incorporate assumptiuon about arbitrary large sequences so the user will be able to operate on >3Gbps records without any additional steps.
- Provide layer of abstractions written using latest C++26 standard, so the algorithms can be reused in separate software or extended to different types of new sequences and indexes.
- Provide optimised kernels for SIMD and GPU architectures written using architecture-agnostic, but performant code.

Tight integration with Postgres also means that as a side effect, we can utilise some benefits of persistent state: deferring expesive operations, keeping large indexing structures in memory in case use reruns query and more effective memory-disk caching schemes.

Chapter 3

The BioSeqDB alignment pipeline

3.1. Alignment as a database execution pipeline

BioSeqDB retains the broad minimap2 sequence of seeding, chaining, and extension, but places it inside a database execution pipeline. An SQL expression is first assigned to a PostgreSQL custom scan. The scan buffers compatible rows, resolves the sequence representation and target-index source, divides the work into memory-admitted waves, and selects a scalar CPU, vectorized CPU, or Vulkan route. Only then does a query enter the biological mapper. The mapper returns typed match records which the scan projects into SQL rows.

This organization separates two related flows. The logical flow transforms sequence symbols into minimizers, anchors, chains, candidate regions, and final alignments. The physical flow decides how sequences and indexes are obtained, which work can share a prepared target, how much memory a wave may reserve, and which processor executes each supported stage. The distinction is important because a fast kernel does not give a fast database operator if the same target index is reconstructed for every row. In the same way, a GPU sketching kernel does not define an end-to-end GPU route if seed lookup and chaining still occur on the CPU.

Figure 3.1 places the long-read route used by BioSeqDB within the conventional read-mapping flow. The upper lane shows a short-read pipeline from another mapper for comparison. Both lanes use the broad stages of seeding, chaining, extension, and traceback. BioSeqDB concentrates on the lower lane, where minimizers reduce the number of positions considered by the selected chaining route.

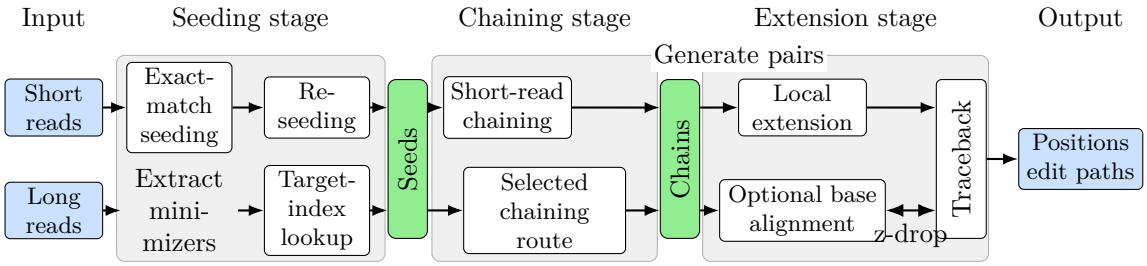


Figure 3.1: Read-alignment stages from input to reported coordinates and edit paths. The upper lane is an external short-read comparison. The lower lane is BioSeqDB's minimizer-based route. The lanes follow the same broad stages without sharing literal interfaces.

Algorithm 3 Logical mapping of one query against a prepared target

Require: Query Q, target T, target index I, parameters P
Ensure: Ordered mapping results
if the entry point begins with sequence symbols then
    $M \leftarrow \mathsf{Sketch}(Q, P.w, P.k, P.minimizer)$
else
    $M \leftarrow \mathsf{ValidatedPrebuiltMinimizers}(Q, P)$
end if
$A \leftarrow \mathsf{LookupAndCanonicalize}(M, I, P.maxOcc)$
$(C, A') \leftarrow \mathsf{Chain}(A, P.chainer)$
$R \leftarrow \mathsf{GenerateRegions}(C, A', |Q|)$
$R \leftarrow \mathsf{ClassifyAndReduce}(R, P)$
if divergence is required then
    $R \leftarrow \mathsf{EstimateDivergence}(R, M, A', P.k)$
end if
if an exact edit predicate permits conservative prefiltering then
    $R \leftarrow \mathsf{RejectImpossibleGapJobs}(R, A', Q, T, P)$
end if
if base-level fields or an exact edit predicate require extension then
    $R \leftarrow \mathsf{Extend}(R, A', Q, T, P.aligner)$
end if
if an exact edit predicate is active then
    $R \leftarrow \mathsf{ApplyExactEditPredicate}(R, P)$
end if
$R \leftarrow \mathsf{ClassifyAndReduce}(R, P)$
if mapping quality is required then
    $R \leftarrow \mathsf{ComputeMapq}(R, P)$
end if
return $\mathsf{ToMatchResults}(R, |Q|, |T|)$

3.2. The logical alignment chain

The generic CPU and SIMD mapper is assembled as a typed push pipeline. Each stage accepts a batch view and emits the next representation while retaining any required data or index owner. The stage type also fixes the selected algorithm. Runtime choices are resolved before the main loop, after which each batch follows one concrete route. The forwarding contract requires exactly one downstream emission, which prevents a stage from dropping or duplicating a batch.

This pipeline has two entry points. The usual entry begins with sequence symbols and computes query minimizers. A second entry accepts authenticated prebuilt minimizers and starts at the index hand-off. The second form is used only when the minimizer representation, source identity, alphabet, w, and k match the requested operation. It is a structural omission of the sketch stage, not an empty sketch operation.

Algorithm 3 gives its common order. Optional work is governed by the SQL projection and predicates. Divergence and mapping quality are not calculated when neither projection nor filtering needs them. Base-level extension is skipped when chain coordinates and scores are sufficient.

3.2.1. Target preparation and minimizers

Target preparation occurs outside the per-query chain. A target may be an in-memory compact sequence, a lazy PostgreSQL sequence, a fragmented value, or a persisted minimizer-index generation. The scheduler first resolves whether the index will be recomputed or read from persistent storage. A prepared CPU target or resident GPU target can then be shared by all compatible queries in a dispatch wave. This placement removes target sketching and index construction from repeated query work.

A persisted index is accepted only when its generation, source identity, storage geometry, alphabet, and minimizer parameters match the request. The pipeline may decode bounded fragments or select partitions, but seed lookup must observe complete occurrence counts and position lists for every minimizer it consumes. A fragment-native CPU or SIMD preparation can go one step further and produce ordered anchors for each query. The later seed stage then performs a checked zero-copy hand-off while the owning batch remains alive.

When minimizers are not prebuilt, the minimizer stage reads the lazy query view and invokes the scalar, SIMD, or Vulkan extractor selected by its trait. CPU implementations write into the current arena. The Vulkan form can return a host view together with a device-resident span, which lets GPU seed lookup reuse the sketch without another upload. An empty query or a query shorter than k produces a valid empty minimizer span.

3.2.2. Seeds, chains, and candidate regions

Seed lookup maps each query minimizer to target occurrences. It discards absent and over-repetitive values according to the occurrence threshold, rejects incomplete persisted postings, converts reverse-strand positions to a common query coordinate, and establishes one canonical anchor order. The position domain remains 64 bit so that a large reference is not narrowed at the stage boundary. Every chaining implementation receives this same order.

The chain stage selects scalar or SIMD dynamic programming, range-minimum-query chaining, or tiled Vulkan dynamic programming. It returns chains and a span of anchors reordered into chain order. Region generation then produces one candidate region for each retained chain, sorts candidates by score, and derives half-open reference and strand-correct query coordinates from the first and last anchors. It also records seeded match and block lengths used by later heuristics.

Figure 3.2 shows the transition on a small example. Exact two-base seeds locate several occurrences in the reference. Chaining compares their order and displacement, then retains the collinear pair. Dynamic programming fills the sequence interval between the anchors. The example uses a compact score matrix to make the extension boundary and traceback explicit.

3.2.3. Reduction, extension, and output

Candidate reduction first assigns primary and secondary relationships among overlapping regions, then compacts the survivor prefix according to score and overlap parameters. Divergence estimation is a demand-driven annotation derived from seed coverage. Neither operation changes the coordinate conventions established by region generation.

The optional prefilter is a BioSeqDB addition. For a region with an active exact edit predicate, it constructs mandatory seed-to-seed gap jobs and uses SneakySnake or GateKeeper to obtain a conservative lower bound on the number of edits. A region is rejected only when the accumulated lower bound already exceeds the predicate threshold. Unsupported geometry

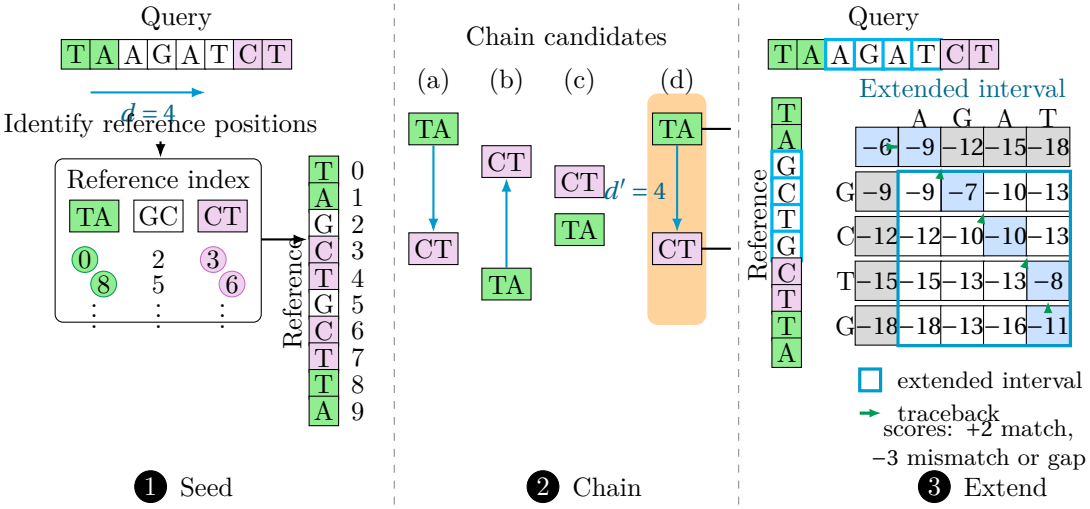


Figure 3.2: Worked seed, chain, and extend example. Two exact seeds identify candidate reference positions. The selected pair has consistent order and displacement. Dynamic programming then aligns the interval between the anchors, with the green arrows marking traceback.

proceeds to exact alignment. The prefilter therefore avoids some extension work without replacing the chosen-path edit count.

The extension stage selects KSW2, Block Aligner, an inter-region batched KSW2 route, or an admitted Vulkan CIGAR route. CPU batching groups short fill problems by length, bandwidth, and orientation, while long fills use an intra-sequence kernel. The Vulkan route forms device jobs and chooses a short script or bounded dynamic programming with traceback according to the admitted shape. Packed edit operations are decoded into owned result fields after device execution.

After extension, any exact edit predicate is applied. The primary and secondary classifier then runs again. If filtering removed a candidate, reclassification ensures that parent choice, secondary scores, and mapping quality refer only to the final candidate set. The sink converts each region to a match record with 64-bit coordinates, scores, seed and chain metadata, and only the optional divergence, mapping quality, edit count, or CIGAR fields required by projection or filtering.

Table 3.1 summarizes the stage boundaries. The table groups adjacent control stages where they share one semantic contract.

3.3. Compile-time algorithm selection and runtime routing

The runtime parameter object contains minimizer dimensions, scoring limits, output requirements, prefilter policy, and algorithm selectors. The dispatcher resolves these values into a tuple containing the sequence shape, base aligner, chainer, minimizer implementation, and pipeline entry point. Registered common combinations receive compile-time specializations. Other valid combinations use a typed generic implementation. This arrangement removes repeated selector branches from high-frequency stage calls without instantiating every possible product of w, k, sequence length, and algorithm choice.

Scalar and SIMD execution share the same logical pipeline. A SIMD selection replaces the relevant minimizer, chaining, or alignment kernel, while region generation and candidate

Table 3.1: Contracts of the BioSeqDB logical alignment pipeline.

Stage	Input	Output	Required property
Minimizer	lazy query symbols	host minimizers and optional device span	Deterministic representation for w, k, alphabet, and position domain. Short queries yield an empty valid span.
Index hand-off	query minimizers	minimizers with a target-index reference	The index identity and lifetime match the target and cannot alias another segment.
Seed lookup	minimizers and index	canonical anchors	Occurrence pruning uses complete postings and preserves strand-correct full-width coordinates.
Chaining	canonical anchors	chains and reordered anchors	Every kernel consumes the same scoring parameters and anchor order.
Region reduction	chains and anchors	classified candidate prefix	Coordinates are valid and removed candidates are not visible downstream.
Prefilter	candidates and sequence windows	conservative survivor prefix	Only a lower bound above the exact threshold can reject a candidate.
Extension and exact filter	candidates, anchors, and sequences	scored regions with optional edit paths	The selected aligner supplies alignment-derived fields, including the chosen-path edit count.
Reclassification and sink	post-filter candidates	ordered match records	Final classification excludes removed regions and optional fields are computed only when required.

classification remain scalar control stages. An "all SIMD" route therefore means that every biological stage with a SIMD implementation used it. It does not imply that input handling, allocation, or result construction consists of vector instructions.

CPU concurrency is controlled at statement scope. Independent queries can be distributed through a oneAPI Threading Building Blocks (TBB) task arena. Coarse inner work, such as alignment regions or fragment preparation, may also be parallelized, but only one nested coarse region is active at a time. A nested request that would create another fan-out runs serially. This rule bounds the effective worker count and prevents a query-level parallel loop from multiplying a region-level loop.

The dedicated Vulkan route has a narrower contract. A full GPU selection requires compatible Vulkan minimizer extraction, resident index lookup, tiled chaining, and coordinate reduction. CIGAR output additionally requires an admitted device extension shape. A GPU work wave contains at least two jobs so that fixed submission costs are not charged to a single comparison. If the runtime, target representation, or requested output does not satisfy the selected mandatory route, execution returns an error. It does not perform the biological operation on a CPU while retaining the GPU selection.

A standalone Vulkan prefilter does not select the full GPU route. It executes one conservative filter inside an otherwise CPU or SIMD pipeline. This distinction prevents a device-accelerated filter from changing the meaning of the mapper's backend label.

Persisted GPU indexes also have a specific boundary. A compatible resident partition can support direct coordinate projection without rebuilding target minimizers. Persisted CIGAR projection is not supported because extension needs target windows in a device-readable representation. The pipeline rejects that combination instead of hydrating the target and completing the alignment on the CPU.

Table 3.2: Selectable biological stages and their execution routes.

Stage	Available routes
Minimizer extraction	scalar minimap-compatible sketching, SIMD rolling-hash variants, and Vulkan minimizer extraction
Seed lookup	in-memory or persisted CPU providers, fragment-native prepared anchors, and resident Vulkan index lookup
Chaining	scalar dynamic programming, SIMD dynamic programming, range-minimum-query chaining, and tiled Vulkan dynamic programming
Final alignment	KSW2, Block Aligner, inter-region batched KSW2, and an admitted Vulkan CIGAR route
Prefilter	scalar or SIMD SneakySnake, Vulkan SneakySnake, and Vulkan GateKeeper within their supported input domains

3.4. Batching, memory, and target reuse

The PostgreSQL custom scan does not invoke the mapper once for each input tuple. Its dispatch buffer retains eligible scan slots together with their query and target operands. The buffer is drained when it reaches its bounded capacity or when input ends. Each batch is homogeneous in alphabet and packed storage width, so the typed processor resolves that pair once instead of switching representation in the per-row loop. During the read phase, result cursors restore the association between a mapping result and its buffered SQL row.

Rows are grouped by query mode, physical target identity, and index policy. This grouping also supports a broadcast query, a query expression evaluated per row, and several independent query predicates. When several predicates use the same active wave, their matching row identifiers are intersected. The physical work can therefore share target preparation without changing the logical conjunction expressed by SQL.

Sequence operands may be flat, fragmented, or lazy. Some kernels consume a lazy compact range directly, while others require a flat payload. Before a wave is dispatched, the scheduler estimates its required hydration bytes and requests a memory permit. If the reservation succeeds, only operands required by the selected kernel are materialized. Their storage is destroyed before the permit is released. If hydration is not admitted and the kernel supports streaming, the operands remain lazy and are read through the PostgreSQL scanner. Reducing materialization does not change the selected biological algorithm.

The memory controller combines the process footprint, current system availability, a process soft limit, a minimum free-memory reserve, and any index-specific limit. Parallel work first attempts the requested fan-out, then reduces the wave size until it fits. If one parallel unit cannot be admitted, the same algorithm runs serially when its minimum working set fits. This is a scheduling fallback, not a scalar algorithm substitution.

Prepared targets define the main reuse scope. A CPU target index, persisted reader, fragment-native anchor owner, or resident Vulkan index is constructed once for a compatible group of queries. Query-local arena scopes are released after each serial query, while the outer prepared target remains alive for the wave. On a CPU, sufficiently large groups of shareable queries may run in parallel. A selected Vulkan route suppresses this CPU fan-out because the device scheduler must observe the complete work set and form resident jobs. Both routes return results in input-query order.

Algorithm 4 states the wave-level policy. The essential ordering is admission before materialization, target preparation before query execution, and result scattering before wave storage is released.

Algorithm 4 Memory-aware dispatch of buffered SQL rows

Require: Buffered rows B, typed sequence layout S, parameters P
Ensure: Mapping results associated with their input row identifiers
Group B by query mode, target identity, and index policy
Estimate materialization bytes for each group
Partition groups into waves under the live memory limit
for each wave W do
$permit \leftarrow \mathrm{TryReserve}(\mathrm{Bytes}(W))$
if $permit$ exists then
    Materialize only operands required by the selected route
else
    Retain lazy operands if the selected route supports streaming
end if
Prepare or reuse one target object for each target identity
if a mandatory Vulkan tuple is selected then
    Validate wave size, runtime support, index form, and output shape
    Run resident query jobs on the device
else
    Run typed scalar or SIMD queries serially or in the TBB arena
end if
Scatter result collections to buffered row identifiers
Destroy wave storage and release $permit$
end for

3.5. Relation to minimap2 and GPU mappers

Upstream minimap2 Li, "Minimap2: pairwise alignment for nucleotide sequences" organizes long-read mapping as an imperative sequence. It sketches the query, retrieves minimizer occurrences from a reference index, prepares and sorts anchors, applies dynamic-programming or range-minimum-query chaining, generates regions, performs post-chain processing and extension, and calculates mapping quality. Parallelism is mainly applied across input fragments, with a worker-local allocation buffer. Figure 3.3 presents this three-module structure and the main substitution points used by vectorized variants.

BioSeqDB preserves this recognizable algorithmic backbone but changes the execution context. Queries are packed, alphabet-aware ranges rather than only text strings. The target index may be in memory, persisted with a database generation, selectively decoded, or resident on a device. Runtime selectors are resolved into typed stages before query execution. Work is grouped by a custom scan rather than by a command-line reader. Parallelism is controlled by both a TBB arena and memory admission. Results are typed records inserted into a relational projection rather than only mapping-format lines.

The scope is also narrower in some biological dimensions. BioSeqDB focuses on database-resident long-read and single-target mapping. It does not claim all paired-read, splice, rescue, or multi-segment modes supported by minimap2. The pipeline adds database-oriented features instead, including lazy sequence views, persistent index identity, projection-driven optional fields, conservative edit prefiltering, and reuse of one prepared target across many SQL rows.

G3SA Han et al., "G^{3}SA: A GPU-Accelerated Gold Standard Genomics Library for End-to-End Sequence Alignment" provides a useful comparison for accelerator design. Its minimap2 route treats sketching, seed lookup, chaining, and extension as a resident GPU work graph. Anchors are stored in a device-friendly layout, tiled chaining reduces repeated global-memory access, and

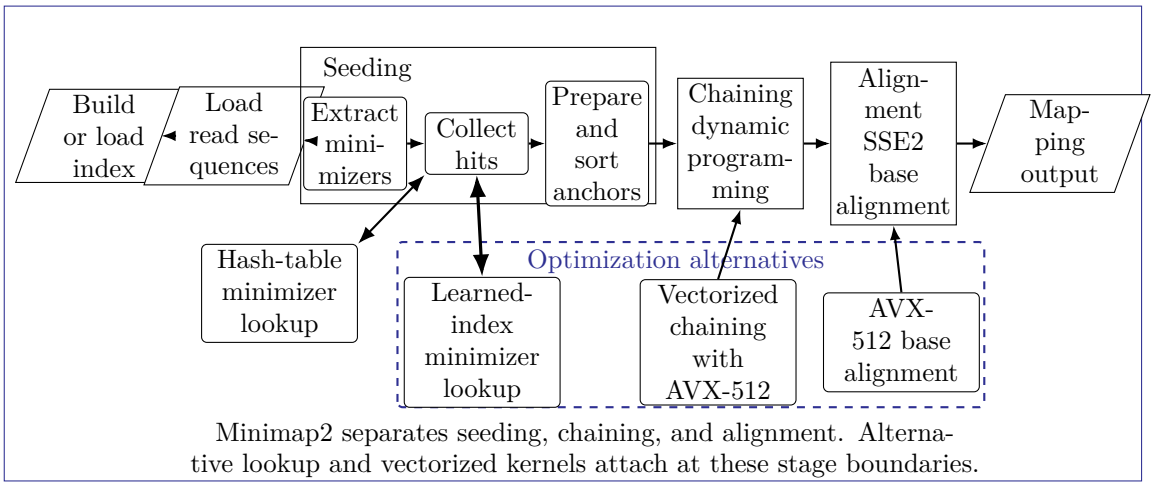


Figure 3.3: The minimap2 mapping workflow and its principal replacement points. The main line contains seeding, chaining, and base-level alignment. The lower boxes show alternative lookup and vectorized kernels attached to the same interfaces. They illustrate modular optimization choices and do not denote BioSeqDB-specific implementations.

gridded traceback stores boundaries instead of a complete direction matrix. These choices show why batching and residency matter at least as much as an isolated fast kernel.

BioSeqDB applies the same systems observation through Vulkan, but database execution imposes different constraints. Query batches are grouped by exact target identity. A target index can be retained across jobs and intermediate device spans can pass directly between sketching and seed lookup. Tiled chaining produces candidate coordinates, while an admitted extension route can generate packed CIGAR operations. Host code still constructs the final SQL-visible objects. Memory permits bound host materialization and device work is divided into bounded waves.

The failure policy also differs. A GPU mapper may send exceptional reads to a CPU worker. BioSeqDB does not silently do so after a mandatory Vulkan route has been selected for an SQL statement. Unsupported tuples, one-job waves, unavailable device operations, and incompatible persisted output shapes are reported as errors. The scheduler may choose a CPU or SIMD route before it enters the mandatory device path. Once the device route is selected, backend substitution would change the stated execution contract.

3.6. Final invariants and current boundaries

The resulting pipeline is governed by six main invariants.

1. The generic CPU and SIMD mapper uses one typed stage order. The resident GPU route uses a separate graph.

2. Runtime algorithm selectors are consumed before the hot loop. Common tuples are specialized and valid uncommon tuples use a typed fallback.

3. Arena spans, lazy providers, persisted readers, and device handles carry ownership that outlives downstream use.

4. Prepared targets have an explicit reuse scope across compatible query waves. Query-local storage can be released without invalidating the target.

Table 3.3: Pipeline-level differences between minimap2, G3SA, and BioSeqDB.

Aspect	minimap2	G3SA minimap2 route	BioSeqDB
Work unit	input read fragment	fixed GPU read batch	PostgreSQL custom-scan wave
Sequence form	text query with an external reference index	pinned host input and device encodings	packed lazy, fragmented, or scanner-backed ranges
Index lifetime	shared by worker calls	retained on the GPU	in-memory, persisted, fragment-native, or resident with database identity
Control flow	imperative mapper with runtime options	stage-specific GPU batches	runtime-to-template routing followed by typed push stages
Parallelism	worker loop across fragments	GPU batches with host support threads	TBB tasks and Vulkan waves constrained by memory admission
GPU failure	no upstream GPU route	exceptional work may move to CPU	a selected mandatory Vulkan route fails if it cannot execute
Result surface	mapping-format records	packed alignments and mapping output	typed match fields projected into SQL rows

5. Worker concurrency and memory admission jointly bound parallel work. Reducing fan-out or materialization does not change the biological kernel.

6. A mandatory Vulkan route is fail-closed. An independent GPU prefilter remains a local acceleration inside a CPU or SIMD pipeline.

Several limits remain. The CPU stage chain and the resident GPU batch route share algorithms and result contracts, but they are not the same instruction graph. Direct use of a persisted GPU index currently supports coordinate projection, not persisted CIGAR extension. Final result objects are built on the host. Small query groups may also be slower on a GPU because submission, allocation, and transfer costs are not amortized.

Within these limits, BioSeqDB turns a procedural mapper into a bounded database operator. It retains the seeding, chaining, and extension structure of minimap2 while moving target preparation outside repeated query work. Typed stages define the data exchanged by algorithms. Lazy sequence access, memory-admitted batching, controlled CPU parallelism, and resident Vulkan waves connect those algorithms to PostgreSQL execution. The following chapters examine the principal biological stages in detail, beginning with minimizer extraction.

Chapter 4

Minimizer extraction

4.1. From every base to a sparse, comparable sketch

The first computational stage of the BioSeqDB mapper replaces a sequence with a smaller set of representative substrings. These representatives are called seeds. A target seed becomes a key in an index. A query seed is looked up in that index, and each compatible occurrence becomes an anchor for the chaining stage described in Section 5.1. The seed set must be sparse enough to reduce index and lookup costs, but dense enough that related sequences retain common seeds.

BioSeqDB primarily uses minimizers. A minimizer is the lowest ranked k-mer in a window of consecutive k-mers Roberts et al., "Reducing storage requirements for biological sequence comparison." This definition is compact, but it leaves several choices open. The hash function, reverse-complement convention, tie rule, position convention, treatment of ambiguous symbols, and fragment boundary rule all affect the emitted seed stream. A query sketch and a stored target index must make the same choices. Equal machine widths do not make two different hash schemes compatible.

This chapter first defines the minimizer contract and its role in seed lookup. It then compares the scalar minimap2 algorithm, the data-independent method of SIMD-Minimizers, and the BioSeqDB CPU and GPU variants. Spaced seeds, strobemers, and fuzzy hashing are included to distinguish minimizers from related seed constructions. These alternatives clarify which properties belong to seed selection and which belong to hashing or index lookup.

4.1.1. Canonical k-mers and minimizer windows

Let $S = s_0 s_1 \dots s_{n-1}$ be a nucleotide sequence. The k-mer beginning at zero-based position i is

$$K_i = S[i, i + k).$$

For DNA, the forward and reverse-complement encodings are

$$F_i = \mathrm{enc}(K_i), \qquad R_i = \mathrm{enc}(\overline{K_i}).$$

A canonical k-mer uses the same key on both strands. A common choice is

$$C_i = \min(F_i, R_i), \qquad z_i = \mathbf{1}[R_i < F_i],$$

where z_i records the selected strand. Applying a hash function H gives a candidate

$$x_i = (h_i, p_i, z_i), \qquad h_i = H(C_i).$$

The position p_i may denote the first or final base of K_i. BioSeqDB's minimap-compatible schemes use the final base because that is the convention used by minimap2 Li, "Minimap2: pairwise alignment for nucleotide sequences."

For a window of w consecutive candidates,

$$W_j = (x_j, x_{j+1}, \ldots, x_{j+w-1}), \qquad m_j = \min_{x \in W_j} h(x).$$

Every identity attaining m_j is a possible minimizer. A complete algorithm must state which equal minimum wins and whether different positions with the same minimum hash are emitted. It must also avoid emitting the same identity again when that candidate remains minimal as the window moves. For independent random hashes, the expected density is approximately $2/(w+1)$ minimizers per candidate position.

Figure 4.1 follows one k-mer through hashing, window selection, and adjacent deduplication. The per-window minimum positions are calculated from the displayed hashes. In particular, the window beginning at hash position five has minimum position eight, not position five.

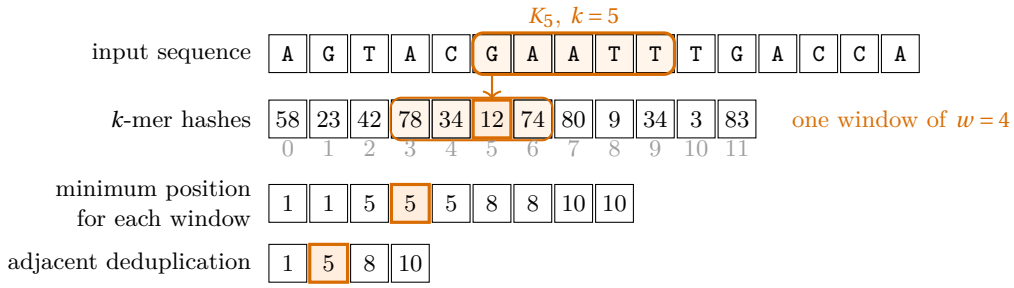


Figure 4.1: Minimizer extraction for $k = 5$ and $w = 4$ with illustrative hashes. Positions in this walkthrough are zero-based k-mer starts. BioSeqDB's packed minimap-compatible record converts them to final-base positions. The highlighted candidate has hash 12 and remains minimal for three consecutive windows, so adjacent deduplication emits its position once.

Equal hash values require a distinction between hash equality and event identity. Two candidates at different positions can have the same hash. They are tied minima, not duplicate records. BioSeqDB therefore carries position and strand alongside the hash until the tie rule has been applied. Only identical adjacent events can be removed without changing the seed set.

An ambiguous symbol invalidates every k-mer spanning that symbol. The rolling encoding and the count of consecutive valid symbols must be reset. The window state is handled according to the selected scheme. This distinction matters when fragments are processed independently or when an ambiguous symbol occurs near a fragment boundary.

4.1.2. From minimizers to candidate anchors

Sketching and lookup are separate operations. The target side stores each seed key together with an occurrence payload. At minimum, that payload identifies the target sequence, position, and strand. The query side computes a compatible key and retrieves its occurrences. A retrieved occurrence paired with the query seed position forms an anchor. Hash equality alone is not proof of seed equality because collisions are possible. An implementation may

store enough key material to verify the seed immediately, or it may let later filtering and chaining reject false candidates.

Figure 4.2 uses fixed four-base seeds on both sides. The symbols h_1, h_2, and h_3 are illustrative fingerprints rather than outputs of a particular hash function.

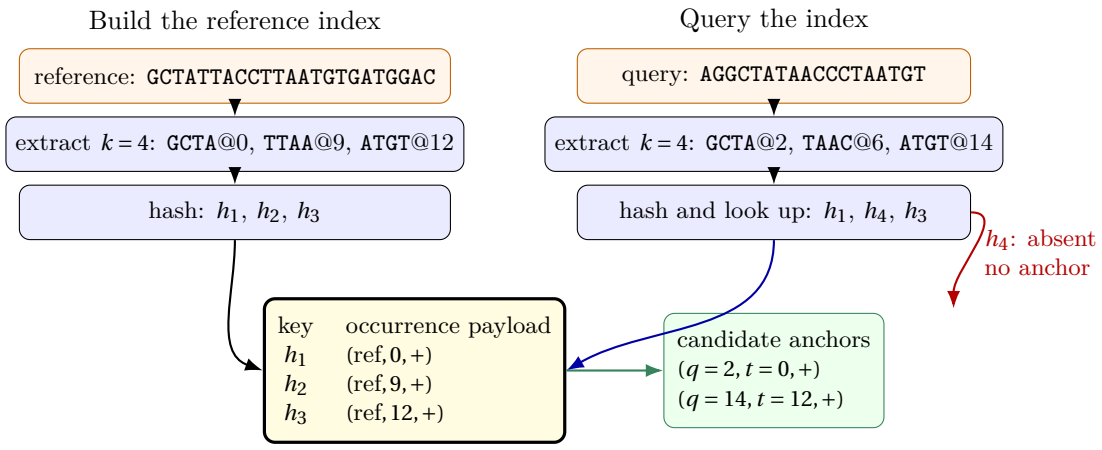


Figure 4.2: Reference-index construction and query lookup. Positions are seed starts in this example. A usable seed index stores occurrence payloads rather than hash values alone. A matching hash produces a candidate anchor. Seed verification or downstream filtering is still required when the stored key does not exclude collisions.

Frequent seeds can produce very large occurrence lists. BioSeqDB can exclude seeds whose target frequency exceeds a configured threshold before anchors are constructed. This policy changes runtime and sensitivity, but not the minimizer hashes. For a capped persistent index, the requested threshold must not exceed the stored count cap.

4.1.3. Minimizers, spaced seeds, and linked seeds

Spaced seeds, linked seeds, and BLEND are comparative designs in this chapter. The current BioSeqDB backends described later implement minimizer schemes.

The terms selection and construction describe different parts of seeding. A minimizer selects one or more complete k-mers from a window by hash rank. A spaced seed applies a binary mask to one fixed span and hashes only retained positions. A linked seed, including a strobemer, chooses components from separated downstream windows and combines them into one key. The intervening sequence is not part of the linked key.

These choices respond differently to mutations. A substitution can change all overlapping contiguous k-mers. A spaced mask ignores changes at omitted positions. A strobemer can retain its component strings when mutations occur between the selected windows. The linking rule must nevertheless be specified. Without selection windows and a deterministic rule, concatenating arbitrary k-mers does not define a reproducible strobemer. One possible rule selects component s_j by minimizing $H(s_1 \| \cdots \| s_{j-1} \| u)$ over candidates u in the jth downstream window.

Figure 4.3 compares the three operations. The numeric hash ranks are local examples. They do not imply that unrelated seed schemes share a hash domain.

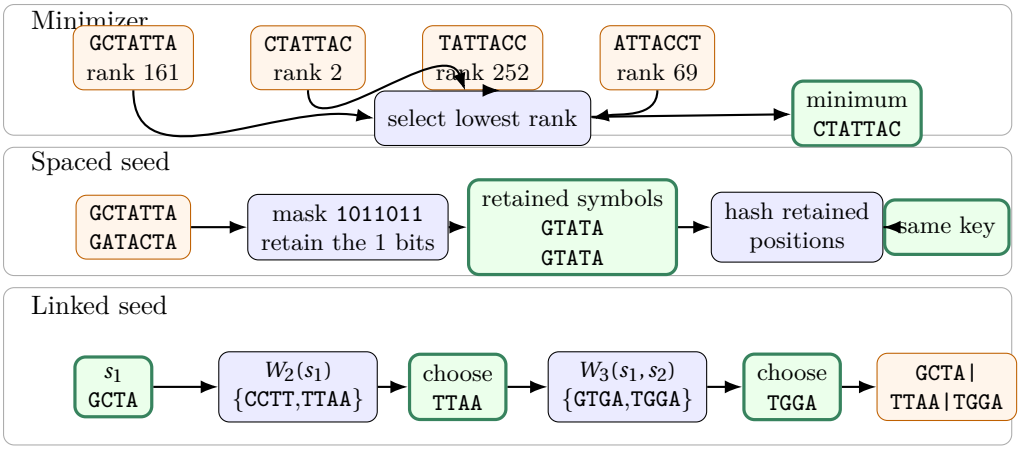


Figure 4.3: Three seed constructions. A minimizer selects the lowest ranked k-mer in a window. A spaced seed retains positions marked by a binary mask. A linked seed chooses components from explicit downstream windows, where each choice can depend on earlier components. The final linked key omits the bases between the selected components.

4.1.4. Fuzzy seed hashing with BLEND

Conventional hashes are designed so a small input change produces an unrelated output. That behavior is useful for uniform minimizer sampling, but it prevents approximate seeds from meeting in a hash table. BLEND instead uses a SimHash construction to make related seeds more likely to share a fingerprint Firtina et al., "BLEND: a fast, memory-efficient and accurate mechanism to find fuzzy seed matches in genome analysis." BLEND-I tokenizes a contiguous seed into overlapping substrings. BLEND-S applies the same principle to linked seed components.

Let a seed produce token hashes $g_1, \ldots, g_r$, each represented by b bits. Bit j of the fuzzy fingerprint can be written schematically as

$$f_j = \mathbf{1}\left[\sum_{q=1}^{r} \left(2g_{q,j} - 1\right) > 0\right].$$

Thus each token contributes a positive or negative vote to every output bit. With an odd token count, the displayed rule has no zero vote. A concrete implementation also defines its behavior for even counts. Similar seeds share many token hashes, so many votes remain unchanged. This raises the probability of an equal fingerprint. It does not guarantee equality, and unrelated seeds can still collide.

When BLEND supplies the rank for a minimizer, every candidate k-mer first receives a fuzzy fingerprint and the window selects the lowest fingerprint. Equal fingerprints must follow the minimizer implementation's stated tie rule. For a linked seed, BLEND-S hashes the selected components and uses the result directly as the lookup key.

Figure 4.4 separates fingerprint construction from index lookup. The bit strings are schematic. They illustrate voting rather than a fixed BLEND parameter set.

4.2. The minimap2 event contract

Minimap2 uses two-bit nucleotide ranks

$$\eta(A) = 0, \qquad \eta(C) = 1, \qquad \eta(G) = 2, \qquad \eta(T) = 3.$$

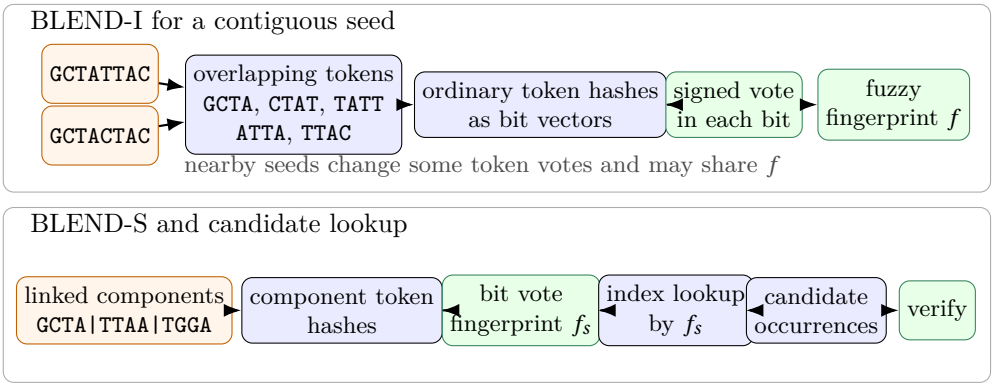


Figure 4.4: Fuzzy hashing in BLEND. Token hashes vote independently on each fingerprint bit. The upper panel shows a possible shared fingerprint for two nearby seeds, not a guaranteed outcome. Index lookup therefore returns candidates that still require verification or downstream filtering.

For a valid base code c_i, it updates forward and reverse-complement words without rebuilding the k-mer

$$F_i = ((F_{i-1} \ll 2) \mid c_i) \,\&\, (2^{2k} - 1),$$
$$R_i = (R_{i-1} \gg 2) \mid ((3 \oplus c_i) \ll 2(k-1)).$$

The smaller word is canonical. A reversible integer mixing function maps it to a well-distributed hash in the $2k$-bit domain. The candidate position is the final base of the k-mer, and the strand bit records which orientation provided the canonical word Li, "Minimap2: pairwise alignment for nucleotide sequences."

The sliding minimum is maintained with a circular buffer of w candidates and one saved minimum. A new candidate replaces the saved minimum when its hash is less than or equal to the current value. The less-than-or-equal rule makes the entering candidate the rightmost equal minimum. When the saved candidate expires, the ring is scanned from oldest to newest with the same rightmost preference. The first complete window and the expiry branch also report distinct identities tied with the selected hash. The final live minimum is emitted when input ends.

These details make minimap2 output an event stream rather than one value for each mathematical window. Algorithm 5 gives the essential state transitions. The event logic is shown separately from packed record widths because the same semantics can be represented by different integer layouts.

An ambiguous base and a storage-fragment boundary are not the same event. An ambiguous base ends the run of valid k-mers. A storage boundary should be invisible when the two fragments are adjacent parts of one logical sequence. The collector must either continue its rolling and window state or replay enough context to reconstruct the same state.

There is one deliberate difference between upstream minimap2 and BioSeqDB's minimap-compatible scalar definition. Upstream minimap2 skips a palindromic k-mer iteration because its strand is undefined. BioSeqDB advances the window with a maximum-valued sentinel. The palindrome cannot become a minimizer, but it consumes one chronological window slot. The CPU and GPU implementations that claim the BioSeqDB minimap-compatible scheme follow this sentinel rule.

Algorithm 5 Minimap-style scalar minimizer collection

Input: sequence (S), (k), and window width (w)
Output: minimizer events in chronological order

1. Initialize forward and reverse rolling words. Fill a circular buffer of (w) candidates with sentinels and set the current minimum to a sentinel.
2. For each input symbol at position (i), construct one candidate.
   (a) If the symbol is ambiguous, reset both rolling words and the valid-run length. Use a sentinel candidate.
   (b) Otherwise update (F) and (R). Once the valid run has length at least (k), hash the smaller word and attach position and strand. A palindromic word is represented according to the selected compatibility rule.
3. Replace the next circular-buffer slot with the candidate.
4. If the candidate hash is no greater than the current hash, emit the old live minimum when required and make the entering candidate the new minimum.
5. Otherwise, if the overwritten slot held the current minimum, emit that event, rescan the ring, and retain the rightmost equal minimum. Emit other distinct identities tied in the required chronology.
6. Handle the first complete window explicitly. After the final symbol, emit the last live minimum.

4.3. Data-independent sliding minima

The scalar ring is efficient because a full rescan occurs only when the saved minimum expires. Direct lane-wise vectorization changes the cost. If the probability of one stream requesting a rescan is approximately $1/w$, then L independent vector lanes request at least one rescan with probability

$$1-\left(1-\frac{1}{w}\right)^{L}.$$

For $w = 10$ and four lanes this is about 34.4%. A vector implementation must then mask unaffected lanes or move their state through a scalar slow path. The vector instruction computes several hashes, but the control flow remains irregular.

Three common sliding-minimum structures expose different tradeoffs.

Method	Update rule	Consequence
Circular buffer with rescan	Keep the current minimum and scan (w) candidates when it expires	Small scalar state, but data-dependent work
Monotone deque	Remove expired values at the front and no-smaller entering values at the back	Amortized linear work, but a data-dependent number of pops per candidate
Block prefix and suffix minima	Precompute minima from each block boundary and combine two values per window	Fixed work per block and regular lane-wise execution

Table 4.1: Alternative structures for a sliding-window minimum. The block prefix and suffix form is useful for SIMD because each crossing window uses one fixed comparison.

Partition the hash array into blocks of exactly w values. For a position i in block B_q, define

$$P_i = \min(a_{qw}, \ldots, a_i), \qquad Q_i = \min(a_i, \ldots, a_{qw+w-1}).$$

Here P_i is a prefix minimum from the block start and Q_i is a suffix minimum to the block end. A window beginning at j and ending at $e = j + w - 1$ satisfies

$$\min(a_j, \ldots, a_e) = \min(Q_j, P_e).$$

A tie-aware implementation carries endpoint identities with each value. It can therefore retain the leftmost and rightmost occurrences of an equal minimum without rescanning the window.

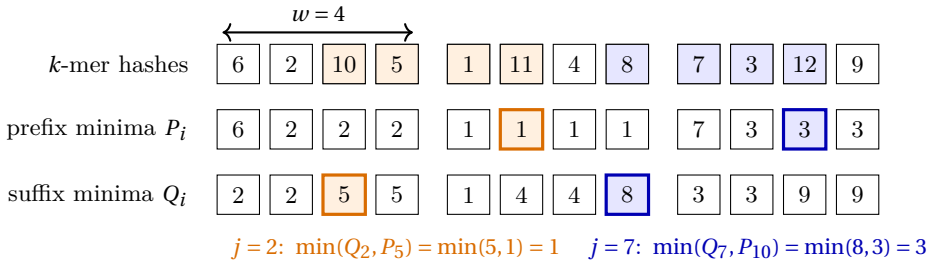


Figure 4.5: Sliding minima from blockwise prefixes and suffixes. Blocks have the same width as the minimizer window. The two shaded crossing windows each use one suffix value from the starting block and one prefix value from the ending block.

4.3.1. Rolling ntHash

SIMD-Minimizers combines blockwise minima with the ntHash family Mohamadi et al., "ntHash: recursive nucleotide hashing"; Groot Koerkamp and Martayan, "SimdMinimizers: Computing Random Minimizers, fast." Associate each base with a pseudorandom word $f(b)$ and let $\mathrm{comp}(b)$ denote its complementary base. For a rotation distance r, define

$$H_i^F = \bigoplus_{j=0}^{k-1} \mathrm{rotl}\big(f(s_{i+j}), r(k-1-j)\big),$$
$$H_i^R = \bigoplus_{j=0}^{k-1} \mathrm{rotl}\big(f(\mathrm{comp}(s_{i+j})), rj\big).$$

After initialization, both states advance in constant time

$$H_{i+1}^F = \mathrm{rotl}(H_i^F, r) \oplus \mathrm{rotl}(f(s_i), rk) \oplus f(s_{i+k}),$$
$$H_{i+1}^R = \mathrm{rotr}(H_i^R, r) \oplus \mathrm{rotr}(f(\mathrm{comp}(s_i)), r) \oplus \mathrm{rotl}(f(\mathrm{comp}(s_{i+k})), r(k-1)).$$

The canonical ntHash value used by SIMD-Minimizers is the wrapping sum of the forward and reverse states. This hash family differs from minimap2's integer mix of a canonical two-bit word. The selected minimizers can therefore differ even when both implementations calculate their own algorithms correctly.

The published SIMD-Minimizers method divides the sequence into eight logical chunks with enough overlap to cover every owned window. Corresponding positions from the chunks occupy vector lanes. Rolling hashes and prefix or suffix minima are updated in lock step.

The algorithm uses a reduced hash key for selection and resolves equal extremes through a canonical orientation of the complete $(k+w-1)$-base window Groot Koerkamp and Martayan, "SimdMinimizers: Computing Random Minimizers, fast." These choices define a seed contract rather than a transparent optimization of minimap2.

Property	Canonical minimizer abstraction	Minimap2	SIMD-Minimizers
Candidate key	Hash of one canonical k-mer	Integer mix of $\min(F_i, R_i)$	Rolling forward and reverse ntHash sum
Window minimum	Any lowest ranked candidate	Circular ring with expiry rescans	Block prefix and suffix minima
Equal minima	Requires an explicit policy	Rightmost entering winner with additional chronological tie events	Extreme endpoint selected from the canonical orientation of the complete window
Parallel structure	Not specified	Sequential candidate stream	Eight overlapped logical chunks processed lane-wise
Compatibility	Defined only after all policies are fixed	Minimap2 hash, position, and event rules	Separate hash width, endpoint, and orientation rules

Table 4.2: The abstract minimizer definition does not determine an interoperable seed stream. Minimap2 and SIMD-Minimizers complete the definition in different ways.

4.3.2. Chunked vector extraction

A chunked extractor assigns each logical chunk an owned interval of candidate positions. The chunk also reads a right context of at least $k+w-2$ bases when ownership is defined by window starts. The first $k-1$ extra bases complete a k-mer near the boundary. The remaining $w-1$ candidates complete windows that begin in the owned interval. Equivalent left-context formulations are possible, but ownership and context direction must agree.

Each vector lane advances through one logical chunk. The hash and sliding minimum stages return chunk-local positions. Adding the chunk origin converts them to absolute sequence positions. Collection then restores chunk order and removes only repeated identities caused by neighboring windows or overlap. Equal hashes at different positions remain distinct.

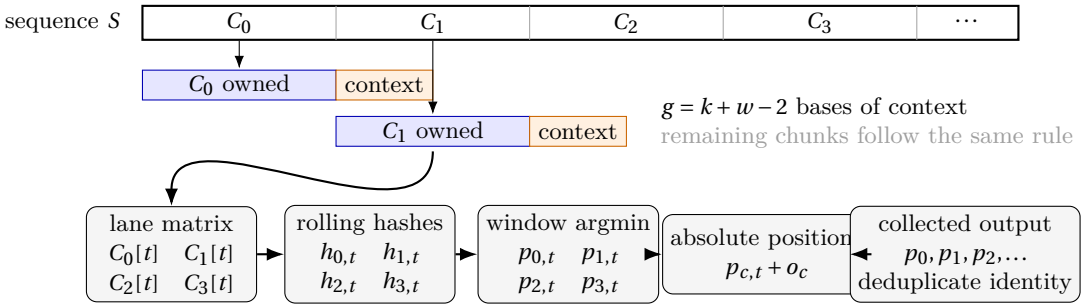


Figure 4.6: Vectorized minimizer extraction over logical chunks. Each chunk owns a disjoint range and reads enough context to complete its boundary windows. Lane-wise rolling hashes and minima produce local positions. Adding each chunk origin and deduplicating full identities yields a global sketch. The symbolic chunk size is an execution parameter rather than part of minimizer semantics.

Lane-wise output is naturally ordered by vector step rather than by logical chunk. A transpose converts vectors of simultaneous events into one sorted list per chunk. Adjacent

deduplication is valid within each list because its positions are ordered. The last emitted position for each chunk is an independent carry value. It must not be mistaken for one value shared across all rows.

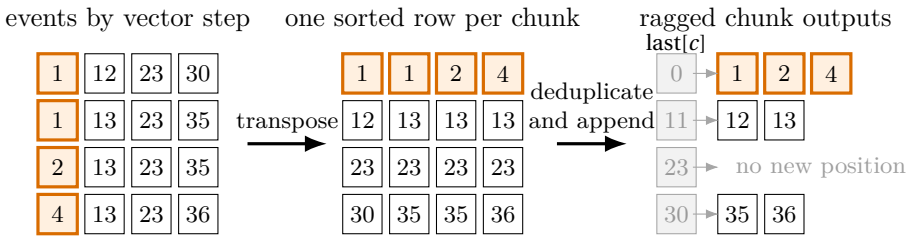


Figure 4.7: Transposition and boundary deduplication for four logical chunks. The input rows are vector steps and the columns are chunks. Transposition makes each chunk a sorted row. Each row has its own previous-position carry, shown in gray. Only new positions are appended, so padding cells are not part of the output.

4.4. BioSeqDB minimizer backends

BioSeqDB separates a seed scheme from an execution backend. The scheme determines the candidate hash, tie policy, ambiguity behavior, position encoding, and fragment semantics. The backend determines where and how those rules are executed. The ExactSimd label covers two distinct selection routes. Its continuation route uses the Scalar candidate order, while its resident vector-stream route has a different ordering contract. PaperNtHashSimd defines another incompatible scheme.

Persistent-index metadata records the seed-scheme identity and the producing backend identity and revision. Query preparation must match these values. Rebuilding only the query sketch cannot repair a mismatch.

4.4.1. Scalar reference

Scalar is the portable implementation of BioSeqDB's minimap-compatible event contract. It uses the two-bit canonical word, the minimap-style integer mix, rightmost equal-minimum selection, chronological tie events, sentinel palindromes, ambiguity resets, and a final flush. It is also the default CPU selection when no more specific backend is requested.

The scalar algorithm is important even when a faster backend is available. It states the complete event order without lane transposition or tile ownership. The ExactSimd continuation route targets the same output. VulkanSegment targets the same DNA4 output on a device. The resident ExactSimd route and PaperNtHashSimd have separate contracts.

4.4.2. ExactSimd continuation

The ExactSimd continuation route uses vector operations to reproduce the BioSeqDB stream that is compatible with minimap across a fragmented sequence. A fragment boundary does not reset biological state. The continuation retains at most the following bounded information:

- the final $k-1$ normalized symbols needed by the rolling k-mer
- fewer than one pending vector group and its absolute positions

- the last $w-1$ candidates needed to complete crossing windows
- the current minimum and the chronology needed for event emission

The bound depends on k, w, and vector width, not on the total sequence length or number of fragments. When the next fragment arrives, it completes the pending group and continues the same logical stream. An ambiguous symbol instead clears the rolling k-mer context and inserts the scheme's sentinel candidate into the existing window chronology.

The common $k = 15, w = 10$ specialization processes eight consecutive candidate positions per block. It needs $k + 8 - 1 = 22$ base codes to form those overlapping k-mers. The codes are packed into a machine word, and fixed shifts extract the eight forward words. Reverse-complement words are constructed in a second vector stream. The lower word in each orientation pair is mixed with the minimap hash, while position and strand remain attached as separate metadata.

After minimap-style scrambling, window reduction uses the minimap-compatible less-than-or-equal rule. This is more work than selecting only one prefix and one suffix value, but it preserves the required rightmost winner and the event order around expiry. Ordinary windows emit through a compact vector path. Equal-hash candidates at different positions enter a separate tie-aware path that reconstructs all required chronological events. The split is an execution choice and does not change observable results.

Algorithm 6 ExactSimd block with bounded continuation

Input: the continuation state and the next vector group of bases
Output: chronological minimap-compatible events and updated state

1. Join saved k-mer context with the new symbols and construct the overlapping forward and reverse-complement words.
2. Select the canonical word in each lane. Replace palindromes and candidates crossing ambiguity with maximum-valued sentinels.
3. Apply the minimap integer hash and attach absolute position, strand, and chronological identity.
4. Join the saved $w-1$ candidate tail with the new candidates. Reduce each complete window with the rightmost equal-minimum rule.
5. Derive entering-minimum, expiry, first-window, and final-event masks.
6. Compact ordinary events directly. For an actual equal-hash, different-identity tie, recover and compact the required events in chronological order.
7. Save the final $k-1$ symbols, $w-1$ candidates, pending symbols, and current minimum for the next fragment.

ExactSimd is selected explicitly. If its checked path cannot adapt or materialize an admitted input shape, an unsupported representation or position range is reported as an error. The request does not fall back to Scalar and is not reinterpreted by a different seed scheme.

4.4.3. PNT1 as a separate ntHash scheme

PNT1 applies the data-independent SIMD-Minimizers structure but defines a BioSeqDB-specific persistent scheme. Its candidate key is the full wrapping 32-bit sum

$$h_i = H_i^F + H_i^R \pmod{2^{32}},$$

and its k-mer strand records whether $H_i^R < H_i^F$. It does not use minimap2's two-bit canonical word or integer mix.

PNT1 also differs from the published SIMD-Minimizers endpoint rule. The published algorithm chooses one equal extreme from the orientation of the complete window. PNT1 retains the leftmost and rightmost identities attaining the minimum. It emits the left identity first, followed by a distinct right identity, then removes identical adjacent events. This rule is deterministic for event identity and remains symmetric with respect to the two extremes. It does not reproduce minimap2 chronology.

Eight logical streams are retained independently of the physical vector width. A target with four 32-bit lanes uses two vector registers per step. A target with eight lanes uses one. Logical stream count is therefore part of the algorithm, while register count is a dispatch property. The streams overlap enough to complete boundary windows. Bounded first- and last-event state suppresses duplicate records at the seven stream joins.

Persisting a PNT1 index over fragmented storage uses bounded replay. For an owned core interval $[c_\ell, c_h)$, the producer reads a halo on each side, runs the PNT1 collector, translates local positions to global positions, and retains only positions in the half-open core. Neighboring cores have disjoint ownership, so one event cannot be published twice. Sorting by hash and encoded position establishes a canonical persistent order even though append order across lock-step logical streams is not an API guarantee.

PNT1 currently specializes the common $k = 15, w = 10$ DNA4 case and requires a 32-bit hash field. These constraints are visible at admission. A PNT1 index is never queried through the Scalar or ExactSimd scheme merely because all three outputs are called minimizers.

4.4.4. Vulkan tile extraction

VulkanSegment applies the BioSeqDB minimap-compatible hash and event state machine to packed sequences on a compute device. The main benefit is not an alternative seed definition. It is the ability to keep packed bases, minimizers, seed hits, and later mapping stages on the same device. Avoiding a host round trip can matter more than an isolated hash-loop improvement.

Candidate positions within one tile are processed in sequence because the rolling word and window minimum depend on prior candidates. Parallelism comes from many tiles scheduled across device invocations. Each tile starts before its owned trigger interval and replays enough context to reconstruct the rolling and window state. It emits only events whose trigger position lies in its half-open owned interval. Replay can overlap, but ownership does not.

Output cardinality is data dependent because ties and expiry can produce more than one event. The compact route therefore uses three stages:

1. Run the state machine in count mode to obtain the number of admitted events for every tile.

2. Apply an exclusive prefix scan to the tile counts and allocate one exact output interval for each tile.

3. Replay the same state machine in emit mode and write each event to its assigned interval.

This process gives deterministic allocation and tile order without reserving one output record per input base. A persistence-oriented route can instead append to a proved capacity and establish order with a stable device sort.

For DNA4, VulkanSegment is intended to reproduce the scalar BioSeqDB event stream exactly. The current DNA5 and generic packed routes have a weaker contract. They retain every scalar event but may add a bounded number of events near tile boundaries and invalid-symbol transitions. This distinction affects downstream workload and is therefore part of backend selection. It is not described as exact minimap compatibility.

4.4.5. Comparison and selection

Table 4.3 summarizes the four choices. A stored index fixes the seed scheme, producer identity, and revision before backend admission is evaluated.

Backend	Seed contract	Main execution method	Principal constraints
Scalar	BioSeqDB minimap-compatible	Rolling two-bit words and circular window buffer	Portable default, sequential candidate stream
ExactSimd continuation	Same as Scalar	Vector candidate construction with bounded fragment continuation and exact tie recovery	Indexed and fragmented route, with the main specialization at $k = 15, w = 10$
ExactSimd resident	Resident vector-stream contract	Full-resident vector streams rank candidates before minimap scrambling	Transient no-index use only when both full-resident sides use this route
PNT1	Versioned PNT1 contract	Eight logical ntHash streams with block minima and bounded stream-boundary deduplication	Packed DNA4, $k = 15, w = 10$, full 32-bit hash, compatible PNT1 index
VulkanSegment	BioSeqDB minimap-compatible for DNA4	Device tiles with context replay, half-open ownership, and count or emit passes	Device and buffer admission, exact DNA4 behavior, bounded conservative DNA5 behavior

Table 4.3: BioSeqDB minimizer backends. Scalar and the ExactSimd continuation route share one seed scheme. The resident ExactSimd route is distinct. PNT1 is a different persistent scheme. VulkanSegment shares the minimap-compatible candidate definition but currently has a representation-dependent boundary contract.

The practical selection rules follow directly:

1. Require the query to match the target index's seed scheme, producer identity and revision, and hash, tie, position, and ambiguity contract.

2. For the minimap-compatible CPU scheme, use Scalar by default. Indexed ExactSimd work requires the continuation route. The resident route is valid only when one transient operation owns two compatible resident inputs.

3. Use PNT1 only with an index produced by PNT1. Treat changing to PNT1 as an index migration.

4. Use VulkanSegment when the device route is admitted and subsequent stages can consume its resident output. Apply the documented DNA4 or DNA5 semantic contract rather than treating both as identical.

5. Reject a failure after a specific vector or device backend has been selected. Runtime substitution must not hide a scheme or semantic change.

4.4.6. Limitations

The current backends do not form one uniform performance ladder. ExactSimd's two selection routes are not lookup-compatible even though they share one backend label. Its most regular continuation specialization is limited to the common $k = 15, w = 10$ shape. PNT1 has the same parameter restriction and additionally requires packed DNA4, the complete 32-bit hash, and a supported vector width. VulkanSegment obtains parallelism across tiles rather than across dependent positions within one tile. Its compact output route repeats computation in count and emit mode.

The remaining representation difference is also significant. DNA4 has an exact device contract. DNA5 currently has a conservative superset contract, which can increase lookup and chaining work even when no scalar seed is lost. Applications requiring identical event streams across all representations must select an exact CPU backend until the device contract is strengthened.

No seed scheme avoids the sensitivity and frequency tradeoff. Increasing w reduces sketch density and index cost, but raises the length of a conserved region needed to retain a shared minimizer. A larger k reduces random and repetitive hits, but makes seeds less tolerant of substitutions and small indels. Fuzzy hashes and linked seeds offer different tradeoffs, but introduce their own collision, verification, and parameter requirements. These choices should be evaluated as complete seed and index configurations rather than as isolated hash functions.

4.5. Conclusion

Minimizer extraction consists of three distinct decisions. The seed scheme defines candidate hashes and event semantics. The execution algorithm maintains sliding minima and fragment state. The index maps selected keys to target occurrences. Correct query processing requires all three decisions to agree with the stored target index.

Minimap2 uses canonical two-bit k-mers, an integer mixing function, and a rightmost chronological event rule. SIMD-Minimizers replaces rescans with rolling ntHash and block-wise prefix or suffix minima, while also defining a different endpoint and orientation policy. BioSeqDB preserves the minimap scheme in Scalar and the ExactSimd continuation route, retains a separate resident ExactSimd route for transient resident inputs, provides PNT1 as a versioned ntHash scheme, and executes minimap-style extraction on Vulkan tiles when the input representation is admitted. Persisted scheme, producer, and revision fields prevent an optimization from silently changing the seed set consumed by lookup and chaining.

Chapter 5

Chaining seed anchors

5.1. From local seed hits to candidate mappings

Minimizers identify short agreements between a query and a reference, but a set of minimizer hits is not yet an alignment. Repetitive sequence can produce many incompatible hits. A plausible mapping is usually represented by a sparser subset that advances in both coordinate systems and stays near one diagonal. Chaining selects this subset before base-level alignment.

BioSeqDB supports scalar dynamic programming, range minimum query (RMQ) chaining, two single instruction, multiple data (SIMD) policies, and a Vulkan graphics processing unit (GPU) path. These choices share an anchor model, but do not all use the same edge score or stopping rule. The scalar algorithm defines the main forward-pass semantics. Exact SIMD preserves that contract. Bounded SIMD and GPU execution use stated approximations. RMQ changes both candidate selection and part of the pair score.

5.2. Anchor geometry and ordering

A seed hit is represented by an anchor

$$a_i = (r_i, q_i, \sigma_i, \ell_i, g_i, \phi_i),$$

where r_i and q_i are inclusive seed-end coordinates on the reference and query, σ_i is the strand, ℓ_i is the query span, g_i identifies a query segment, and ϕ_i stores additional ordering flags. Reference and query positions are stored as 64-bit values. The scalar and SIMD pair scorers nevertheless require local coordinate differences to fit signed 32-bit arithmetic. Full-width storage therefore permits large absolute coordinates, but it does not remove the bound on a single predecessor gap.

Figure 5.1 shows the geometric interpretation. Valid predecessors lie below and to the left of the current anchor. A chain can move between nearby diagonals to represent an insertion or deletion, but large diagonal changes receive a penalty or fall outside the chaining bandwidth.

The diagonal coordinate $r_i - q_i$ is constant for anchors from an exact, gap-free segment. An increasing diagonal coordinate indicates that reference advance has exceeded query advance, as expected across a deletion relative to the query. A decreasing value indicates the converse. Chaining does not decide the base-level gap placement. It tests whether the displacement is plausible, charges the corresponding edge cost, and passes the interval between selected anchors to extension. This separation lets sparse seed geometry restrict the expensive alignment problem without treating each seed as an independent mapping.

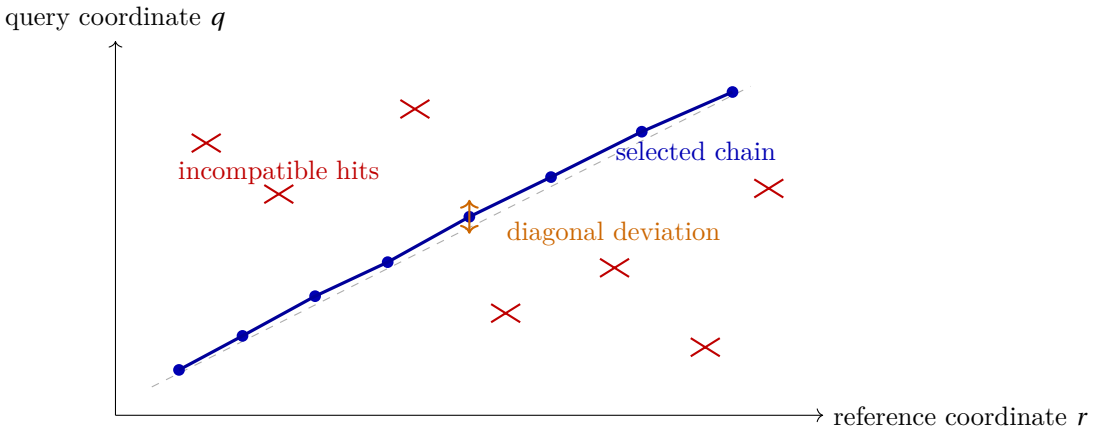


Figure 5.1: Anchor geometry. Chaining selects an increasing, approximately collinear path. Base-level alignment later explains the intervals between anchors.

The dynamic program requires a deterministic topological order. Anchors are sorted first by strand and reference coordinate, then by query metadata and coordinate. Two packed comparison keys express the established order:

$$\begin{aligned} K_r(a_i) &= (\sigma_i \ll 63) \mid \left(r_i \,\&\, (2^{63}-1)\right), \\ K_q(a_i) &= (g_i \ll 48) \mid (\phi_i \ll 40) \mid (\ell_i \ll 32) \mid (q_i \bmod 2^{32}). \end{aligned} \tag{5.1}$$

Comparison is lexicographic in K_r, K_q, and the full q_i. The final comparison resolves equal packed low words. The reference ordering reserves its high bit for strand, so canonical reference positions must be below 2^{63}. The query projection is a compatibility order rather than a globally increasing full-width order once query coordinates exceed 2^{32}. Ordinary per-query mapping remains well below this boundary.

The first and last anchor endpoints are converted to half-open mapping coordinates after chaining. On the forward strand,

$$q_{\text{start}} = q_{\text{first}} + 1 - \ell_{\text{first}}, \qquad q_{\text{end}} = q_{\text{last}} + 1.$$

For a reverse-strand query of length L, the normalized interval is mapped back to $[L - q_{\text{end}}, L - q_{\text{start}})$.

5.3. Scalar chaining recurrence

The scalar algorithm follows the single-segment, non-complementary-DNA branch of minimap2 chaining Li, "Minimap2: pairwise alignment for nucleotide sequences." For a current anchor i and an earlier anchor j, define

$$d_r(i,j) = r_i - r_j, \qquad d_q(i,j) = q_i - q_j, \qquad \delta(i,j) = |d_r(i,j) - d_q(i,j)|,$$

and

$$h(i,j) = \min\{d_r(i,j), d_q(i,j)\}.$$

A candidate is admissible when both differences are positive, both anchors have the same strand, the coordinate differences fall inside their active windows, and $\delta(i,j) \le B$, where B is the chaining bandwidth.

The reward for linking j to i is

$$\alpha(i,j) = \min\{\ell_j, h(i,j)\} - \begin{cases} 0, & \delta(i,j) = 0 \ \wedge \ h(i,j) \le \ell_j, \\ \left\lfloor \lambda_g \delta(i,j) + \lambda_s h(i,j) + \frac{1}{2}\,\mathrm{fastlog}_2(\delta(i,j)+1) \right\rfloor, & \text{otherwise.} \end{cases} \tag{5.2}$$

The first term rewards newly covered query sequence. The remaining terms penalize diagonal drift and long skips. The default coefficients are derived from minimizer length k:

$$\lambda_g = 0.01 k\, s_g, \qquad \lambda_s = 0.01 k\, s_s, \qquad s_g = 0.8, \qquad s_s = 0.$$

The best score ending at anchor i and its predecessor are

$$\begin{aligned} f_i &= \max\left(\ell_i, \max_{j \in W_i}\{f_j + \alpha(i,j)\}\right), \\ p_i &= \arg\max_{j \in W_i}\{f_j + \alpha(i,j)\}. \end{aligned} \tag{5.3}$$

If no predecessor improves the seed span, the chain starts at i and $p_i = -1$. Sorting makes the recurrence a sparse longest-path calculation in a directed acyclic graph.

Candidate search is bounded in three ways. Reference and query windows are raised to at least the bandwidth, so their effective limit is $M = \max(\text{max-gap}, B)$. At most the newest I anchors in the active window are examined, with a default $I = 5000$. The ordered scan can also stop after the visited-generation skip counter exceeds its limit, which is 25 by default. These bounds make dense repetitive regions tractable, but they make control order part of the algorithm.

The standard reference-gap limit is 5000 bases and the standard bandwidth is 500 bases. Increasing the gap limit expands the active predecessor set. Increasing the bandwidth admits larger insertion and deletion candidates. Changing the iteration or skip limit changes only how much of that admissible set is inspected. These parameters are related, but none can replace the others without changing the selected graph edges.

The three bounds serve different purposes. The reference window permits an ordered expiry pointer because anchors are sorted by reference position. The query window and diagonal bandwidth are pairwise tests because query coordinates need not increase monotonically across all active anchors. The iteration limit restricts the number of newest candidates considered after geometric expiry. The skip rule can end this restricted scan even earlier when recent predecessor paths repeatedly indicate that older candidates are unlikely to improve the score.

Equation 5.2 also distinguishes seed overlap from a true gap. When two anchors lie on one diagonal and their displacement does not exceed the predecessor span, the new contribution is only the non-overlapping coverage and no gap penalty is charged. For a diagonal change or a larger displacement, the linear diagonal term is supplemented by a logarithmic correction. The optional skip term depends on the smaller coordinate advance. An edge contribution may be negative. It can still be selected when the predecessor's accumulated score makes the total exceed the current best.

Strict improvement means that the newest candidate wins a score tie. The visited-generation rule is not a count of arbitrary non-improving anchors. It increments only for marked candidates, marks p_j rather than j, and decrements after a later improvement. Reordering candidates or reducing a batch before replaying these events can change the stopping point. The final step in Algorithm 7 is the global-best rescue. It preserves a strong older

Algorithm 7 Scalar chaining with bounded predecessor search

Input: canonically sorted anchors and chaining parameters
Output: endpoint scores f and predecessor indices p

1. For each anchor i, expire anchors outside the strand and reference window, then set $f_i \leftarrow \ell_i$, $p_i \leftarrow -1$, and $c \leftarrow 0$.
2. Visit at most I active candidates j from newest to oldest. Ignore candidates that fail the geometric tests.
3. For an admissible candidate, set $v \leftarrow f_j + \alpha(i, j)$.
   (a) If $v > f_i$, set $f_i \leftarrow v$, $p_i \leftarrow j$, and $c \leftarrow \max(0, c-1)$.
   (b) Otherwise, increment c when j carries the current generation mark. Stop when c exceeds the skip limit.
   (c) If the scan continues and $p_j \geq 0$, mark p_j for the current generation.
4. If the strongest completed endpoint was outside the scanned suffix, evaluate it once and accept it only on a strict improvement.

Table 5.1: Material differences between BioSeqDB chaining and the corresponding minimap2 path.

Aspect	BioSeqDB behavior
Feature scope	Uses one sequence segment and omits complementary-DNA branches.
Coordinates	Stores full 64-bit positions, while scalar and SIMD pair differences remain valid only in the stated signed 32-bit domain.
Backtracking	Uses a shared chain recovery procedure without minimap2's full drop parameter and final target-order compaction behavior.
RMQ	Uses a related augmented-tree formulation, but its simplified scorer and query boundaries do not define the same candidate set.
SIMD and GPU	Add BioSeqDB policies whose exact or approximate contracts are stated separately. They are not upstream minimap2 modes.

endpoint that early stopping would otherwise omit, but it does not make the search exhaustive.

5.3.1. Relation to minimap2

The recurrence and scalar stopping behavior are close to minimap2's single-segment, non-complementary-DNA path, but BioSeqDB chaining is not a literal copy. Table 5.1 lists the differences that affect interpretation.

5.4. Endpoint selection and backtracking

The forward pass computes a possible chain endpoint at every anchor. Backtracking ranks endpoints by (f_i, i) in descending order, then follows predecessors until the path ends or reaches an anchor already claimed by a stronger path. A path is retained only if it meets the minimum chain score and minimum anchor count, whose defaults are 40 and 3.

When a new walk joins a claimed suffix, its attributed score is reduced by the score at the merge endpoint. This prevents a weak prefix from inheriting the complete score of a stronger

Algorithm 8 Ranked chain recovery

Input: scores f, predecessors p, and chain thresholds
Output: compacted chains in descending endpoint order

1. Rank all endpoint indices by descending (f_i, i).
2. For each unclaimed endpoint in that order, follow p and collect anchors until the walk terminates or reaches a claimed anchor.
3. If the walk reaches a claimed suffix, subtract the score at the merge endpoint from the candidate prefix score.
4. Mark the traversed anchors. Retain the path only when its adjusted score and anchor count meet their thresholds.
5. Reverse each retained predecessor walk into forward coordinate order and append it to the compact chain array.

path. Claimed anchors also make endpoint order observable. Path discovery may proceed in parallel, but commits must respect the endpoint ranking when paths overlap.

Rejected walks remain claimed. This detail prevents later, weaker endpoints from traversing the same prefix and changes which alternatives survive. The forward peak-score vector is not used to reproduce every upstream drop rule. The resulting contract is therefore a ranked recovery of BioSeqDB predecessor paths, not the complete minimap2 chain-compaction procedure.

Recovered paths are compacted into chain records that contain the score, anchor count, and offset into the ordered anchor array. The mapping stage then classifies the highest-scoring region as primary. For a later region, let o be query overlap with existing primaries, u be uncovered query length, and $m_{\min}$, $m_{\max}$ be the smaller and larger region spans. The region is secondary when

$$\frac{o}{m_{\min}} - \frac{u}{m_{\max}} > \text{mask-level} \quad \wedge \quad u \le \text{mask-length}.$$

A secondary region is retained when its score is sufficiently close to its parent:

$$s_{\text{secondary}} \ge \rho\, s_{\text{parent}} \quad \vee \quad s_{\text{secondary}} + \Delta_{\min} \ge s_{\text{parent}}.$$

The default ratio is $\rho = 0.8$. A global secondary limit and parent-coordinate duplicate removal follow this comparison. Primary regions are retained. This classification occurs after chaining, yet a change in endpoint or backtracking order can still change which region becomes primary.

5.5. Range minimum query chaining

The scalar method can evaluate $O(nI)$ pairs for n anchors and a maximum of I predecessors per anchor. RMQ chaining reduces candidate lookup to $O(\log n)$ operations by isolating the part of the gap cost that is linear in predecessor coordinates. Each active predecessor receives the priority

$$\pi_j = -\left[f_j + \frac{\lambda_g}{2}(r_j + q_j) \right]. \tag{5.4}$$

An augmented balanced tree is keyed by (q_j, j). Every node caches the minimum priority in its subtree. A query interval can therefore combine the nodes on its boundary with cached values from fully covered subtrees.

For each reference-coordinate group, RMQ chaining performs four operations:

1. Remove anchors whose reference positions are outside the active window.
2. Query the tree over the permitted query-coordinate interval and obtain the index with minimum cached priority.
3. Re-evaluate the proposed edge with the RMQ pair score, then compare it with starting a new chain.
4. Insert the newly scored group and update subtree minima during tree rebalancing.

The delayed insertion preserves the strict positive reference difference. Including the anchor index in the tree key gives deterministic behavior when query coordinates are equal. Expiry and insertion each cost $O(\log n)$. A covered-subtree query also costs $O(\log n)$, while the stored tree and its augmentation require $O(n)$ space in the worst case.

A capacity limit bounds the active tree in highly repetitive input. The tree result is only a predecessor proposal. The RMQ pair scorer assigns the edge value, and starting a new chain remains an alternative.

The current RMQ path is an algorithmic variant rather than a faster execution of Equation 5.3. Its simplified penalty condition tests the query distance against predecessor span, while the scalar condition uses the minimum of reference and query distance. Its primary range query can admit an equal-query predecessor and does not independently reject every candidate outside the diagonal bandwidth. An optional second tree is inactive in the standard configuration. RMQ is therefore useful for dense anchor sets, but it does not provide scalar-equivalent endpoint selection.

5.6. SIMD chaining on the CPU

For one target anchor, pair-score arithmetic over several predecessors is independent. BioSeqDB stores reference positions, query positions, and spans in structure-of-arrays buffers and evaluates these pairs with portable SIMD. Coordinate subtraction uses low 32-bit words, so the full differences must satisfy

$$0 < d_r(i,j) \le \text{INT32__MAX}, \qquad 0 < d_q(i,j) \le \text{INT32__MAX}.$$

The reference window supports the first bound. The query bound must also hold for the input. Admissibility masks suppress invalid lanes, and explicit masks protect the final partial vector.

5.6.1. Exact ordered decisions

The exact policy vectorizes score production and preserves scalar decision order. A batch that cannot improve the current best can process its generation marks together. If any lane can improve, candidates are replayed from newest to oldest. This replay preserves strict ties, skip-counter decrements, marked predecessors, and early termination. The global-best rescue is found with a vector scan and then applied through the scalar eligibility rule.

This division is necessary because the arithmetic is lane-independent while the stopping state is ordered. Exact SIMD therefore means equality with the BioSeqDB scalar score and predecessor contract, not that every decision is a vector reduction.

For each vector batch, the exact policy first computes coordinate masks, coverage rewards, penalties, and candidate scores. A comparison mask identifies lanes that can exceed the current best. If this mask is empty, only the marked lane count can affect the ordered state. If the mask is nonempty, scalar replay consumes the batch in descending predecessor order and can stop before reaching its oldest lane. This two-level procedure avoids scalar pair arithmetic while retaining the event sequence that determines p_i.

5.6.2. Bounded batch approximation

The bounded policy processes candidates in newest-first batches of L lanes. Its non-improving batch budget is

$$C_{\max} = \max\left(1, \left\lfloor \frac{\text{max-skip}}{L} \right\rfloor + 1\right). \tag{5.5}$$

Any improvement resets the counter. When the budget is reached, older batches are omitted. Equal maxima within the examined interval select the largest predecessor index. The global-best candidate is tested only if it lies beyond the cutoff.

This rule does not reproduce visited-generation events. A valid older predecessor can lie beyond the omitted interval, so bounded SIMD cannot guarantee the same endpoint for every anchor set. Its name and policy boundary make this approximation explicit.

The bounded rule gains regularity by replacing per-candidate marks with one counter per completed vector batch. It can reduce all lane maxima before making a control decision. The trade-off is structural rather than numerical. Pair scores inside the visited interval use the same arithmetic, but the set of visited predecessors can differ from both scalar and exact SIMD.

Both SIMD forward policies use vector-assisted backtracking. Independent ranked starts occupy lanes and contiguous predecessor runs use a fast path. Commits still follow endpoint rank because paths can meet through the shared claimed-anchor state. A selected SIMD policy also fails when no valid multi-lane target is available instead of silently switching algorithms.

5.7. Memory locality in GPU chaining

GPU chaining is limited by dependency order and memory traffic. A source-at-a-time organization repeatedly reads one completed source score and scatters updates across separated target scores. Tiled chaining first resolves the triangular dependencies within a tile. It then keeps completed sources in shared memory and target accumulators in registers while evaluating the tile-to-tile pairs. Figure 5.2 reconstructs these two access patterns. The tiled organization follows the locality argument used by G3SA Han et al., "G^{3}SA: A GPU-Accelerated Gold Standard Genomics Library for End-to-End Sequence Alignment."

The G3SA CUDA design applies this decomposition with different kernels for short, medium, and long ranges. Its larger paths stage tiles in shared memory and rotate source assignments across threads. BioSeqDB adopts the goal of reducing memory traffic, but its current Vulkan score generator has a different execution structure.

5.7.1. Vulkan score generation

BioSeqDB divides anchors into hard ranges when strand changes or the reference gap between adjacent anchors exceeds the configured maximum. One 32-lane subgroup processes each range. Targets advance sequentially because target $i+1$ may depend on the score for target

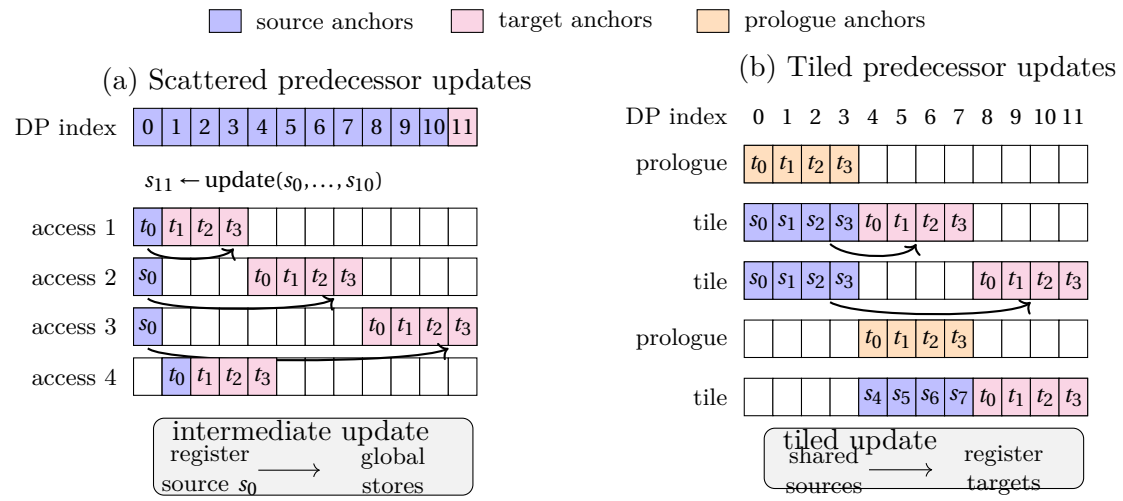


Figure 5.2: Scattered and tiled chaining memory access. The scattered method reuses one source while writing separated target groups. The tiled method resolves within-tile dependencies in prologue rows, then reads contiguous source blocks from shared memory and retains target scores in registers.

Table 5.2: G3SA and BioSeqDB GPU chaining use different execution contracts.

Aspect	G3SA CUDA	BioSeqDB Vulkan
Parallel unit	Range-dependent kernels, with large workgroups and shared tiles for longer ranges	One 32-lane subgroup for each hard range
Score order	Within-tile prologue followed by a rotated tile-to-tile tail	Sequential targets with newest-first predecessor batches
Anchor data	Fixed assumptions in the principal dynamic-programming path and relative predecessor offsets	Per-anchor span, full-width distance checks, and absolute predecessor indices
Search control	Fixed predecessor cap without minimap2 generation-mark replay	Device cap, batch-based stopping, and no global-best rescue
Completion	GPU-oriented range processing, backtracking, and compaction	CPU recovery for rich output or compact GPU reduction for coordinate output

i. For each target, the subgroup scores predecessor batches from newest to oldest, reduces the batch maximum, and chooses the largest predecessor index on a tie. A workgroup barrier follows each completed target.

The shader reads each anchor span and checks 64-bit coordinate differences before narrowing. It stores absolute predecessor indices. These choices differ from fixed-span and relative-predecessor layouts used by the original CUDA design. Chaining remains independent of alphabet after anchors have been formed.

Table 5.2 separates the shared memory-locality idea from the execution details. G3SA uses a prologue and a rotated tiled tail for its larger ranges. The Vulkan implementation uses sequential target completion with 32-wide predecessor batches. It therefore should not be described as a direct translation of the CUDA kernels.

GPU candidate search is approximate. The device examines at most

$$I_{\text{GPU}} = \min(I_{\text{requested}}, 3000)$$

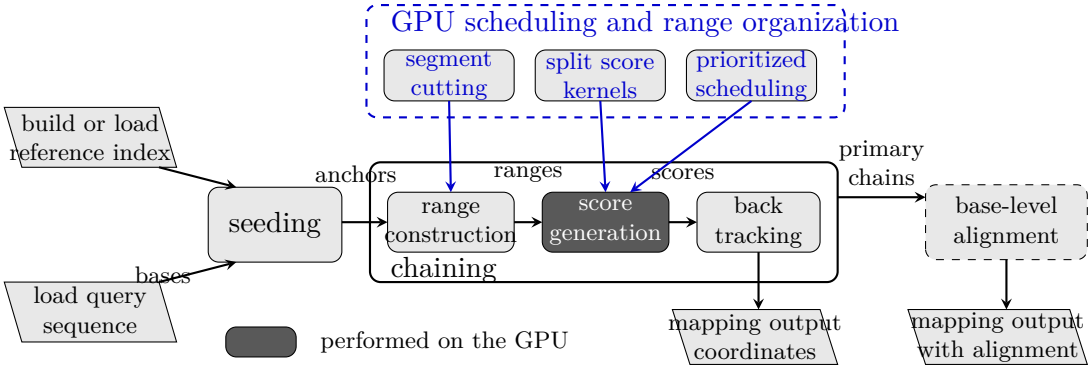


Figure 5.3: Vulkan chaining for rich projections. CPU range construction precedes GPU scoring and CPU backtracking. Coordinate-only resident execution can retain all three chaining stages on the device.

predecessors. When the skip limit is positive, it stops after $\lfloor \text{max-skip}/32 \rfloor + 1$ non-improving batches. It does not replay visited-generation events and has no global-best rescue. A hard range longer than 131072 anchors is sampled toward 65536 anchors while retaining its endpoints. These rules bound GPU work and memory, but they can alter a chain.

Figure 5.3 compares the CUDA design that motivated the path with BioSeqDB's Vulkan execution. The upper boxes summarize the CUDA scheduling techniques. In the lower rich-projection path, dark boxes identify GPU stages. BioSeqDB uses CPU hard-range construction and GPU score generation without reproducing the CUDA kernel topology.

5.7.2. Completion and result projection

Rich result projections expose the complete score and predecessor arrays to a CPU backtracker after GPU completion. Coordinate-only projections can instead use a GPU reducer that repeatedly chooses the strongest unclaimed endpoint and traces one chain. The standalone reducer examines at most 16 endpoint starts, so a valid later start can be missed after 16 rejected traces. An eligible resident path keeps seed-hit processing, sorting, range construction, score generation, and coordinate reduction on the device. It returns only a compact mapping result.

GPU selection is explicit. It requires a suitable device, coherent mapped buffers, and successful memory admission. Failure does not cause an implicit switch to the CPU. This behavior makes the selected execution contract visible to a database query and prevents an approximate GPU request from being reported as a scalar execution.

5.8. Execution policy

No chaining method is best for every anchor distribution and output projection. Table 5.3 summarizes the available contracts. The scalar policy is the default. A direct SIMD request can select exact ordered decisions, while the database SIMD policy may select bounded batches. RMQ and Vulkan remain explicit choices.

Table 5.3: Chaining policies and their semantic boundaries.

Policy	Search structure	Semantic relation	Main limit
Scalar	Newest-first bounded dynamic program	Closest BioSeqDB form of single-segment minimap2 forward chaining	$O(nI)$ pair search and signed 32-bit local differences
RMQ	Augmented balanced tree	Related scoring variant, not scalar-equivalent	Simplified pair test and broader query boundary
SIMD exact	Vector pair scores with ordered replay	Exact to BioSeqDB scalar f, p behavior	Ordered events restrict full-lane execution
SIMD bounded	Vector batches with an early cutoff	Explicit approximation	May omit an older valid predecessor
Vulkan	One subgroup per hard range	Explicit GPU approximation inspired by G3SA	3000-candidate cap, batch stopping, range sampling, and projection-specific backtracking

5.9. Summary

Chaining turns local seed hits into a small set of collinear candidate mappings. The scalar recurrence rewards newly covered sequence and penalizes diagonal drift. Its bounded windows, visited-generation rule, strict tie handling, and global-best rescue make scan order observable. Ranked backtracking then recovers competing chains for primary and secondary region selection.

RMQ replaces repeated candidate scans with an augmented tree but has distinct pair and boundary semantics. Exact SIMD vectorizes independent arithmetic and replays ordered decisions. Bounded SIMD trades scalar equivalence for a fixed batch cutoff. GPU chaining reduces memory traffic through range-oriented score generation and compact result routes, while candidate caps and long-range sampling make it approximate. Keeping these contracts separate allows callers to choose an execution method without confusing algorithmic ancestry with equivalent behavior.

Chapter 6

Final alignment and extension

6.1. From a seed chain to a base-level alignment

Chaining identifies an ordered set of anchors with plausible query and target coordinates. It does not determine the substitutions, insertions, and deletions between those anchors. Final alignment resolves this remaining ambiguity. It refines the outer coordinates, assigns a dynamic-programming score, and reconstructs an edit path when the query requests a CIGAR or an exact edit count.

BioSeqDB treats final alignment as an optional pipeline stage. Coordinate-only queries can return regions derived from seeds and chains. A request for a base-level path enables traceback, while an exact edit predicate also requires the traced path because a seed-derived divergence estimate is insufficient. Avoiding dynamic programming when these outputs are absent is important for database workloads that project many candidate coordinates.

The stage has three execution routes. KSW2 supplies two-piece affine scoring, a static band, Z-drop, and minimap-style endpoint rules. Block Aligner uses a single affine model and an adaptive SIMD block. A restricted GPU route aligns windows produced by GPU seeding and chaining. These routes do not have identical scoring or endpoint semantics. Route selection is explicit, and each route has a distinct scoring and result contract.

The alignment modes in Section 2.1 determine which matrix borders are fixed and which endpoints are free. BioSeqDB applies these conditions at the level of subproblems. Inter-anchor rectangles are global fills with fixed endpoints. The two outer rectangles are extension problems whose endpoint may be selected before the end of the available flank. This distinction is more accurate than describing an entire chained region as one global or local matrix.

6.2. Affine-gap dynamic programming

Let $Q = q_1 \dots q_n$ be a query and $T = t_1 \dots t_m$ a target interval. For substitution score $s(q_i, t_j)$, an affine gap of length $k \geq 1$ has penalty

$$g(k) = q + ke,$$

where $q > 0$ is the gap-open penalty and $e > 0$ is the extension penalty. The first gap symbol therefore costs $q + e$. With query positions on the vertical axis and target positions on the

horizontal axis, the global Needleman, Wunsch, and Gotoh recurrence is

$$I(i,j) = \max\{H(i-1,j) - q - e,\ I(i-1,j) - e\}, \tag{6.1}$$
$$D(i,j) = \max\{H(i,j-1) - q - e,\ D(i,j-1) - e\}, \tag{6.2}$$
$$H(i,j) = \max\{H(i-1,j-1) + s(q_i,t_j),\ I(i,j),\ D(i,j)\}. \tag{6.3}$$

The insertion and deletion states record whether a path already ends in a gap. A global fill initializes both axes with cumulative gap costs. An extension changes the boundary and endpoint rules while retaining the same cell recurrence. A full matrix takes $O(nm)$ time and $O(nm)$ trace memory when every decision is retained Gotoh, "An improved algorithm for matching biological sequences."

The default nucleotide model uses match score $a = 2$, mismatch penalty $b = 4$, and two affine families

$$(q_1, e_1) = (4, 2), \qquad (q_2, e_2) = (24, 1).$$

KSW2 assigns a gap the lower of the two costs,

$$g(k) = \min\{4 + 2k,\ 24 + k\}.$$

The short family is cheaper below length 20, both are equal at 20, and the long family is cheaper above 20. The recurrence keeps two insertion states and two deletion states:

$$I_r(i,j) = \max\{H(i-1,j) - q_r - e_r,\ I_r(i-1,j) - e_r\}, \tag{6.4}$$
$$D_r(i,j) = \max\{H(i,j-1) - q_r - e_r,\ D_r(i,j-1) - e_r\}, \qquad r \in \{1,2\}, \tag{6.5}$$
$$H(i,j) = \max\{H(i-1,j-1) + s(q_i,t_j), I_1, I_2, D_1, D_2\}. \tag{6.6}$$

This model charges short sequencing errors and long gaps differently. Block Aligner instead uses only the first affine family. Selecting Block Aligner is therefore a scoring choice as well as a performance choice.

6.2.1. Banding and stopping criteria

Anchors suggest an expected diagonal d_0. KSW2 can restrict evaluation to cells satisfying

$$|(i-j) - d_0| \le w,$$

where w is the half-width of the band. Work is then approximately $O(w(n+m))$, but an optimum that leaves the band cannot be recovered. The ordinary alignment width is derived from the chaining width as

$$w_{\text{align}} = \lfloor 1.5\, w_{\text{chain}} + 1 \rfloor.$$

Z-drop terminates an extension whose score has deteriorated sufficiently from the best cell. If $S^\star$ is the best score at $(i^\star, j^\star)$, termination is permitted when

$$S^\star - S(i,j) > Z + e_z \left| (i - i^\star) - (j - j^\star) \right|. \tag{6.7}$$

The displacement term allows a long gap to move away from the best diagonal without receiving the same stopping penalty as a score loss along that diagonal. In the two-piece kernel, e_z is the long-gap extension penalty. Block Aligner does not use this rule. Its outer extensions apply X-drop and require two consecutive steps below the threshold before stopping.

For mandatory KSW2 fills used by an exact edit predicate, banding and Z-drop are disabled. This prevents the path between two fixed anchors from ending early. The outer candidate window and the selected chain still constrain the search, so this setting does not turn the complete mapping task into an unbounded full-matrix comparison.

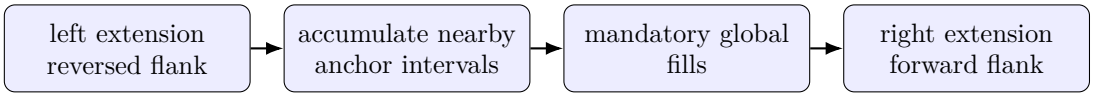
6.3. Alignment around a chain

BioSeqDB follows the ordinary DNA extension structure of minimap2 Li, "Minimap2: pairwise alignment for nucleotide sequences." It solves three kinds of subproblem rather than one matrix over the complete region. The left flank is reversed and aligned as an extension. Rectangles between selected anchors are global fills. The right flank is a forward extension.

Table 6.1: Boundary conditions of the three alignment phases.

Phase	Orientation	KSW2	Block Aligner
Left flank	reversed	extension endpoint, band, and Z-drop	native X-drop extension
Between anchors	forward	global two-piece affine fill	global single-affine adaptive block
Right flank	forward	extension endpoint, band, and Z-drop	native X-drop extension

Short distances between consecutive anchors are accumulated. A fill is started only after both the query and target spans reach a minimum length, or when the last anchor forces the remaining rectangle. This reduces a long region to a sequence of moderate subproblems. It also prevents the overhead of running dynamic programming separately for every closely spaced seed.



ordered anchors preserve the query and target coordinate direction

Figure 6.1: Decomposition of a chained region into two extensions and a set of global inter-anchor fills. Several short anchor intervals may form one fill.

Algorithm 9 Final alignment of one chained region

Input. Query Q, target T, ordered anchors A, parameters P, and a selected backend.

1. Derive inner anchor bounds and bounded outer extension intervals.
2. If both left intervals are nonempty, align them in reverse orientation and update the left endpoint.
3. Place a cursor at the first anchor.
4. Visit each later anchor in order.
   (a) Accumulate query and target distance while either span is below *P.minLen*, except at the final anchor.
   (b) Solve each resulting positive rectangle as a global fill.
   (c) Append its CIGAR and move the cursor to the current anchor.
5. If both right intervals are nonempty, run the permitted forward extension and update the right endpoint.
6. Merge adjacent equal CIGAR operations and project reverse-strand coordinates when required.

Each phase returns a score, consumed lengths, endpoint information, and an optional trace. A non-dropped global fill consumes both slices. Ordinary KSW2 fills may instead stop under Z-drop and use their maximum-scoring positions as truncated endpoints. Exact-edit KSW2 disables Z-drop. An extension may consume only a prefix of the available oriented slices, which moves the reported outer coordinate to its selected endpoint. The region result is formed by adding phase scores and concatenating phase traces in coordinate order. Reversing the left input changes the direction of computation, but not the final order of the reported path.

Stopping state is also phase-specific. A KSW2 Z-drop in an earlier phase can suppress a later right extension because the minimap-style control flow treats the region as dropped. Block global fills do not use Z-drop. Its left and right X-drop extensions are separate searches, so the right extension remains eligible after an earlier Block phase. This is a material behavioural difference and is retained as a backend rule rather than normalised through a shared option.

The port differs from minimap2 in its supported policy surface. It implements the ordinary DNA seed-gap and end-extension structure, but not the splice, RNA, inversion-retry, or long-join policies. It also obtains slices from lazy sequence representations, including reverse-oriented slices, without first copying a complete region. These differences concern the surrounding pipeline. The KSW2 recurrence remains recognisable, while Block Aligner uses a separate phase contract instead of imitating KSW2 options that it does not support.

6.4. KSW2 with portable SIMD

6.4.1. Difference representation

A direct SIMD implementation stores absolute H, I, and D scores. Absolute values grow with sequence length and may require wide lanes. KSW2 uses the Suzuki and Kasahara observation that differences between neighbouring scores remain bounded by the scoring parameters even when the absolute score is large Suzuki and Kasahara, "Introducing difference recurrence relations for faster semi-global alignment of long sequences." Representative differences are

$$U(i,j) = H(i,j) - H(i-1,j), \qquad V(i,j) = H(i,j) - H(i,j-1).$$

The main recurrence operates on signed byte values derived from these differences. Wider scores are reconstructed where exact maxima and endpoints are needed. More cells therefore fit in each SIMD vector.

BioSeqDB expresses this recurrence with portable SIMD operations. Lanes hold cells from one alignment problem, usually along an anti-diagonal. This is intra-sequence parallelism. It differs from placing several independent sequence pairs in the lanes of one vector. A single-affine specialization is used when the two penalty pairs are equal. The default parameters select the two-piece specialization, with four gap states and their continuation bits. Trace decisions are stored for cells inside the admitted band.

6.4.2. Endpoint and traceback rules

KSW2 tracks the best internal cell, the best score at each sequence end, and the bottom-right score when that cell is reached. A global fill traces from the bottom-right cell. An extension normally traces from the best admitted endpoint, with the query-end score and end bonus considered by the selected mode. A left extension is computed on reversed slices, after which its CIGAR runs are restored to source order.

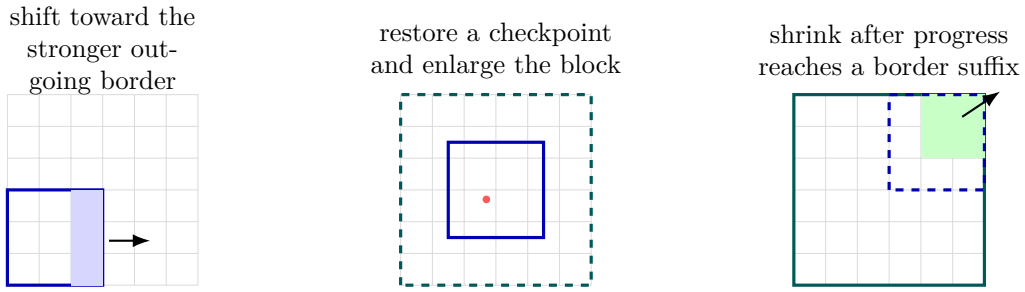
The BioSeqDB port changes data access and allocation rather than the scoring model. It encodes only each requested subrange from a lazy view, reuses operation-local scratch, and dispatches the same recurrence to supported CPU targets. The second affine family, band, Z-drop, endpoint bonus, and tie rules remain reasons to select KSW2. They are not projected onto Block Aligner.

6.5. Adaptive block alignment

6.5.1. Block movement and growth

Block Aligner evaluates the single-affine recurrence in a moving $B \times B$ square rather than a fixed diagonal band Liu and Steinegger, "Block Aligner: an adaptive SIMD-accelerated aligner for sequences and position-specific scoring matrices." Each step computes a strip of width S on the right or lower edge. The maximum over the two outgoing border prefixes chooses whether the block moves right or down. The block can therefore follow a path that moves far from the original diagonal.

Greedy movement can miss a path after a long gap or a low-scoring interval. When several steps fail to find a new maximum, the algorithm restores a checkpoint and increases B. The larger region admits more alternatives. After progress resumes near an outgoing border suffix, the block may shrink and move diagonally. The minimum and maximum block sizes bound this adaptation.



(a) move (b) grow (c) shrink

Figure 6.2: Adaptive block movement. Border scores guide each shift. Lack of progress causes replay with a larger block, while later progress permits a smaller block.

6.5.2. SIMD recurrence within a block

Vectors are laid out vertically or horizontally inside the block. One gap state then has a dependency between lanes. For vector values $x_0, \ldots, x_{L-1}$, the required affine prefix operation is

$$y_i = \max_{0 \le k \le i} \{x_k + (i-k)g\}, \qquad g = -e. \tag{6.8}$$

Shift-and-maximum stages at distances $1, 2, 4$, and then higher powers of two compute this scan in $O(\log L)$ vector operations. A carry connects the end of one vector with the next. The diagonal transition and the other gap state use ordinary vector dependencies.

Score computation retains only the block borders and checkpoint copies, so its working score memory is $O(B)$. Traceback needs additional strip records along the adaptive trajectory. A checkpoint includes the trace position, which allows speculative strips to be discarded when the algorithm grows and replays a region.

BioSeqDB keeps the reference step geometry and admits SIMD lane shapes that preserve it. Global mode fills the fixed rectangles between anchors. X-drop mode handles the outer extensions. Input ranks are obtained from the same lazy views as KSW2. SIMD masks encode direction and gap continuation decisions, and traceback decodes these masks directly into packed runs. The main material difference from KSW2 remains the adaptive search with one affine family.

Algorithm 10 Adaptive block alignment

Input. Encoded slices, affine scores, block bounds, and either global or X-drop mode.

1. Place a minimum-size block at the initial matrix corner and save a checkpoint.
2. Compute the next outgoing strip with the vector prefix recurrence.
3. Record score and trace borders, then update the best score.
4. If progress continues, move toward the stronger outgoing border.
5. If progress stalls and growth is possible, restore the checkpoint, enlarge the block, and recompute that region.
6. If the best score reaches a suitable border suffix, shrink the block and advance diagonally.
7. Repeat until the global endpoint or the X-drop condition is reached.
8. Trace backward through the retained strips and emit packed runs.

6.5.3. Exact diagonal certificate

An equal-length fill can sometimes be resolved without constructing a block. Let its common length be n, let D be the exact score of the coordinate diagonal, and let M be the largest substitution score. Any different global path must contain the same positive number of inserted and deleted symbols. Its score is at most

$$U = (n-1)M - 2(q+e).$$

It follows that

$$D > U \quad \Longrightarrow \quad \text{the coordinate diagonal is the unique optimum.} \tag{6.9}$$

Strict inequality excludes a competing path with an equal score. When the condition holds, BioSeqDB can return the diagonal score, a single match run, and the exact mismatch count. Otherwise it runs the adaptive dynamic program. The certificate proves only this specific geometry and does not make Block Aligner exact for every admitted input.

6.6. Traceback, CIGAR, and edit counting

All alignment routes cross their low-level boundary using packed CIGAR runs of the form

$$\mathit{word} = (\mathit{length} \ll 4) \mid \mathit{opcode}.$$

The common opcodes include M for a diagonal step, I for an insertion, D for a deletion, and the more explicit equality and mismatch variants = and X. A run records one opcode and its length. Adjacent equal runs are merged within a traceback and across phase boundaries.

The forward calculation and backward trace have separate roles. The forward pass selects scores and endpoints. The trace reconstructs the path through diagonal and gap states. KSW2 records decisions for its admitted band. Block Aligner records decisions for computed strips. The GPU route either retains a bounded direction matrix or reconstructs directions tile by tile. All three then produce the same run-oriented interface.

A generic M run does not reveal whether its diagonal symbols are equal. The edit count is therefore computed from the emitted path and the corresponding sequence symbols:

$$E_{\text{path}} = \#\{\text{unequal diagonal steps}\} + \sum_{I \text{ runs}} \mathit{length} + \sum_{D \text{ runs}} \mathit{length}.$$

Block Aligner counts unequal diagonal steps during traceback, while KSW2 can replay diagonal runs against the sequence ranks. These CPU routes return the unit-cost edit count of the chosen path. It is distinct from the affine score and from a seed-derived estimate.

The CIGAR also defines coordinate consumption. A diagonal run consumes query and target symbols, an insertion consumes query symbols, and a deletion consumes target symbols. Replaying these rules checks the reported span and allows CPU routes to derive aligned lengths and edit count. The complete GPU route currently returns CIGAR and match counts, but it does not expose the chosen-path edit-count field or accept an exact edit predicate. Reverse-strand coordinates are converted only after the phase CIGARs have been assembled in alignment order.

6.7. Accelerator boundary and gridded traceback

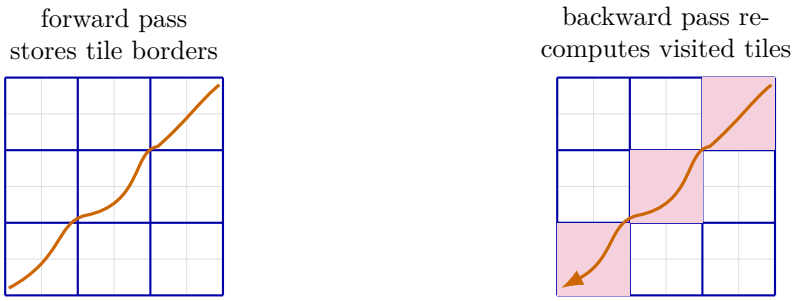
The GPU route begins after candidate windows have been chosen by GPU seeding and chaining. If a query needs coordinates only, those windows can be returned without base alignment. If it needs CIGAR, each selected query and target window is packed as symbol ranks and submitted as one job in a batch. The result contains score, endpoints, status, and a slice of packed CIGAR runs.

This route has an explicit ownership boundary. Its configuration must select the complete GPU frontend and an admitted base-alignment identity. The latter does not mean that the CPU Block Aligner recurrence runs on the device. It is a route selection condition. The general CIGAR kernel uses its own two-piece affine global recurrence over the selected coordinate window. Unsupported geometry or an unavailable GPU component rejects the route instead of continuing with a CPU alignment under the same label.

Conventional GPU traceback writes one direction value for every computed cell. G3SA reduces this traffic by retaining score states at tile boundaries and recomputing directions only for tiles reached by the final path Han et al., "G^{3}SA: A GPU-Accelerated Gold Standard Genomics Library for End-to-End Sequence Alignment." BioSeqDB applies the same memory principle in Vulkan. The forward pass stores five affine score states along tile borders. Starting from the global endpoint, traceback reloads a border, recomputes one tile in fast memory, traces to the next boundary, and discards the temporary directions. For tile dimensions b_q and b_t, border storage is proportional to

$$\left\lceil \frac{n}{b_q} \right\rceil \left\lceil \frac{m}{b_t} \right\rceil (b_q + b_t),$$

apart from the constant number of affine states. One reusable tile needs $O(b_q b_t)$ direction storage. The scheme replaces a full $O(nm)$ direction matrix with tile borders and additional arithmetic during traceback. It is most useful when writing the full trace matrix would dominate device memory traffic.



(a) boundary state (b) local replay

Figure 6.3: Gridded traceback. Blue tile borders persist after the forward pass. Only the purple tiles crossed by the selected path are recomputed during traceback.

Short or structurally simple windows first use specialized scripts, including greedy gap scripts that are not guaranteed to maximize the affine score. Moderate windows can retain a bounded direction matrix. Longer admitted windows use gridded traceback. The current GPU CIGAR recurrence is global over the chosen window. It does not apply the KSW2 band, Z-drop, or end bonus even if related values are present in the broader request. It should therefore be described as a GPU-specific alignment contract, not as a device implementation of KSW2 or Block Aligner.

6.8. Runtime selection and design limits

The alignment route is selected explicitly before region alignment. The selected CPU kernel is then specialized for KSW2 or Block Aligner, so the inner recurrence does not branch on the backend for each cell. Output needs are resolved separately. A coordinate-only query may omit base alignment, while CIGAR or exact edit output makes traceback mandatory.

The GPU route additionally requires the complete accelerated frontend and an admitted feature tuple before alignment begins.

Several limits follow from these contracts. First, KSW2 and Block Aligner can return different scores because their gap models and search regions differ. Second, the ordinary DNA extension port does not implement every specialised minimap2 mode. Third, the GPU route uses a global coordinate window and lacks the current CPU endpoint rules. Fourth, adaptive search and finite bands can both omit cells that an unrestricted matrix would examine.

These differences are exposed through route selection rather than hidden behind a single claim of algorithmic equivalence. KSW2 is appropriate when two-piece affine scoring and minimap-style stopping rules are required. Block Aligner is appropriate when its single-affine adaptive search matches the requested semantics. The GPU path is appropriate only when the complete accelerated route and its global window contract are admitted. When available, CIGAR and edit count describe the path produced by the selected route.

Chapter 7

Alignment prefiltering

7.1. Why filter after chaining but before alignment?

Seed matches and chains identify sequence intervals that may align well. They do not establish that the complete intervals satisfy an edit-distance bound. Repeated sequence, short seeds, hash collisions, and sequencing errors can all produce plausible chains whose intervening gaps are expensive to align and ultimately fail the requested bound. Applying affine-gap dynamic programming to every candidate is correct, but it spends time on candidates that a cheaper procedure may be able to disprove.

A pre-alignment filter addresses this case. It runs after chaining has defined candidate regions and before the base aligner processes chain-defined fills. Its result is deliberately limited. The filter may show that a candidate cannot meet the requested threshold, or it may leave the question unresolved. It does not produce an alignment, a CIGAR string, or an edit script.

This distinction is required by the SQL interface. Consider a predicate of the form

```
WHERE mmsearch(sequence, query)
  AND bioseq.edit(sequence) < 10
```

The strict bound $B = 10$ means that the unit-cost edit count reconstructed from every returned alignment traceback must be below ten. The prefilter receives the equivalent inclusive threshold

$$E = B - 1 = 9.$$

For mandatory fill i, let D_i be the minimum global Levenshtein distance between its query and target strings. Let C_i be the edit count on the traceback selected by the base aligner. A safe filter computes L_i such that

$$0 \le L_i \le D_i \le C_i.$$

The second inequality holds because no particular edit script can use fewer edits than the minimum over all scripts. For disjoint mandatory fills,

$$\sum_i L_i \le \sum_i C_i \le C_{\text{region}},$$

where C_{region} is the edit count exposed by `bioseq.edit`. A region may therefore be rejected when $\sum_i L_i > E$. A retained candidate still proceeds to the selected aligner, which constructs

a traceback before the original strict predicate is applied. The approximation can remove work, but it cannot replace exact evaluation.

The terms false positive and false negative are easy to misread in this setting. Here, the positive filter outcome means that a candidate is retained for alignment. The two errors have different effects.

Outcome	Meaning	Effect
False positive	A pair above the threshold is retained	Additional alignment work, with no change to the query result
False negative	A pair within the threshold is rejected	A valid result is lost, which violates the SQL predicate

Table 7.1: Consequences of prefilter errors. Safe filtering permits false positives but not false negatives.

BioSeqDB applies this contract to the geometry created by chaining. For a multi-seed chain, mandatory fill windows extend from the first anchor start to successive anchor ends. These windows include the ranks covered by anchors. An exact or hash-supported seed is not assigned zero edit cost merely because it formed an anchor. Its symbols are compared within the mandatory fill, so a hash collision can loosen candidate generation but cannot remove edits from the filter bound. Single-seed spans are handled separately by exact edit accounting.

The terminal windows have different semantics. Local extension and X-drop may omit an expensive prefix or suffix. A large global distance in such a window does not prove that the permitted local alignment fails. The prefilter therefore examines mandatory fill windows and leaves optional end extensions to the aligner. Reverse-strand fills are first expressed in the orientation in which they will be aligned.

Unequal lengths also require a restricted rule. Every transformation from Q to T needs at least

$$\Delta_{\text{len}} = ||Q| - |T||$$

insertions or deletions. BioSeqDB uses this exact lower bound for unequal lengths rather than extending an equal-length approximation beyond its stated domain. Unsupported sizes, thresholds, and execution geometries are retained conservatively. Resource or device failures are ordinary query errors, since silently changing the selected execution route would make performance and diagnostic behavior difficult to interpret.

7.2. SneakySnake chip-maze filtering

SneakySnake treats approximate matching as routing through a chip maze Alser, Shahroodi, et al., "SneakySnake: a fast and accurate universal genome pre-alignment filter for CPUs, GPUs and FPGAs." For equal-length strings of length m and an inclusive threshold E, the maze has $2E+1$ rows. Row d, where $d \in [-E, E]$, represents one displacement between the query and target. At target position j, its cell is

$$Z_d(j) = \begin{cases} 0 & 0 \le j + d < m \text{ and } Q[j+d] = T[j] \\ 1 & \text{otherwise.} \end{cases}$$

A zero is a match on that diagonal and a one is an obstacle. Starting at a checkpoint, the algorithm measures the matching prefix on every available diagonal. It follows the longest

run, crosses the obstacle at its end when the right boundary has not yet been reached, and starts another search from the new checkpoint. The number of crossed obstacles is the filter value. If it exceeds E, the pair cannot satisfy the threshold in the safe equal-length domain.

Figure 7.1 shows three escape segments. The highlighted route stays on the unshifted row. It follows exact runs at positions 0–2, 4–6, and 8–10, crossing obstacles at positions 3, 7, and 11. The third crossing raises the count above $E = 2$, so this example is rejected. The maze abstracts away the edit script. In particular, it records neither the operation responsible for a change of row nor the vertical distance between selected segments.

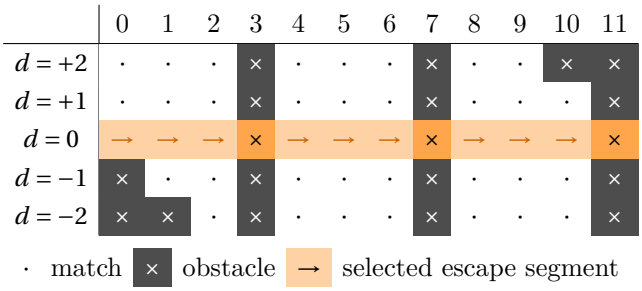


Figure 7.1: A rejecting SneakySnake chip-maze walk for Q = `AAAAAAAAAAAA`, T = `AAACAAACAAAC`, and $E = 2$. Each row represents a query-to-target displacement. Amber cells show the successive escape segments and three crossed obstacles on the unshifted row.

The corresponding procedure is compact.

Algorithm 11 Unpartitioned SneakySnake core for one equal-length submaze

$c \leftarrow 0$
$r \leftarrow 0$
while $c < m$ do
$b \leftarrow \max_{d \in [-E,E]} \text{matchingPrefix}(Q, T, c, d)$
if $c + b \geq m$ then return retain
$r \leftarrow r + 1$
if $r > E$ then return reject
$c \leftarrow c + b + 1$
end while
return retain

The lower-bound direction follows from the relation between a maze route and an edit script. Between two edits, an alignment follows one diagonal through a run of equal symbols. A substitution ends a run on the same diagonal. An insertion or deletion moves the next run to another diagonal. An edit script of cost k therefore induces matching segments separated by at most k obstacles. The chip maze also permits transitions whose cost an edit script would represent more precisely, so it is a relaxation of edit distance.

SneakySnake chooses the furthest reachable checkpoint under this relaxed model. The greedy escape rule minimizes the number of obstacles in the routing model Alser, Shahroodi, et al., "SneakySnake: a fast and accurate universal genome pre-alignment filter for CPUs, GPUs and FPGAs." If $D(Q, T) \leq E$, the edit script induces a maze route with at most E obstacles, and the greedy route cannot use more. Consequently, a greedy count above E proves that the global distance also exceeds E. The converse does not hold because the relaxation can omit costs that an edit script must pay.

The maze is less expressive than edit-distance dynamic programming. This loss of information is intentional. The obstacle count can be smaller than the true distance, which retains some pairs that later fail alignment. It must not be larger in the domain used for rejection. Partitioning a long gap into bounded submazes may weaken the result further, since a new submaze forgets some preceding routing state. That change can add false positives, but it does not create false negatives. Bounds from non-overlapping mandatory pieces can still be accumulated because those pieces require separate edits.

BioSeqDB compares normalized alphabet ranks rather than DNA characters. The same equality operation can therefore handle nucleotide and amino-acid alphabets once both operands have a common representation. A scalar CPU path scans matching runs directly. A vector CPU path compares contiguous groups of ranks and locates the first mismatch. Both execute the same checkpoint rule. On a GPU, separate candidates can be assigned to independent invocations at small thresholds. Larger thresholds expose more diagonal work, so a small cooperating group can evaluate shifts for one candidate. Fixed submaze sizes bound memory use and keep dispatch geometry predictable.

Long gaps are divided into consecutive bounded submazes, and Algorithm 11 is applied to each one. The lower-bound counts are added. Restarting at a submaze boundary discards routing state and can reduce the resulting count. It cannot increase the bound above the value for the complete maze, so partitioning weakens filtering rather than changing accepted query results.

At each checkpoint, an exact pruning rule can reduce prefix scans. Suppose the best run found so far has length b. Another diagonal can improve the maximum only if its first b positions match, including the position at offset b. Testing that position first discards only a diagonal that cannot win. Every possible improvement still receives a complete matching-prefix comparison.

These mappings change how comparisons are scheduled, not the rejection criterion. If a vector lane, work group, or bounded representation cannot express a job, the job is retained for exact alignment. This rule keeps backend selection separate from query correctness.

7.3. GateKeeper shifted-mask filtering

SneakySnake must repeatedly search diagonals at each checkpoint. Very short gaps with small thresholds admit a more regular bit-vector computation. GateKeeper derives such a filter from Shifted Hamming Distance methods Alser, Hassan, et al., "GateKeeper: a new hardware architecture for accelerating pre-alignment in DNA short read mapping"; Bingöl et al., "GateKeeper-GPU: Fast and Accurate Pre-Alignment Filtering in Short Read Mapping." For equal-length strings, the unshifted mismatch mask is

$$H_0[j] = [Q[j] \neq T[j]].$$

For each displacement $d = 1, \ldots, E$, two further masks compare the target with shifted query positions.

$$H_{-d}[j] = [j \geq d \wedge Q[j-d] \neq T[j]]$$
$$H_{+d}[j] = [j + d < m \wedge Q[j+d] \neq T[j]].$$

The complete representation contains $2E + 1$ masks. A zero denotes a possible match. A run of true matches displaced from the main diagonal can therefore appear in one of the shifted masks even when H_0 contains mismatches.

Isolated zeros are often accidental matches rather than evidence of a useful run. GateKeeper amends each mask before the masks are combined. BioSeqDB fills an internal zero run of length one or two only when ones bound it on both sides. Longer runs and runs touching an edge remain unchanged. For an unbounded bit vector M, with positions outside the valid interval treated as zero, this operation can be written as

$$\begin{aligned} S &= \neg M \wedge (M \ll 1) \wedge (M \gg 1) \\ P &= \neg M \wedge (\neg M \gg 1) \wedge (M \ll 1) \wedge (M \gg 2) \\ \mathsf{amend}(M) &= M \vee S \vee P \vee (P \ll 1). \end{aligned}$$

The amended result is restricted to the valid sequence positions. When a mask spans several machine words, neighboring bits must be carried across word boundaries. Otherwise an identical two-zero run would receive different treatment depending on its position in memory.

A shift also creates a vacant position at one end. If this position remained zero, the later intersection could interpret it as a match and hide an end error. After amendment, BioSeqDB sets the boundary-adjacent bits

$$\widehat{H}_{-d}[d-1] \leftarrow 1, \qquad \widehat{H}_{+d}[m-d] \leftarrow 1.$$

The index identifies the position next to the vacant prefix or suffix for displacement d. The ordering is significant because compensation before amendment could turn an edge run into a bounded run and fill it.

All amended and compensated masks are intersected. The result is divided into four-bit groups. A zero group contributes no cost. The patterns

$$0101, \quad 0110, \quad 1001, \quad 1010, \quad 1011, \quad 1101$$

contribute two, while every other nonzero group contributes one. Accumulation can stop at $E+1$, since larger values have the same threshold outcome. This grouped estimate is not a popcount and it is not an exact edit distance.

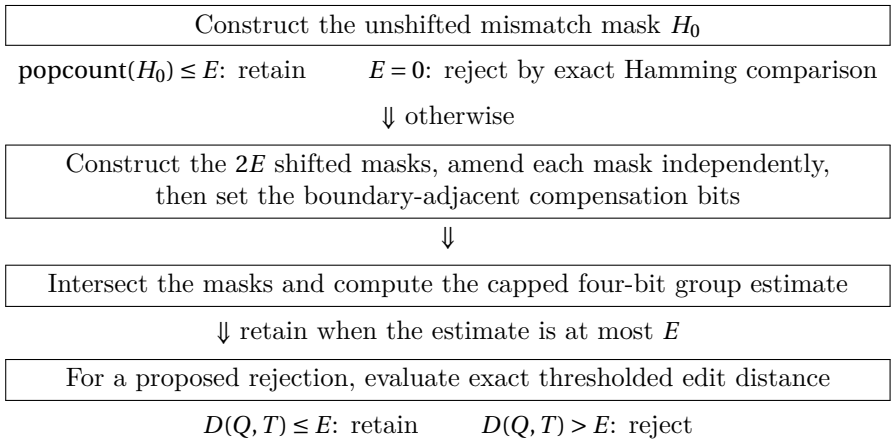


Figure 7.2: GateKeeper decision flow in BioSeqDB. The unshifted mask provides an exact rejection certificate when $E = 0$. For positive thresholds, shifted masks can retain a candidate or propose rejection, and the exact threshold guard confirms the proposal.

7.3.1. The exact rejection guard

Mask amendment changes zeros to ones, and the grouped estimate is an approximation. An observed absence of false negatives on a data set would not prove safety for every sequence. BioSeqDB therefore treats an estimate above E as a rejection proposal. A thresholded exact Levenshtein calculation confirms whether $D(Q,T) > E$. If it does not, the candidate is retained.

The exact guard changes the role of the published filter. GateKeeper was designed to predict whether a pair is likely to satisfy an edit threshold, and its filtering accuracy is evaluated empirically Alser, Hassan, et al., "GateKeeper: a new hardware architecture for accelerating pre-alignment in DNA short read mapping"; Bingöl et al., "GateKeeper-GPU: Fast and Accurate Pre-Alignment Filtering in Short Read Mapping." An empirical false-negative rate is not a sufficient contract for a database predicate, where one rejected valid pair changes the result set. BioSeqDB therefore uses shifted masks to decide whether exact threshold evaluation is needed. The masks can retain a candidate immediately, but for $E > 0$ they cannot reject one without the guard. When $E = 0$, a nonzero unshifted mismatch mask is itself an exact certificate because no insertion or deletion can produce a zero-edit alignment.

Algorithm 12 Guarded GateKeeper decision for one equal-length gap

$H \leftarrow \mathsf{unshiftedMismatchMask}(Q, T)$
if $\mathsf{popcount}(H) \le E$ then return retain
if $E = 0$ then return reject
$A \leftarrow \mathsf{amend}(H)$
for $d \leftarrow 1$ to E do
$H_{-d} \leftarrow \mathsf{shiftedMismatchMask}(Q, T, -d)$
$H_{+d} \leftarrow \mathsf{shiftedMismatchMask}(Q, T, +d)$
$M_- \leftarrow \mathsf{amend}(H_{-d})$
$M_-[d-1] \leftarrow 1$
$M_+ \leftarrow \mathsf{amend}(H_{+d})$
$M_+[m-d] \leftarrow 1$
$A \leftarrow A \wedge M_- \wedge M_+$
end for
if $\mathsf{groupedCost}(A) \le E$ then return retain
if $\mathsf{exactDistanceExceeds}(Q, T, E)$ then return reject
return retain

The exact guard evaluates only the threshold question. The scalar and generic GPU routes use banded dynamic programming. A path that leaves the band already requires more than E insertions or deletions. Values are capped at $E+1$, and only the preceding and current rows are needed. Specialized GPU variants instead use an exact bounded wavefront with longest-common-prefix extension. Both forms answer whether $D(Q,T) > E$, use bounded memory, and omit traceback. They run only when the mask calculation proposes rejection.

This design changes how GateKeeper results may be combined. A retained mask estimate is not a proved lower bound, so it must not be added across fills. A confirmed rejection is sufficient on its own because one mandatory fill already exceeds the region threshold. The mask remains useful as a cheap screening step that limits calls to the exact guard.

The rank representation again removes assumptions about nucleotide encoding. On CPUs, masks fit into one or a small number of machine words for the intended short-gap domain. Boolean operations then evaluate many sequence positions at once. On GPUs, each invocation handles a candidate with regular register operations. Specialized kernels are useful for common short lengths and low thresholds, while a generic kernel covers other supported

mask geometries. Longer gaps or larger thresholds are routed to SneakySnake or retained for the base aligner.

7.4. Dispatch and query-pipeline integration

Algorithm selection depends on both candidate geometry and execution backend. The scalar and vector CPU routes use SneakySnake across their supported range. A GPU batch uses GateKeeper when all jobs are short, equal in length, and have small thresholds. Other supported GPU batches use SneakySnake. Resolving the algorithm once for a batch avoids separate transfers and result merging. A mixed batch that does not satisfy GateKeeper's domain follows the broader SneakySnake route.

The mapping is summarized in Table 7.2. The table describes policy rather than query semantics. Every route must satisfy the same rejection contract.

Execution route	Preferred filter	Reason
Scalar CPU	SneakySnake	Direct prefix scans have low setup cost and cover general supported gaps
Vector CPU	SneakySnake	Contiguous rank comparisons accelerate matching prefix detection
GPU, short regular gaps	Guarded GateKeeper	Bit masks map to regular per-candidate operations
GPU, broader supported gaps	SneakySnake	Parallel diagonal evaluation covers larger thresholds without large mask state
Unsupported geometry	No rejection	Exact alignment preserves query semantics

Table 7.2: High-level mapping of prefilter algorithms to execution routes.

The relevant stage order is

$$\text{seeding} \rightarrow \text{chaining} \rightarrow \text{preliminary classification} \rightarrow \text{mandatory-fill prefiltering} \rightarrow \text{base alignment} \rightarrow \text{exact edit predicate} \rightarrow \text{final result classification}$$

This order has two important consequences. First, rejected regions do not consume base-alignment work. Retained regions are unchanged, including scoring parameters, gap costs, local-extension rules, and traceback. Second, the exact predicate runs before final primary and secondary classification. A region that violates the SQL bound cannot alter the final classification or mapping quality of a region that satisfies it.

Each fill result is associated with its owning region. SneakySnake lower bounds for disjoint mandatory fills are added with saturation at $E{+}1$. A region is removed when that total exceeds E. A GateKeeper mask estimate is not composable and contributes no numeric lower bound. GateKeeper rejects a region only through the exact $E = 0$ Hamming certificate or an exact-guard confirmation. Survivor compaction retains region order so downstream classification sees the same ordering it would have seen without prefiltering.

The prefilter is activated only when the query asks for exact edit accounting and supplies a usable positive strict bound. Bounds that cannot be represented are not converted by unsigned wraparound. The exact predicate still handles zero, negative, and otherwise unsupported cases according to SQL semantics. Backend choice follows the query's execution plan. Merely having a GPU available does not move an otherwise CPU query to a device.

7.5. Limitations

Retention is not evidence of similarity. SneakySnake may undercount edits because it omits vertical movement between escape segments and resets state at bounded submazes. GateKeeper may combine zeros from different shifted masks. Both effects can retain a pair whose true distance exceeds E. Exact alignment and the final predicate remain necessary.

Only mandatory fills are filtered. The method does not reject from a costly local prefix or suffix because the aligner may legally omit it. This restriction reduces the amount of work that can be filtered, but it preserves local-extension semantics.

Unequal lengths receive a weaker bound. The length difference can reject an impossible threshold, but it says little about substitutions within unequal strings. Such candidates often proceed to alignment even when their true distance is large.

Algorithm domains are finite. Bit masks, bounded submazes, and GPU work groups impose length and threshold limits. Inputs outside a proved and representable domain are retained. A new kernel or encoding cannot be used for rejection until it satisfies the same lower-bound or exact-guard contract.

Filtering has a break-even point. A filter adds rank comparisons, batching, and dispatch overhead. If nearly all candidates survive, this cost may exceed the saved alignment work. GPU launch latency is particularly relevant for small batches. Dispatch policy therefore uses candidate geometry, but performance selection never weakens the exact post-alignment check.

The edit model is specific. The filter bounds minimum unit-cost Levenshtein distance, while the SQL predicate counts unit-cost edits on the selected traceback. The latter count can be larger, which is the safe direction established by $D_i \le C_i$. The aligner may select that traceback using affine gap scores or other scoring choices. A bound for another distance model must not be treated as a proof for this predicate.

7.6. Conclusion

BioSeqDB places prefiltering between chaining and base alignment, where it can remove regions whose mandatory fills cannot satisfy an exact SQL edit bound. SneakySnake provides a general lower-bound route through shifted matching diagonals. GateKeeper provides a compact shifted-mask calculation for short regular gaps, with an exact threshold guard before rejection. CPU and GPU implementations schedule these calculations differently while retaining the same contract. Unsupported inputs and inconclusive estimates proceed to alignment, and every survivor is checked by the exact predicate. This asymmetry permits extra work from false positives while preventing result loss from false negatives.

Chapter 8

Sequence representations and lazy evaluation

8.1. One logical sequence, several physical forms

The algorithms described in the preceding chapters are easiest to express over an ordered array of nucleotide ranks. PostgreSQL does not always present a sequence in that form. A value may be an in-memory packed object, a compressed or out-of-line PostgreSQL value, a set of columnar fragments, or a view over a sequence source. It may also be the result of slicing, concatenation, reversal, or complementation. These cases have the same logical content but different costs and lifetime constraints.

Converting every input to one byte per symbol would give algorithms a simple interface. It would also decode complete values when a query needs only a small interval. For a sequence of length N and a requested interval of length m, that choice changes the immediate work from $O(m)$ to $O(N)$. It also creates a second representation that may be several times larger than the packed source. This cost is particularly undesirable for repeated SQL expressions and for alignment against long references.

BioSeqDB separates four properties of a sequence:

1. The logical sequence is a length and an ordered stream of alphabet ranks.

2. The packing geometry maps ranks to bits and physical storage units.

3. The representation owns or locates the data and defines its lifetime.

4. The access requirement states whether a consumer can stream symbols or requires a contiguous packed interval.

The same logical interface can therefore serve minimizer extraction, alignment, SQL operators, index construction, and accelerator uploads. A lazy value may allocate metadata and retain a source handle. It is still lazy when it postpones decoding and repacking the complete symbol payload until a consumer requests a representation that requires them. The resident interface is exposed by `compact_sequence_range`. The non-owning logical facade is `lazy_sequence_view`. Figure 8.1 shows this separation.

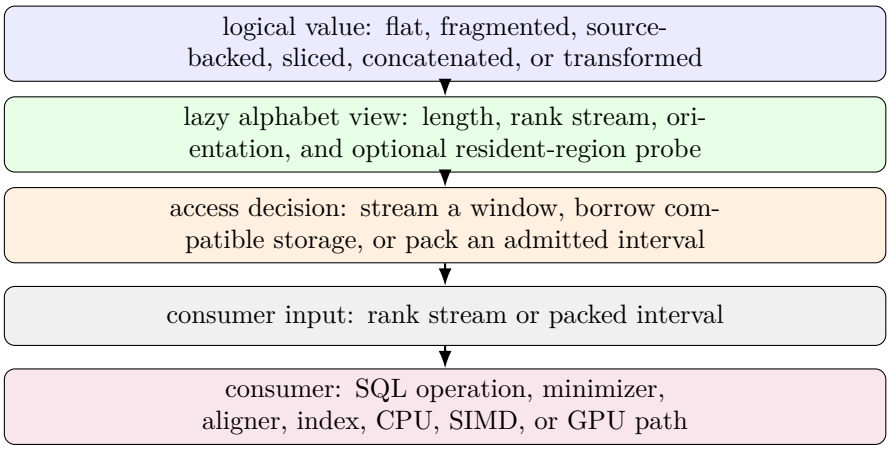


Figure 8.1: Sequence abstraction layers. Ownership remains with the value or an associated read session. Packed storage is acquired only when required by the consumer.

8.2. Alphabet-generic compact ranges

8.2.1. Logical and physical traits

The compile-time compact range represents the resident end of the design. Its alphabet type determines rank cardinality and biological operations such as complementation. A separate storage-unit type and bit width determine packing. Algorithms can therefore operate on DNA4, DNA5, or another registered alphabet without embedding a particular code table or machine-word width.

Listing 8.1 gives a conceptual interface. It is an API sketch rather than an implementation listing.

Listing 8.1: Conceptual compact-range interface.

```
compact range traits
    alphabet type
    storage unit type
    bits per symbol
    symbols per unit
    unused high bits per unit

compact range operations
    logical size
    physical unit count
    rank at logical position
    symbol iterator
    copy ranks from [first, last) into destination
```

The distinction between logical size and physical unit count is essential. If a unit contains L symbols and the last unit contains P empty symbol slots, then the logical length is

$$N = UL - P,$$

where U is the number of units. Code that treats U as a symbol count is correct only for unpacked storage with $L = 1$.

Packing width must also remain separate from algorithmic arithmetic. A 32-bit DNA4 storage unit holds 16 ranks, but a 27-symbol rolling key requires more than that unit can

represent. Hash state, score types, and coordinates are selected from their mathematical ranges. They are not inferred from the physical load width.

The range family has a common vocabulary but more than one ownership model. Table 8.1 summarizes the principal forms.

Table 8.1: Principal compact-range forms.

Input form	Physical form	Ownership rule
ordinary alphabet range	one value per symbol	borrows a stable lvalue from the caller
packed owning range	bit-packed vector	owns the packed payload
validated PostgreSQL value	typed trailing payload	borrows the resident datum and its established lifetime

A borrowed range must not be constructed from a temporary. Likewise, a range over a PostgreSQL value is valid only while the detoasted datum or read session remains live. The shared interface does not erase these obligations.

8.2.2. Ranks, symbols, and orientation

Rank iteration returns compact integers suited to inner loops. Symbol iteration converts each rank to the alphabet type and is useful when an operation needs alphabet-aware semantics. Keeping both forms avoids repeated symbol construction in numeric kernels while retaining a typed interface for validation and text conversion.

Interval extraction accepts logical coordinates $[from, to)$, source orientation, and destination order. Source orientation determines coordinate mirroring and complementation for a reverse-complement view. Destination order determines how the selected ranks are written. These are independent choices. A canonical minimizer cursor, for example, can inspect reverse-complement coordinates while retaining the order expected by its rolling state.

An allocating full conversion remains available at byte-oriented boundaries. It is not the default for alignment. The interval form seeks to the first relevant packed unit and decodes only the selected symbols. For resident data, its work is $O(to - from)$. The corresponding compact-range operation is `to_encoded_into`. Lazy and fragmented representations expose the same bounded rank-copy contract.

A logical slice has its own length but does not change the coordinate system of the underlying payload. If a forward view covers $[a, b)$, view position j maps to source position $a + j$. In a reversed view it maps to $b - 1 - j$. Nested slices first normalize their bounds against the current view, then compose this mapping with the parent. This rule avoids accumulating rounding or padding adjustments at each SQL operation. Padding belongs to the physical final unit, while slice bounds remain exact symbol coordinates.

Iterator categories reflect what the representation can actually provide. A resident compact range supports direct rank access. A source-backed view may offer the same logical operation through a bounded read rather than through a stable reference. Generic consumers should therefore express traversal in terms of ranks and windows. They should request a resident range only when their kernel truly depends on typed contiguous storage.

8.3. Packing and resident PostgreSQL values

8.3.1. Packing geometry

For an alphabet with σ ranks, the default bit width is

$$b = \lceil \log_2 \sigma \rceil.$$

A registered packing policy may choose a wider code, provided that every rank is representable. For a unit of W bytes, BioSeqDB stores

$$L = \left\lfloor \frac{8W}{b} \right\rfloor \quad \text{symbols and leaves} \quad r = 8W - Lb$$

unused high bits. For logical position i,

$$u(i) = \lfloor i/L \rfloor, \qquad s(i) = b(i \bmod L),$$

$$rank(i) = \big(unit[u(i)] \gg s(i)\big) \,\&\, (2^b - 1).$$

The first logical symbol occupies the least significant b bits of the first unit. Figure 8.2 shows the layout for eight DNA4 symbols in a 16-bit unit.

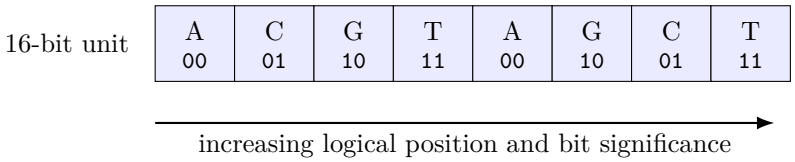


Figure 8.2: Least-significant-symbol-first packing for DNA4. Sequential extraction masks the low rank and shifts toward higher positions.

DNA4 uses two bits per rank and fills all standard unit widths. DNA5 needs three bits, so each unit can contain high bits that are not symbol slots. Those unused bits differ from the empty symbol slots in the final unit. Both values must be recorded correctly when computing length or converting widths.

A resident rank access uses one unit load, one shift, and one mask. Repacking an interval of m symbols takes $O(m)$ work. Cross-width views require more care. Matching symbol widths and divisible unit sizes are not enough to justify a typed pointer cast. Alignment, object lifetime, permitted type access, and byte order must also agree. BioSeqDB consequently distinguishes native typed borrowing from byte-compatible upload. A byte-compatible descriptor may be copied to an accelerator buffer, but it does not authorize a CPU to dereference the bytes through another unit type.

8.3.2. A flat trailing-payload representation

A C++ container is not a PostgreSQL datum. BioSeqDB therefore has a standard-layout packed object whose fixed header is followed by aligned data. The header records the alphabet code, storage-unit width, logical length, and padding in the last unit. Runtime metadata is checked before it is dispatched to an alphabet and width specialization. The inner rank loop then receives compile-time masks, complement rules, and lane types.

The trailing payload permits a zero-copy compact range when the PostgreSQL value is resident, uncompressed, aligned, and already validated. The adapter does not create a vector.

This statement does not mean that access to every SQL value is copy-free. PostgreSQL may first detoast or decompress a value according to its normal storage rules PostgreSQL Global Development Group, PostgreSQL 17 Documentation: TOAST. It also does not mean that a pointer remains valid after its memory context or read session ends.

Boundary validation checks header metadata, payload size, unit count, padding, and logical length for consistency. Only after this check does runtime metadata select a typed compact range. This places dynamic dispatch outside the symbol loop while preserving one SQL sequence type for several alphabets and storage widths.

8.4. A lazy view over flat and fragmented storage

8.4.1. The reader-facing facade

A packed pointer cannot describe a sequence that crosses fragments or draws its symbols from a sequence source. Joining all pieces before reading the first rank would defeat bounded access. BioSeqDB instead exposes a small alphabet-specific view whose backing representation is selected at run time. Its public behavior consists of:

- reporting the logical length
- writing a requested rank interval into caller-provided storage
- iterating bounded chunks in either orientation
- probing whether a compatible packed interval is already resident
- reporting whether the backing state is eligible for worker sharing

The facade stores a state pointer and a table of representation operations. It does not contain or own the complete sequence. A flat value, a fragmented scanner, a source-backed value, and a composition tree can therefore present the same rank-stream contract. Algorithms depend on that contract rather than on scanner or source types.

When a requested window lies inside one fragment, a backend can read it directly. A request that crosses boundaries is assembled in bounded scratch storage. Reverse traversal maps the logical window to source coordinates before changing write order and applying complement semantics. Thus a terminal slice does not require a prefix scan through earlier fragments. Source lookup and I/O costs still depend on the source, but the interface limits the requested coordinate range.

The resident probe is deliberately stricter than rank streaming. It succeeds only when the requested interval has stable storage, compatible packing, and a valid lifetime. A source cache page is not exposed as a resident region if the facade cannot retain its pin. This prevents a local optimization from creating a dangling pointer.

Listing 8.2 summarizes a window read. The pseudocode places coordinate normalization before backend access so that each backend receives one valid half-open interval.

Listing 8.2: Bounded rank extraction through the lazy facade.

```
READ_WINDOW(view, first, last, orientation, output)
    require 0 <= first <= last <= view.length
    mapped = map_to_source(first, last, orientation)
    for each intersecting piece in mapped order
        if piece is a literal
            decode only the intersecting packed ranks
```

```
    otherwise
        request only the intersecting source ranks
    apply output order and complement mapping
require output.count = last - first
```

The output buffer is supplied by the caller, which controls temporary memory. The backend may divide a large interval into smaller chunks, but it must preserve rank order and report the same logical result independently of piece boundaries. A consumer can therefore choose a chunk size suited to its cache or vector kernel without changing sequence semantics.

8.4.2. Ownership and failure handling

The lazy facade is non-owning. A call owner or PostgreSQL read session retains its state and releases source handles, detoasted values, arenas, and admitted allocations. Code that invokes the facade must finish before that owner is destroyed. Callback interfaces restrict results so that pointers, iterators, references, and range views cannot be returned accidentally. A callback must also avoid storing the view through a side effect.

This rule matters because PostgreSQL errors may transfer control without C++ stack unwinding. A read session registers cleanup actions and drains them on the PostgreSQL error path before rethrowing the database error. Relying only on ordinary destructors would leave resources live when an error crosses C++ frames PostgreSQL Global Development Group, PostgreSQL 17 Documentation: Using C++ for Extensibility.

Worker eligibility is a capability, not an ownership token. A flat resident value may be worker-shareable. A composition is shareable only when every source leaf permits it. Any CPU task or accelerator submission that can outlive the current call first obtains a stable resident owner. The owner and its memory permit remain live until task completion or the accelerator fence.

8.5. Persistent slices and lazy composition

8.5.1. Immutable piece trees

SQL values must survive nested slicing, concatenation, and transformation. A raw pointer into another PostgreSQL value cannot supply that lifetime. It also cannot describe a slice that begins inside a packed unit. BioSeqDB represents such expressions as expanded PostgreSQL objects backed by immutable balanced piece trees.

There are three node kinds. A source leaf stores a retained source handle and stable logical coordinates. A literal leaf stores owned bytes with validated type information. A branch joins two subtrees and caches their combined length, piece count, and height. Reverse and complement states are attached to edges rather than stored as payload-bearing nodes.

Figure 8.3 illustrates the representation. Existing subtrees can be shared by later expressions because their nodes are immutable and their arenas retain dependencies.

Concatenation joins two roots and rebalances the tree. Slicing splits at the two boundary coordinates, retains the middle subtree, and shares unaffected nodes. Reversal changes an edge transform and visits branch children in the opposite logical order. At a leaf, the same transform maps coordinates, changes output order, and complements ranks when required.

Transform composition follows the biological operation rather than the tree shape. Two reversals cancel. Two complements cancel for alphabets whose complement mapping is an involution. Reverse-complement can therefore be recorded as a small edge state even when it is applied to a deeply nested concatenation. During traversal, the accumulated state

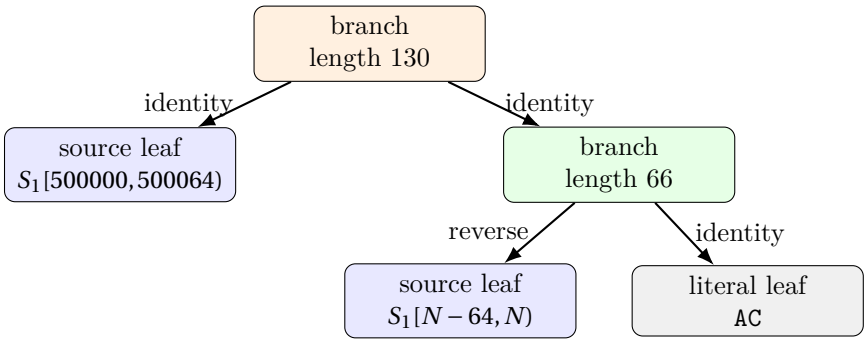


Figure 8.3: Persistent composition of source slices and a literal. Reversal is edge metadata. Neither source interval is copied into the tree.

determines child order and leaf decoding. This keeps repeated transformations independent of sequence length and avoids inserting long chains of unary nodes.

Joins of already compatible trees and all splits follow height-bounded paths. Concatenation that binds an unbound literal must validate and pack it. The SQL boundary may perform additional structural and size checks. The first expansion of a flat or out-of-line value can copy its packed bytes once without decoding ranks. Reading a selected interval later costs $O(m)$ for m emitted ranks plus fragment or source overhead.

The tree's arenas make sharing compatible with PostgreSQL memory contexts. They retain older arenas and source dependencies, and they release those dependencies through registered cleanup. As a result, a value returned by one SQL operator remains valid for the next operator without copying the original symbol payload.

8.5.2. Bound and unbound values

An expanded value records whether its alphabet is bound. A bound sequence has an alphabet and packing width. An unbound SQL literal contains canonical text but has not adopted an alphabet. Length, slicing, reversal, comparison, and unbound concatenation do not require complement semantics. Combining a bound value with an unbound literal validates the literal and adopts the bound alphabet. Two distinct bound alphabets are not silently recoded.

Element access returns a letter value containing both alphabet code and rank. This preserves identity in arrays and prepared statements. An in-range element of an unbound sequence is an error because its physical alphabet is unknown. An out-of-range element yields SQL `NULL`.

Equality and hashing follow logical content. For a bound value, the key contains a bound-state tag, alphabet code, and rank stream. Storage width, padding, flat or expanded form, source backing, and piece boundaries do not affect the result. An unbound value uses a different state tag and canonical text. It is therefore not equal to a bound sequence merely because both print the same characters.

8.5.3. SQL views and transforms

Sequence subscripting uses zero-based, half-open coordinates. Negative indices are relative to logical length. Omitted bounds select the corresponding end, and bounds are clamped to the valid range. If the normalized upper bound is below the lower bound, the result is an empty typed sequence. Missing elements produce `NULL`. Null bounds propagate `NULL`. Assignment, multiple dimensions, and non-unit steps are rejected.

Listing 8.3: Examples of logical sequence operations.

```
s[0]        returns the first letter
s[-1]       returns the last letter
s[:2]       selects [0, 2)
s[1:-1]     removes the first and last letters
s[4:2]      returns an empty typed sequence

concat(left, right)
reverse(sequence)
reverse_complement(bound_sequence)
```

Concatenation accepts sequence and textual operands as well as arrays of text or letter values. Reverse is defined for bound and unbound values because it only changes order. Reverse-complement requires a bound alphabet with a complete complement mapping.

A cast that keeps the alphabet and storage width can remain metadata-only. A same-alphabet width change records the target geometry and delays repacking until a flat consumer requests it. Binding an unbound literal validates each symbol. Relabeling a bound sequence as another alphabet fails. These rules are the same for flat, out-of-line, expanded, and source-backed values.

8.6. Access planning and materialization

8.6.1. Consumer requirements

Not every algorithm can stream. A scalar scan needs sequential ranks. A windowed algorithm may need bounded random access. A SIMD kernel or accelerator upload may require one contiguous packed interval. BioSeqDB represents this choice with three access categories. Their resulting routes are summarized in Table 8.2.

Table 8.2: Access routes selected from consumer requirements.

Route	Typical case	Copy and lifetime rule
stream symbols	forward scan over a tree, source, or fragment set	decodes bounded chunks into caller scratch with no complete packed allocation
borrow resident	aligned interval in a compatible flat value	performs no payload copy while the datum owner remains live
materialize admitted	unaligned interval or resident-only consumer	packs only the selected interval and retains its allocation permit through use

For sequential access, the consumer supplies bounded scratch space and reads chunks from the facade. For contiguous access, the planner first asks whether the representation can lend a compatible resident interval. The interval must begin at a boundary represented by the receiving range. A start in the middle of a packed unit cannot be passed as an ordinary compact range because the consumer would interpret the wrong first symbol.

If borrowing is impossible, the planner computes the packed size of the selected interval and obtains a memory permit. Materialization then packs only $[from, to)$. The returned owner holds both the allocation and its permit. Moving the owner into asynchronous work also moves the accounting lifetime.

The term zero-copy therefore applies to specific transitions, not to the sequence system as a whole. A validated flat value can be adapted by reference. A slice or transformation of an expanded tree can share payload nodes. A fragmented interval can be streamed without a

complete packed result. In contrast, PostgreSQL may detoast a value, an unaligned interval may require repacking, and text or binary output must create its destination form.

Durable storage is another explicit boundary. A source view is scoped to its transaction and cannot be placed directly into a heap or index tuple. A tree must be flattened before it crosses that boundary. If its logical size exceeds the flat PostgreSQL representation, the operation reports a size error rather than truncating the value.

Figure 8.4 follows the access decision from a bound SQL expression to a consumer.

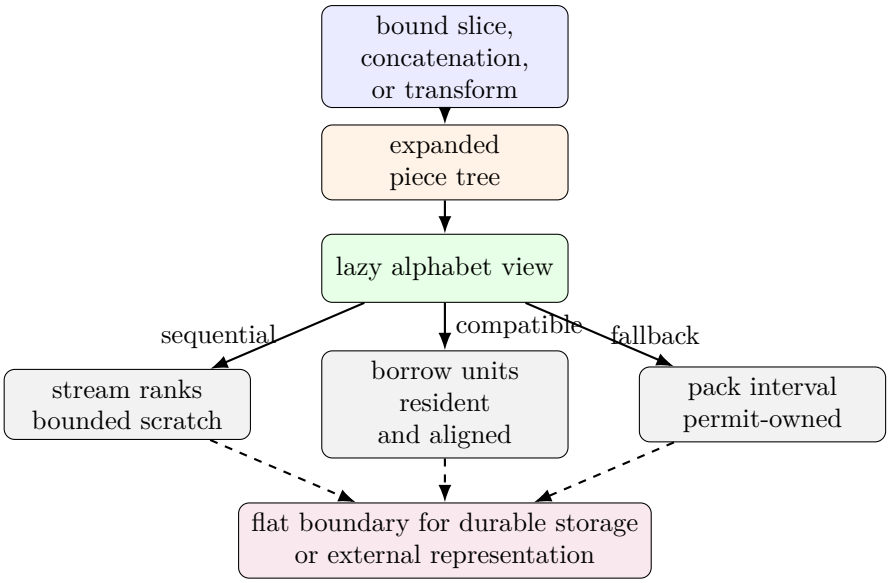


Figure 8.4: A bound lazy expression can finish through any access route. Flattening is performed only when the receiving boundary requires a durable or alternate representation.

8.6.2. Why eager conversion is avoided

Eager conversion has three effects beyond its immediate copy. First, it makes small slices pay for the full source. Second, it expands compact DNA ranks into byte or text buffers that place additional pressure on memory bandwidth. Third, it hides allocation inside a convenience operation, which makes memory admission and asynchronous lifetime difficult to account for.

Lazy composition changes the unit of work. Metadata operations preserve source coordinates and transforms. Streaming consumers request bounded chunks. Resident-only consumers name the interval they need and receive an owner for that interval. Conversions for external byte-oriented APIs remain valid, but they occur at a declared boundary and can be moved outside repeated calls when their input is invariant.

This design does not assume that laziness is always faster. A consumer that reads the complete value repeatedly may benefit from one admitted packed copy. The access planner makes that choice visible. The important property is that a slice or source-backed value is not forced into a complete resident vector before the actual consumer requirement is known.

8.7. Contract for sequence consumers

The sequence model can be summarized by the following rules:

1. Write alphabet-generic algorithms against compact-range traits or the lazy rank facade. Derive rank masks and complement behavior from the selected alphabet.

2. Keep storage width separate from hash, score, and coordinate width. Physical packing does not define an algorithm's numeric range.

3. Request the weakest access form that supports the computation. A forward scan should stream. A resident-only kernel should request and own the selected packed interval.

4. Convert only the required logical interval at a byte-oriented or durable boundary. Do not hide complete conversion in an iterator or convenience wrapper.

5. Treat every borrowed pointer as a capability tied to a named owner. Do not persist pointers into PostgreSQL values, source cache pages, or scratch buffers.

6. Preserve logical equality across flat, fragmented, source-backed, and composed forms. Representation details must not change rank content.

The earlier resident-vector model made one algorithm-friendly layout mandatory for every input. The current model makes logical ranks and ownership mandatory, while packed contiguity is an explicit request. This permits database-resident slices and fragmented sources to enter the alignment pipeline without treating all PostgreSQL and source storage as one kind of memory. The next chapter applies these representation and lifetime rules to PostgreSQL planning and custom scan execution.

Chapter 9

PostgreSQL API and custom scans

9.1. Alignment as a relational operation

The alignment pipeline described in the preceding chapters is not naturally expressed as a scalar SQL function. Its efficient unit of work is a batch of sequence pairs, and one pair may produce several accepted alignments. A scalar predicate instead receives one row, returns one Boolean value, and is normally evaluated independently at each call site. It cannot retain a group of inputs for vector or accelerator execution, and it cannot expand one input row into a set of match rows.

BioSeqDB nevertheless presents search through ordinary typed SQL. A typical query is

```
SELECT t.id,
       bioseq.score(t.id),
       bioseq.target_start(t.id),
       bioseq.target_end(t.id),
       bioseq.cigar(t.id)
FROM targets AS t
WHERE mmsearch(
        t.sequence,
        $1::bioseq.seq('dna5', 64))
```

The projected CIGAR value records the edit path of the current alignment SAM/BAM Format Specification Working Group, Sequence Alignment/Map Format Specification. The argument of each projection helper identifies the row lineage to which the value belongs. It does not contain the score or coordinates. These fields are supplied by the custom executor that owns the corresponding `mmsearch` expression.

Let input row i produce m_i accepted matches. The result cardinality is

$$N_{\text{out}} = \sum_{i=1}^{N_{\text{in}}} m_i, \qquad m_i \geq 0.$$

An input without a match produces no output. An input with one match behaves like a conventional filtering result. An input with several matches is projected several times, once for each match. Table 9.1 summarizes the resulting difference in execution semantics.

The SQL expression therefore acts as a rendezvous between PostgreSQL planning and BioSeqDB execution. It gives the parser a normal function signature and gives the planner a recognizable operation. It is not the location at which alignment is computed.

Table 9.1: Scalar evaluation and custom-scan execution.

Property	Scalar SQL function	BioSeqDB owner
Work unit	One tuple and one call	A bounded batch of sequence pairs
Cardinality	One scalar value per call	Zero, one, or many rows per input
Alignment fields	Independent calls have no shared match identity	All projected fields come from one match record
Execution route	PostgreSQL invokes the function directly	The scan selects the CPU or GPU pipeline for the batch

9.2. SQL surface and ownership contract

BioSeqDB provides a two-argument `mmsearch` form that uses session settings and a three-argument form that accepts typed alignment options. Both return `boolean` at the SQL type level. Match projections include the dynamic-programming score, query and target coordinates, edit count, mapping quality, strand, primary status, divergence, seed counts, and CIGAR. The usual rules for SQL types and null values remain applicable.

The Boolean return type permits `mmsearch` to appear in a `WHERE` or `JOIN ON` clause, but the ordinary function body is a failure boundary. If execution reaches it without a BioSeqDB owner, it reports that the query shape is unsupported. Returning `true` would be unsafe because any plan that missed the custom rewrite would accept every row. A cheap scalar fallback would also lose batching and multiple-match semantics.

Projected helpers follow the same ownership rule. During planning they remain ordinary expressions, so PostgreSQL can place them below an aggregate or on the nullable side of a join. During custom execution they are replaced with typed values associated with one match. A helper attributed to another target is not filled from the current alignment. This check prevents a score from one search from being combined with the coordinates of another search in the same query.

The requested projections also define the work required from the alignment pipeline. For example, CIGAR output requires traceback data, while a query that projects only coordinates need not format a CIGAR string. BioSeqDB snapshots the per-call or session options during planning. At scan initialization, it combines them with a field mask derived from the planned expressions. The batch then uses one consistent configuration.

9.3. Planner support, custom paths, and custom plans

The `mmsearch_support` function participates in PostgreSQL's planner support interface PostgreSQL Global Development Group, PostgreSQL 17 Documentation: Function Optimization Information. It assigns a high startup cost to direct scalar evaluation. This cost discourages a plan that treats alignment as an inexpensive row filter, but it does not by itself create an owner. Correctness cannot depend on a custom path winning a cost comparison against an executable scalar path.

Relation and join path hooks perform the structural work. They locate the exact function expression and its `RestrictInfo`, then check argument types, relation lineage, parameterization, option shape, and expression placement. A base-table predicate can become a `BioseqScan`. A search between two relations can become a `BioseqBatchJoin`. A transformed or lazy sequence expression may require a `BioseqSliceProject` below the owner so that the evaluated sequence has a stable source identity.

Only after these checks does the planner construct a `CustomPath`. Its cost estimates source reading, operand evaluation, and alignment. Join ownership also includes materializing the query side and reading the target side. These are finite nonnegative costs. The large scalar cost identifies an undesirable unowned call, while the custom cost describes the actual work of the owning plan.

Figure 9.1 shows the division of responsibility.

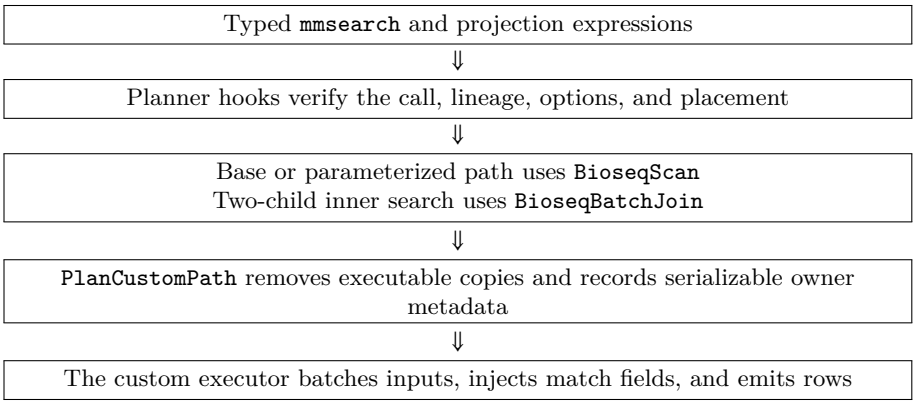


Figure 9.1: Planning and execution of an owned sequence search. The scalar function is retained for typing and planning, not as an alignment route.

PostgreSQL separates a `CustomPath` from its executable `CustomScan` PostgreSQL Global Development Group, PostgreSQL 17 Documentation: Writing a Custom Scan Provider. BioSeqDB uses this boundary to keep process-local state out of the plan. Child paths are carried as PostgreSQL paths. Expression metadata that must pass through the setrefs stage is stored as expression nodes. Relation identities, source recipes, and version tags are stored in copyable plan data. Pipeline objects, buffers, and sequence providers are created later by the executor.

`PlanCustomPath` removes the owned `mmsearch` from ordinary, vectorized, and storage-level qualification lists. Leaving any copy in those locations would provide a second execution site for the same operation. The planner also publishes hidden child outputs before setrefs when an upper node will need an evaluated slice or a parameter. This allows PostgreSQL to bind the value through the normal plan tree rather than through hidden global state.

Plain `EXPLAIN` initializes the scan only for description and submits no alignment work. Prepared statements likewise store expressions and source descriptions, then bind their values when the plan is executed.

9.4. Typed placeholder injection

PostgreSQL evaluates a projection from an expression tree. After planner setrefs have established the final expression locations, a tree mutator replaces each attributable helper with a typed PostgreSQL `Const` node.

The node address is stable throughout execution. Before one output row is projected, the scan writes the current match fields into the node's value and null flag. Coordinates, lengths, and edit count use `BIGINT`. Scores and counters whose source representation is 32 bit use `INT4`. Status fields use `BOOLEAN`, divergence uses a floating type, and CIGAR uses `TEXT`. All placeholders for a row are populated from the same match record.

The injected node is an executor data channel, not a constant in SQL semantics. Injection occurs after planning, so PostgreSQL cannot fold the changing value into the plan. For each

emitted alignment, the custom scan updates the stable nodes, rebuilds the projection state, and then calls `ExecProject`.

Joins require an additional placement mechanism. PostgreSQL may move a helper into a child target list, and an outer join may later null-extend that child. The projection support function can wrap the helper in a `PlaceHolderVar`, preserving its evaluation level, relation set, and nulling relations PostgreSQL Global Development Group, PlaceHolderVar in the PostgreSQL 17.10 source tree. The custom scan replaces the owned expression inside that placement structure. A stable owner identity distinguishes otherwise identical helper functions that belong to different searches.

9.5. Bounded batch execution

The executor alternates between collecting input and draining output. The batch has a fixed capacity, so the complete relation need not be retained. When the capacity is reached, or when the source ends, the current batch is submitted to the selected alignment pipeline. Each input obtains a cursor over its accepted matches. PostgreSQL then pulls projected rows until these cursors are empty.

Algorithm 13 Conceptual execution of an owned `mmsearch`

1. Read source rows, apply ordinary residual conditions, and capture stable sequence references for non-null operands.
2. When the batch is full or the source ends, run the selected alignment pipeline and create a match cursor for each input.
3. For the next match, restore its input row, populate the requested placeholders, and call Project.
4. Return one tuple, then continue step 3 until all match cursors are empty. Release the batch and resume step 1 if input remains.

This order reconciles batching with PostgreSQL's pull executor. The parent still asks for one tuple at a time, but expensive processing occurs once for a group of inputs. An input with no cursor disappears. An input with three cursor entries is restored and projected three times with three coherent sets of match fields.

Tuple lifetime is important during this process. A table scan may reuse its active slot on the next call, and PostgreSQL resets short-lived memory between returned rows. BioSeqDB therefore copies the row representation needed by the batch and retains stable sequence-source descriptors. Large or fragmented sequences can remain locator backed while the provider acquires the regions required for the current dispatch. If a lazy value must be materialized for projection, the stable source is restored before another match from the same input is emitted.

Scalar and vectorized table access converge at the same submission boundary. A vectorized filter may first produce a keep mask, but every retained row is then evaluated for alignment operands and submitted to the common batch. This separation allows ordinary relational filtering to remain vectorized without turning alignment into a Boolean vector predicate.

An optional prefilter can reject candidates only under its stated lower-bound condition. Survivors still pass through alignment. A condition on an exact match field, such as an edit-count limit, is applied to the resulting match record before projection. The prefilter therefore does not replace the user's SQL condition with an approximation.

9.6. Join execution and rescans

A search predicate below a join can remain owned by its base `BioseqScan`. This is the simplest case when the query sequence is a constant or comes from the same row. For correlated nested execution, the query operand arrives as a PostgreSQL execution parameter. On rescan, the node clears retained rows and match cursors, rebinds the parameter, and starts a new batch. Rebinding is required because each outer row may provide a different query sequence.

A join-level search between target and query relations uses `BioseqBatchJoin`. The query child is materialized into stable sequence references, while the target child is streamed. The owner forms batches of target and query pairs, performs alignment once, and emits the ordinary columns of both children together with the current match fields. Residual join conditions are evaluated at the placement required by PostgreSQL.

When an upper owner needs a lazy slice or another supported transformation, a projection provider evaluates it. The provider records the base lineage and the transformation recipe. The batch join and its projection helpers can then refer to the same evaluated value without repeating the transformation.

The join-level rewrite is intentionally limited to supported inner-join shapes with one attributable search operation. The planner declines ownership when it cannot preserve row security, volatility, nulling, or expression order. Such a call reaches the explicit unsupported-query boundary if no lower owner exists. This behavior is preferable to executing a function that cannot produce the required match rows.

9.7. User-visible plan information

The owning node is visible in PostgreSQL `EXPLAIN`. A base search is reported as `Custom Scan (BioseqScan)`, while a two-child search is reported as `Custom Scan (BioseqBatchJoin)`. A lazy transformation may appear as `BioseqSliceProject`. The owned `mmsearch` must not remain as an ordinary `Filter` or `Join Filter`, because either label would indicate an executable scalar copy.

Plain `EXPLAIN` reports planned information such as projected columns, operand ownership, source policy, and the planned pipeline architecture. `EXPLAIN ANALYZE` adds the executor's observed row counts and the architecture that actually ran. It can also report batch submissions, index read or recomputation decisions, storage acquisition, and fallback reasons. Text, JSON, XML, and YAML output use the same structured description rather than separate algorithm names maintained by hand.

Ordinary residual conditions can still appear below the custom node, provided that they do not contain the owned search. Under `ANALYZE`, the custom-node row count is the number of emitted matches rather than the number of source rows. It can therefore exceed the table row count when one target has several accepted alignments. Conversely, a scanned target with no match does not contribute an output row. This interpretation follows directly from the cardinality equation at the start of the chapter.

A useful plan review therefore asks three questions. First, is the expected BioSeqDB custom node present at the correct relation or join level. Second, is the semantic `mmsearch` absent from descendant filters and join filters. Third, under `ANALYZE`, do emitted rows and the executed pipeline agree with the query and configuration. Cost alone does not answer these questions.

The custom interface has deliberate limits. Batching delays the first result until at least one dispatch completes. A join owner may need to materialize one child. Unsupported

security or join shapes are rejected. Each new match projection requires a SQL type, an ownership mapping, and a population rule. These restrictions follow from the requirement that one search operation has one executor owner and that every emitted field refers to one match.

9.8. Conclusion

BioSeqDB keeps a conventional SQL surface while assigning alignment to a custom relational operator. Planner support marks direct scalar evaluation as expensive. Structural hooks build an owning custom path, and plan construction removes every executable copy of the search predicate. Typed placeholders then connect one match record to ordinary PostgreSQL projection expressions.

The executor collects bounded batches and emits zero, one, or many rows for each input. Base scans, parameterized rescans, and inner batch joins share this ownership model. EXPLAIN exposes the selected owner and distinguishes planned from executed pipeline information. The next chapter describes the physical sequence representation that allows these nodes to retain small locators and acquire packed sequence regions only when alignment requires them.

Chapter 10

Physical sequence storage in PostgreSQL

10.1. Storage as part of query execution

A biological sequence can be represented as a PostgreSQL variable-length value, but this representation alone is insufficient for large reference collections. A sequence may exceed the size of one PostgreSQL allocation. An alignment may need only several bounded intervals from that sequence, while a metadata query may not need sequence bytes at all. Storing every sequence as one eagerly reconstructed value would therefore couple the cost of a query to the total payload size even when the requested work is small.

BioSeqDB separates ordinary columns from externalized sequence payloads. The table access method stores scalar values and sequence locators in a columnar layout. Packed sequence fragments reside in relations selected by sequence length and shard capacity. A locator in the parent row identifies the fragment range and the immutable source version. Readers resolve that locator only when an operator requests sequence data. Persistent minimizer indexes form a third relation family, described in the next chapter. They are derived from the sequence source and are not part of the primary payload layout.

This organization makes storage an execution property rather than a passive serialization choice. Projection determines which column streams are read. Scalar predicates can discard chunk groups before a sequence locator is loaded. A lazy scanner bounds the fragment data retained by one execution. PostgreSQL locks, snapshots, write-ahead logging (WAL), and publication rules govern when newly written stripes and fragments become visible.

The columnar layer was adapted from Citus Columnar Storage. BioSeqDB retains its page mapping Citus Data contributors, Citus Columnar storage implementation, stripes, chunk groups, per-column streams, compression metadata, and min/max pruning. It adds sequence-specific indirection and execution interfaces. Table 10.1 summarizes the resulting distinction.

10.2. Columnar relation layout

10.2.1. Logical byte space and reservations

The main relation fork begins with two non-data blocks. Block 0 holds the column-store metapage and block 1 is reserved. Logical stream data begin after the PostgreSQL page header in block 2. Let

$$P = \mathrm{BLCKSZ} - \mathrm{SizeOfPageHeaderData}$$

Table 10.1: Generic column-store functions and BioSeqDB additions.

Concern	Generic column store	BioSeqDB behavior
Physical organization	Stripes, chunk groups, compressed column streams	The same organization for scalar data and compact locators
Large values	One serialized value, possibly managed by PostgreSQL TOAST	Fixed-size packed fragments addressed by a validated locator
Filtering	Min/max pruning over ordered scalar values	Scalar pruning occurs before sequence access, while sequence values are excluded from min/max pruning
Reading	Reconstruct projected rows from selected streams	Publish scalar or batch slots and acquire sequence fragments on demand
Reuse	Backend-local decompression may avoid repeated decoding	Separate scopes for fragment residency, decompressed chunks, persistent indexes, and shared query results

be the usable payload of one database page. Let $x \geq 0$ be an offset relative to the start of the stream-data area. It maps to

$$\mathsf{block}(x) = 2 + \left\lfloor \frac{x}{P} \right\rfloor, \qquad \mathsf{inpage}(x) = \mathsf{SizeOfPageHeaderData} + (x \bmod P).$$

Higher layers can consequently reserve a continuous byte interval without splitting every compressed stream at page boundaries.

The metapage records the format and storage identities together with the first unused stripe number, row number, and byte offset. A writer reserves non-overlapping ranges while holding the relation extension lock. It then serializes its stripe into the reserved range. Metapage changes use PostgreSQL's generic WAL support, so crash recovery follows the database page and buffer model. Initialization and truncation require stronger relation locks because they replace the shared allocation state.

A reservation is not equivalent to a visible stripe. Completion records the actual row count, byte length, and stream metadata in the storage catalogs. Readers use published metadata under their PostgreSQL snapshot. They do not infer valid data from a byte range that a writer reserved but did not finish. This distinction permits concurrent writers to allocate disjoint ranges while preserving transactional visibility.

10.2.2. Stripes, chunk groups, and streams

A stripe is the large append unit. Each stripe is partitioned into chunk groups, and every chunk group covers the same logical row interval for all columns. The default configuration places up to 150,000 rows in a stripe and 10,000 rows in a chunk group. Both values are table options. Their purpose is to balance metadata granularity, compression ratio, and the amount of data handled in one scan step.

For each column in a chunk group, the writer produces two streams. An existence bitmap represents NULL values at one bit per row. A value stream contains aligned serialized values for present rows. The writer may compress the value stream with Zstandard, LZ4, or PostgreSQL PGLZ, depending on the build and table options. It retains the compressed form only when it is smaller than the original. The chunk metadata records the algorithm that was actually used, including the uncompressed case.

Figure 10.1 shows the hierarchy and its relationship to a projected scan. The figure separates physical page mapping from the logical units used by the reader.

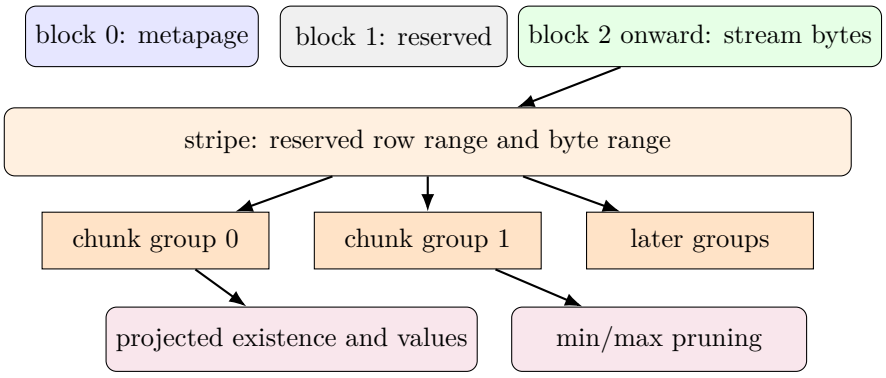


Figure 10.1: Column-store hierarchy. Published metadata maps stripes and chunks to the logical byte space. Projection and pruning determine which streams are decoded.

During a flush, offsets are calculated before any stream is copied. For one column, the physical order is its existence buffers followed by its value buffers. The completed metadata describes stripe and chunk-group row counts, stream offsets, compressed and uncompressed lengths, deletion masks, and compression identifiers. This metadata also supports columns added after an older stripe was written. Such a column is reconstructed from its PostgreSQL default or as NULL.

The metadata is divided by scope. A relation-options record contains the stripe size, chunk-group size, compression method, and compression level. A stripe record maps one storage identity and stripe number to its row interval and logical byte interval. Chunk-group records contain row and deletion counts. Per-column chunk records contain stream locations, value counts, compression details, and optional extrema. Row-mask records represent deleted logical rows without rewriting the immutable stream data. The sequence shard registry is separate from these ordinary column-store records because its key must also identify the parent column, length bucket, and fragment shard. Keeping these scopes distinct allows a narrow metadata change to be recorded without treating a fragment address as column-stream metadata.

The writer maintains min and max values for ordinary columns when PostgreSQL provides an appropriate comparison procedure. These extrema support chunk elimination. Sequence columns are excluded. A locator byte pattern has no biological ordering, and the lexicographic order of packed symbols does not provide a useful range predicate for alignment. A neighboring length, date, or sample category remains eligible for pruning.

Several limits prevent a chunk from becoming an unbounded allocation. A single aligned serialized value of 256 MiB or more is rejected. A stripe is flushed before a value buffer approaches its approximately 1.024 GiB bound. These limits apply to ordinary column chunks. Large biological sequences use the fragment layout described next, so their logical length is not restricted to one chunk buffer.

10.3. Fragmented and packed sequence payloads

10.3.1. Locator and fragment geometry

An externalized sequence is represented in its parent row by a normal `bioseq.seq` header followed by a 40-byte locator payload. The complete stub occupies 56 bytes. The header marks the value as a locator while retaining the alphabet identifier, packed storage width, and padding of the final fragment. The current locator format is version 3. It records the format version, length bucket, shard, fragment count, logical sequence length, first fragment row, and source epoch.

The source epoch identifies an immutable content version. A row address alone states where bytes were written, but it cannot show whether those bytes still belong to the same logical sequence after a storage lifecycle change. Readers therefore require a non-zero epoch and compare it with the registered source. They also recompute the expected fragment count and final padding from the logical length and packed representation. Invalid versions, reserved bytes, geometry, or epochs are reported as data corruption rather than followed as addresses.

The physical unit is alphabet-generic. For a packed storage unit of B bytes that represents L letters, one fragment contains

$$U_f = 1024 \text{ units}, \qquad S_f = 1024B \text{ bytes}, \qquad N_f = 1024L \text{ letters}.$$

For DNA4 stored in 64-bit units, a full fragment occupies 8 KiB and represents 32,768 bases. Other alphabets change L, and another packed unit changes B, without creating a DNA-only fragment path. Only the last fragment may contain padding.

10.3.2. Buckets, shards, and the canonical registry

Known-size insertions select one of 35 length buckets from the bit width of the final sequence length. The first bucket groups the smallest lengths, the middle buckets distinguish bit widths 9 through 41, and the last bucket covers larger values through the 64-bit range. A streaming insertion cannot know its final size when writing starts, so it uses the largest bucket. Bucketing is an operational choice rather than a biological classification. It prevents short and very long values from competing for the same active shard.

A bucket can contain several shards. All fragments of one sequence occupy one shard and receive contiguous row numbers. If r_0 is the first row, the row for fragment i is

$$r_i = r_0 + i, \qquad 0 \le i < n_f.$$

Known-size insertion estimates the required bytes and opens a new shard when the active shard lacks sufficient configured capacity. Streaming insertion checks the per-sequence cap while fragments arrive. A sequence is never split across shards. If it exceeds the configured shard limit, insertion fails with an error instead of creating a locator with ambiguous ownership.

The shard registry maps a parent storage identity, sequence column, bucket, and shard to the corresponding fragment relation and attachment relation. Its identity includes the database, tablespace, parent physical identity, column, bucket, and shard. This scope prevents an address in one database or tablespace from being accepted in another. The attachment relation can refer to derived state, while canonical persistent minimizer generations remain separate from the primary fragment rows.

Figure 10.2 follows one locator from the parent row to a bounded algorithm request.

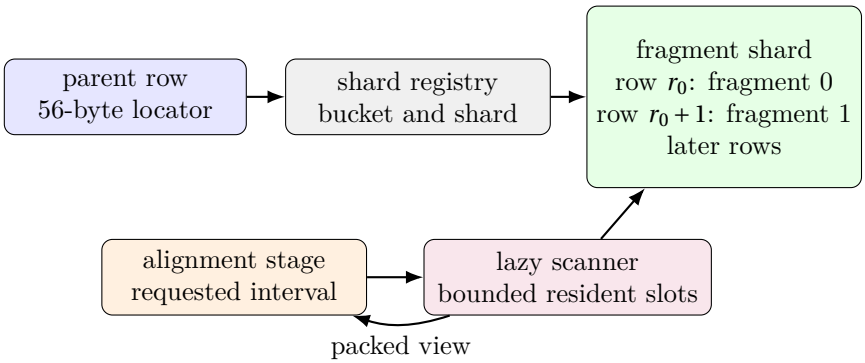


Figure 10.2: Sequence indirection and bounded acquisition. The scanner resolves the validated locator and reads only fragments needed by the requested interval.

10.3.3. Insertion, concurrency, and cleanup

Fragment allocation and writing use transaction-scoped advisory locks. One key coordinates allocation for a parent column and bucket. A second key also contains the shard and serializes writes to that internal resource. The keys are compact coordination identifiers, not full catalog identities. A hash collision can delay an unrelated writer, but it cannot make locators interchangeable because subsequent catalog and epoch checks use the complete identity.

Registry rows become authoritative only through the transaction that creates or updates the storage resources. Byte deltas accumulated during writing are applied when write states close. A reader holds a PostgreSQL snapshot and uses normal relation ownership rules, so it cannot treat an uncommitted fragment set as a complete source. Parent deletion and relation lifecycle handling are also explicit. Fragment relations are registered storage resources rather than permanent members of the extension installation. Parent-table drop processing removes associated resources, and orphan cleanup covers mismatches between registry rows and physical relations.

This separation matters because storage relations have a different lifetime from SQL extension objects. It also avoids changing extension membership from nested database execution contexts. The user-visible rule is simple. A committed parent row either resolves to a validated fragment source or fails with a storage error. It never silently reads a relation found only by a reused physical address.

10.4. Projection, pruning, and lazy reconstruction

A scan begins with the requested columns and predicates. For each visible stripe, the reader obtains its chunk metadata and determines which chunk groups can satisfy eligible scalar predicates. It then loads only selected streams for projected columns. The processing order is:

1. read stripe and chunk metadata under the scan snapshot
2. reject chunk groups contradicted by scalar min/max bounds
3. read existence and value streams for the remaining projected chunks
4. decompress when necessary and reconstruct aligned PostgreSQL values

5. apply deletion masks and residual predicates

6. resolve sequence locators only for rows and operators that need them

Consider a relation containing a packed sequence, an identifier, collection date, length, and sample attributes. A date predicate can reject a chunk group without loading the sequence locator stream. A projection of identifiers and lengths need not enter fragment storage at all. If an alignment operator needs the sequence, it receives a lazy packed range rather than a concatenated allocation.

The PostgreSQL-backed scanner retains a small set of open shard states for one execution. Fragment data are copied into scanner-owned memory with 16 fixed slots. With DNA4 and 64-bit units, the maximum full-fragment payload in those slots is

$$16 \times 1024 \times 8 = 131{,}072 \text{ bytes.}$$

A request contained in a resident fragment can return a direct packed view. A request crossing a boundary is copied into bounded scratch space. Slot replacement therefore decouples the resident sequence payload from the total logical sequence length.

This bound applies only to fragment acquisition. Minimizer anchors, chaining state, persistent index segments, and dynamic-programming matrices have their own memory budgets. Fragmentation prevents an implicit whole-sequence copy, but it does not make every alignment operation constant-memory.

The same column reader can publish scalar tuple slots or batch slots. A batch contains up to one chunk group's rows, projected vectors, NULL masks, a keep mask, and source row numbers. Supported comparisons and Boolean expressions can update the keep mask before scalar row materialization. Supported metadata aggregates may consume the batch interface. This use of the term vectorized describes batched executor calls and does not imply that every transition is a hardware SIMD instruction.

The planner uses the batch path only for a restricted single-relation plan whose child can publish the required slot type. DISTINCT aggregates, aggregate filters, alignment-owned projections, joins, and unsupported types retain the ordinary scalar executor. Empty and all-NULL inputs for some integer and date aggregates are a current limitation because their vector initial states do not yet match PostgreSQL's NULL results. The optimization must therefore be interpreted according to its stated eligibility and semantic limits.

Two local caches may reduce repeated work during these scans. Per-execution fragment state keeps open shard relations and bounded fragment slots. The decompression cache stores decoded compressed column chunks for one PostgreSQL backend. Its key includes relation, stripe, chunk group, and column. It has an explicit byte ceiling, skips chunks currently in use during eviction, and returns borrowed buffers whose lifetime remains tied to the cache entry. Neither mechanism coordinates work between sessions.

10.5. Coordination of repeated query requests

Shared reuse of a completed SQL result is separate from fragment storage and column decompression. The optional query-result cache can coalesce concurrent executions with the same execution identity. Its identity includes the analyzed query shape, parameter values, execution and security context, and a dependency version vector. Normalized SQL text alone is insufficient because two statements with the same text can run with different parameters or visibility rules. The cache must be loaded when the PostgreSQL server starts because its registry and memory areas are shared across backends.

The cache has three user-visible modes. In `disabled` mode, execution does not create a cache claim. In `exact_query` mode, the first executor of an exact identity becomes the producer. Concurrent identical executors wait, and later identical requests may replay the sealed ordered multiset while its dependencies and lifetime remain valid. In `reusable_fragments` mode, exact coalescing remains available. Broader reuse at an expensive operator boundary requires a provider that can prove compatible coverage and retain the lineage needed for residual predicates. Without that proof, behavior reduces to exact reuse. The current BioSeqDB installation does not supply a production coverage provider, so broader operator-level reuse remains unavailable.

The first exact claimant produces the result. A result becomes reusable only after every cache row group and its manifest have been sealed and checked against the schema and dependency versions. Producer failure or invalidation leaves no partial result that another query can replay. A canceled waiter releases its claim without canceling the producer, while a failed producer allows waiting queries to run normally or establish a new generation.

Completed results use checked columnar cache row groups. Memory and file admission have explicit ceilings. If neither tier can accept another row group, the original query continues without publishing a reusable result. Expiration is measured from successful publication, and cached results are cleared across a server restart because they are derived state rather than durable table data.

10.6. Resulting storage contract

The storage design gives each layer a defined responsibility. The Citus-derived column engine manages pages, stripes, chunk groups, scalar projection, compression, deletion masks, and min/max pruning. Sequence indirection adds alphabet-aware packed fragments, length buckets, shard capacity, locator validation, and source epochs. Lazy scanners translate bounded interval requests into fragment access. Shared query-result reuse has its own identity, publication protocol, resource admission, and failure behavior.

These boundaries support database execution without presenting a sequence as an ordinary column-store value. Metadata filters can reduce the candidate set before any sequence read. Alignment stages can request only the intervals they need. Concurrent writers reserve independent physical ranges, while readers observe only committed metadata and validated fragment sources. Repeated queries may share a sealed result, but an incomplete or stale generation is never exposed as complete. The result is a PostgreSQL storage model in which logical sequence length, scan projection, and retained memory are related by explicit contracts rather than by eager materialization.

Chapter 11

Persistent sequence indexes

11.1. Persistent indexes as derived database state

A minimizer index avoids sketching and organizing the same target for every alignment query. In a standalone mapper, the index can be a file that the user rebuilds when the reference changes. A current versioned database index has a stricter identity. It belongs to a physical relation, a sequence column, and a particular version of the source data. Its descriptor records the alphabet, minimizer parameters, hash scheme, occurrence policy, position width, storage format, and any execution interface that consumes the stored shape directly. Older asynchronous payloads have a narrower compatibility identity, as described in Section 11.4.

BioSeqDB therefore treats a persistent sequence index as reproducible database state, rather than as an opaque cache. The source sequence remains authoritative. Replacing or deleting an index does not change it. Conversely, an index built for an earlier source epoch is not eligible after the sequence changes, even if the relation and row identifiers still exist. This rule prevents an old payload from being attached to new sequence bytes by name alone.

The index has a control plane and a data plane. The control plane identifies a visible version and its compatibility contract. The data plane contains hash controls, positions, and navigation metadata. PostgreSQL stores the authoritative descriptor and, in the general path, the payload chunks. A grouped fragment index may additionally have an immutable sealed image for direct reading. That image remains subordinate to the PostgreSQL descriptor. It cannot become current without successful construction and publication of the corresponding generation.

Figure 11.1 summarizes the resulting lifecycle. Readers resolve the current descriptor under their snapshot. A writer builds a new generation without modifying the version already visible to those readers. Only a complete and compatible generation can replace the current descriptor.

11.2. Logical organization of minimizer indexes

BioSeqDB retains two logical minimizer organizations because they optimize different stages of the workload. Minimizers per fragment, abbreviated MPF, matches the fragmented representation of long sequences. Minimizers per sequence, abbreviated MPS, reorganizes the same information by hash range for selective loading. The registered index policy also includes compact symbol counts, which support planning and filtering but do not provide seed positions. The implemented persistent MPF and MPS policies admit the profile $k = 15, w = 10$.

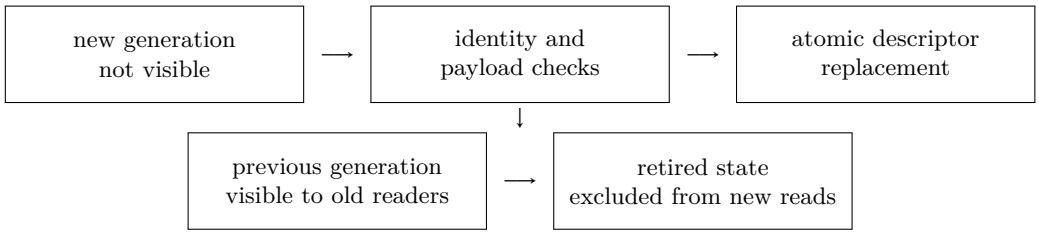


Figure 11.1: Publication of an immutable index generation. A failed build may leave removable unpublished data, but it cannot make an incomplete index visible. The legacy storage mode uses replacement within one relation and does not provide this generation lifecycle.

Other minimizer parameters can be used by transient execution, but these persistent layouts do not store them.

11.2.1. Minimizers per fragment

An MPF object is built for one source fragment. Its main components are a hash table and a contiguous positions array. Each occupied hash entry stores an offset and a count into that array. An occurrence encodes its local target position together with strand information. Readers rebase positions when assembling a queryable view.

Each fragment can be sketched independently. This bounds one construction unit and lets an insertion policy update affected index objects without sorting every minimizer in the logical sequence. A query over a long target must inspect or assemble several fragment images. The default path accepts this cost because it follows the existing storage and ingestion unit.

The grouped MPF representation, called Format 5, stores controls and positions in a specified array layout rather than serializing a language-specific hash container. The layout can be checked before use and can be adapted to a CPU or accelerator view. A sealed MPF5 image offers a direct read path for this grouped layout. It is an immutable reader representation, not an independently managed index provider.

11.2.2. Minimizers per sequence

MPS performs more work during construction so that a query can avoid loading unrelated target hashes. It combines fragment sketches into sequence-wide, hash-ordered buckets. A compact directory maps each stored hash interval to the bucket or stream region that contains its controls and positions. The reader searches this directory with the distinct query hashes, then admits only the intersecting regions.

The selective Format 6 organization separates three logical streams. The directory describes hash ranges and authorized spans. Control blocks store hashes and occurrence counts. Postings extents store positions for hashes that pass the occurrence policy. This order matters. The reader can reject a highly repetitive hash from its count without allocating or reading its full position list. The current Format 6 path requires DNA4 symbols, four-byte hashes, and the registered $k = 15, w = 10$ profile. Other combinations are not interpreted as this selective format.

Table 11.1 compares these organizations with the standalone minimap2 index discussed later in the chapter.

Neither BioSeqDB organization is always preferable. MPF avoids a global merge and follows incremental ingestion. MPS can reduce cold input when the query touches a small part

Table 11.1: Logical roles of persistent minimizer organizations.

Property	BioSeqDB MPF	BioSeqDB MPS	minimap2 `.mmi`
Construction unit	Source fragment	Logical sequence with bounded external merge	Reference index file
Lookup shape	Hash controls with a shared positions array	Hash-range directory, selected controls, and postings	Bucket-local hash tables and position arrays
Cold read	Fragment images needed by the target	Query-selected hash regions	Complete index reconstructed by the loader
Update model	Legacy insertion maintenance or explicit rebuild	Normally an explicit rebuild	User replaces the file
Identity	Relation, column, source epoch, algorithm, and generation	Same database identity plus directory and stream metadata	File parameters under the mapper's load contract
Main use	Default fragment-native path	Selective access for large targets	Reuse by a standalone mapping process

of the target hash space. The relevant factors are source fragmentation, update frequency, query selectivity, construction cost, and the memory available for transient assembly.

11.3. Bounded construction and publication

A sequence-wide index cannot safely be built by appending every $(hash, position)$ pair to one vector and sorting it in memory. The input may contain billions of bases, and one frequent minimizer may have a position list much larger than an ordinary work buffer. BioSeqDB instead uses bounded sorted runs. Pairs enter an admitted memory region. A full run is sorted and either retained under a residency permit or written to temporary database storage. The merge limits its fan-in, so the number of open readers and merge buffers is independent of the total number of runs.

The source is scanned in ranges with the halo required by minimizer extraction. Local positions are converted to global sequence coordinates before entering a run. The final merge groups equal hashes and writes storage units near a target size. A frequent-hash accumulator can also spill, which prevents one posting list from bypassing the general memory bound.

Algorithm 14 gives the high-level construction procedure. It omits physical row sizes and backend-specific calls because they do not change the algorithmic contract.

Algorithm 14 Memory-bounded construction of a sequence-wide minimizer index

Input. Source identity I, the registered minimizer profile $k = 15, w = 10$, and memory budget M.

1. Reserve a build identity and capture the source epoch.
2. Scan bounded source ranges with the required overlap.
3. Extract minimizers and append globally positioned $(hash, position, strand)$ records to the active run.
4. When a run reaches its permit, sort it and retain it only if memory remains available. Otherwise spill it.
5. Merge runs with bounded fan-in until one hash-ordered stream remains.
6. Group equal hashes, spilling a large posting list when necessary.
7. Write buckets or streams with lengths, checksums, and compatibility metadata.
8. Write the navigation directory after all referenced data is complete.
9. Check the unpublished object against I and its manifest.
10. Publish its descriptor atomically, or discard it on failure.

Writing the directory last is a basic completeness rule. A reader cannot infer a usable sequence-wide index from a prefix of buckets because there is no final directory that authorizes them. Immutable generation mode strengthens this rule. All rows and optional sealed artifacts belong to a fresh generation. The writer checks the complete object before replacing the small descriptor that names the current version. Readers that already resolved the previous descriptor may continue until their snapshot is no longer visible.

The older storage mode writes chunked objects in ordinary sidecar relations and replaces their contents within a transaction. It remains the default configuration and provides broad support for the registered index layouts. Grouped immutable generations are an explicit mode. They give stronger copy-on-publication behavior but impose additional manifest, retirement, and artifact management. The thesis distinguishes these modes because directory-last completion in one mutable namespace is not equivalent to snapshot-safe generation replacement.

Grouped derived index relations may be unlogged because their bytes can be rebuilt from durable sequences. This reduces duplicated write traffic, but it also defines a recovery obligation. After an unlogged reset, the prior index must not be advertised as ready. It must be rebuilt and published again.

The sealed direct-reader image follows the same authority rule. Construction first produces an authenticated grouped generation. The image is then written under a private name, made immutable, synchronized, and published without overwriting an existing generation. The database descriptor becomes current only after the final artifact has the expected identity. Retired descriptors are excluded from new reads. This sequence keeps filesystem publication consistent with transaction visibility without treating the image as durable source data.

Index updates are governed separately from reads. An index can be maintained on insertion or rebuilt only on request. MPF is the default minimizer index and is suited to insertion-time work. MPS is normally manual because construction requires a sequence-wide merge. Deferred persistence transfers stable relation identity and parameters to a database worker. It does not transfer live snapshots, relation handles, or memory owners. The worker opens the source under its own database context and rechecks the epoch and policy before writing.

11.4. Compatibility and integrity

A valid byte sequence is not necessarily a usable index. Query and target minimizers must be produced by the same hash scheme and parameter set. The reader also needs a position width capable of representing the source, a known control layout, and occurrence counts with the expected meaning. A direct accelerator view adds an execution interface requirement because its kernels interpret the stored shape without a general conversion step.

For current versioned MPF and MPS formats, the compatibility identity includes at least the following fields:

- source relation, sequence column, source epoch, and generation
- alphabet and encoded symbol width
- minimizer k, window w, hash scheme, and scheme version
- position width, occurrence policy, and persistent layout
- producer identity, and accelerator identity for directly consumed persisted outputs

Legacy asynchronous payloads do not carry the complete scheme identity. They are admitted only through their narrower layout contract. In particular, bounded ntHash reads are excluded from this legacy persisted path.

Manifests bind this identity to section lengths, ordering, checksums, and address ranges. Readers check arithmetic before constructing spans, so a corrupt count cannot wrap around and designate an apparently small allocation. Reserved fields and format versions are checked rather than ignored. Relation chunks are checked at the granularity in which they are admitted. A sealed image checks its fixed header and identity when opened, while complete section checks are part of publication and full inspection.

The SQL read requirement determines the response to absence or mismatch. A request that requires a compatible persisted index fails if no matching index is visible. It does not silently sketch the source. A permissive request may recompute a transient index. A request that requires absence bypasses persisted reads and forces transient recomputation even when stored data exists. These policies separate correct fallback behavior from measurements or operations that depend on a particular representation.

Compatibility checking occurs before a CPU or accelerator view is reused. A resident object is keyed by the persisted assembly identity and relevant algorithm parameters. Matching SQL types are insufficient because two payloads with the same row schema can use different hash schemes or control layouts.

11.5. Selective cold loading and memory accounting

Serialized size does not describe the peak memory of an index read. A backend may simultaneously own a directory, selected controls, postings, decoded positions, a transient lookup, query occurrence counts, input buffers, and an accelerator upload. BioSeqDB uses explicit permits for these significant owners. A component obtains admission before allocating or retaining its destination, and the permit remains held for the lifetime of cached or device-resident data.

Figure 11.2 shows the cold MPS path. The directory is small enough to read and check as a complete navigation object. Distinct query hashes identify the relevant control blocks.

Counts are then compared with the occurrence threshold. Only postings for surviving hashes are authorized, read, and clipped to the requested target interval.

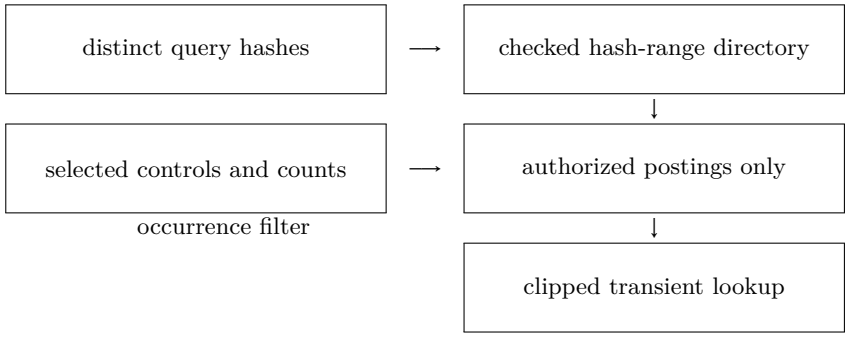


Figure 11.2: Query-selective cold loading. Navigation and counts are checked before position extents are admitted. Unselected and highly repetitive postings do not enter the backend's decoded working set.

Physical access must preserve this logical selectivity. Adjacent authorized ranges are coalesced into bounded read slabs, rather than fetched as many small objects. Checksums are evaluated at the addressed unit. For MPF, the direct sealed-image route avoids reconstructing a large immutable image through a large number of relation tuples. The relation generation remains authoritative for lifecycle and identity even when the read data plane is mapped.

Memory accounting distinguishes mapped address space from owned working data. Opening a large immutable image does not allocate an equally large decoded container, but operating-system page residency is not controlled by the backend's memory permit. The permit covers decoded arrays, assembled lookups, resident chunks, and device allocations. A memory-bound claim must state which of these domains are included. A process-cold measurement must also consider filesystem cache state separately.

Mapped-image lifetime is registered with database resource ownership so normal completion and error handling close it. Backend-local caching is bounded by entry and byte limits. The active relation reader can retain checked chunks, while runtime-specific caches may retain compatible derived views. Transaction abort and relation invalidation remove entries that may no longer be reused. These caches reduce repeated work within one backend, but they do not define visibility across backends. PostgreSQL snapshots and the published descriptor remain the source of coherence.

11.6. Runtime index selection

The alignment dispatcher resolves an index policy for each source sequence column. Presence alone is not enough. The dispatcher combines the requested index name with the read requirement, visible source epoch, compatibility identity, available reader route, memory admission, and execution backend. Algorithm 15 states the decision.

Algorithm 15 Runtime selection of a persistent minimizer index

Input. Source identity I, query minimizers Q, and read requirement R.

1. Resolve the active index policy for the source column.
2. If stored reads are permitted, resolve the visible descriptor.
3. If a descriptor exists, check its source epoch, algorithm identity, layout, lengths, and addressed checksums.
4. For MPF, read compatible fragment images or the configured sealed image, then obtain an admitted runtime view.
5. For MPS, search the directory with Q, load selected controls and postings, and assemble a clipped lookup.
6. Reuse a compatible resident view when its complete identity matches.
7. If no usable index exists and R requires persistence, fail.
8. Otherwise build a transient index and, when policy permits, schedule persistence using stable database identity.
9. Retain the selected descriptor and runtime view until the batch ends.

The default active minimizer path is MPF. When both deferred MPF and MPS policies could write an index, selection is deterministic and prefers MPF. This avoids presenting two incompatible background products as one implicit choice. MPS is selected explicitly when its construction cost is justified by selective reads. A strict persisted requirement turns an absent or incompatible object into an error. A permissive query can continue with recomputation.

Runtime selection also separates persistence from the final execution shape. The stored representation may be adapted into a CPU lookup or a compatible accelerator view. Those derived views have their own memory lifetime and cache identity. Device dispatch does not weaken host-side compatibility checks.

11.7. Differences from minimap2 `.mmi`

The minimap2 index is an appropriate algorithmic reference because BioSeqDB uses the same general family of minimizer seed lookup.[1] A `.mmi` file records minimizer parameters, reference metadata, bucket data, and optionally the packed reference sequence. Each bucket has a hash lookup and a positions array. A singleton position can be stored directly in the hash value, while a repeated minimizer points to a range in the positions array. The loader reads the file and reconstructs the complete in-memory bucket representation.

This design fits a command-line mapper. The caller chooses the file and replaces it when the reference changes. Stored parameters are interpreted by the mapper's own load contract. There is no need for a database snapshot to keep an older file version visible while another transaction publishes a replacement.

BioSeqDB retains the useful offset-and-count lookup idea, but changes the surrounding contract. Sequence-wide layouts use global coordinates, while MPF stores fragment-local positions because long sequence values are stored in fragments. MPS and the selective stream format can read only query-relevant hash ranges. Neither organization is wire-compatible with `.mmi`.

[1] Li, "Minimap2: pairwise alignment for nucleotide sequences"; Li, minimap2.

The larger difference concerns ownership. A versioned BioSeqDB index is bound to a relation, column, source epoch, algorithm identity, and published generation. Legacy updates can participate in insertion policy or an explicit rebuild. A new immutable generation is checked before publication, and the predecessor remains available to older snapshots. Cold loading is subject to backend memory admission, and cached CPU or device views inherit the same compatibility identity. These mechanisms are unnecessary for a standalone `.mmi` file, but they are required when index reuse is part of transactional SQL.

The direct sealed-image route does not remove this distinction. Although both systems can read an ordinary file efficiently, BioSeqDB's image is selected by an authoritative database descriptor, and its retirement is recorded transactionally. It is derived from the database generation and may be regenerated. It is not a user-selected replacement for PostgreSQL index state.

11.8. Operational limits and conclusions

Persistent minimizer storage reduces repeated construction, but it adds recovery and administration requirements. Unlogged payload relations must be rebuilt after reset. Sealed images must either be included in backup and replication procedures with their descriptors or be regenerated from the authoritative source. Operating-system page cache residency lies outside backend memory permits, so warm and cold measurements require separate interpretation.

The two retained organizations address different workload shapes. MPF follows fragment-native ingestion and remains the default. MPS pays for a bounded sequence-wide merge to gain query-selective loading. The versioned forms of both organizations include the source epoch and minimizer semantics in their compatibility identity. MPS construction uses admitted sorted runs and bounded spill. Immutable mode checks an unpublished generation before atomic descriptor replacement.

The resulting design differs from a database wrapper around a minimap2 file. It defines a transaction-aware index lifecycle with explicit logical organization, physical representation, publication, memory admission, and runtime selection. These properties allow persistent seed lookup to coexist with lazy sequence storage, concurrent SQL readers, and multiple execution backends.

Chapter 12

Bounded sequence ingestion and integrity limits

12.1. Bounded sequence ingestion and integrity limits

Alignment and search operate on compact sequences, whereas public archives usually distribute FASTA or FASTQ text. A FASTA record contains a header and a sequence body. FASTQ adds a separator and a quality string whose length must match the sequence. Records may span many input buffers, and one record may be larger than a PostgreSQL value. Ingestion must therefore combine record framing, compact encoding, metadata extraction, relational typing, and transactional storage.

BioSeqDB separates these concerns into a reusable input layer and a PostgreSQL-specific sink. The input layer reads bytes, selects the record grammar, parses incrementally, and emits headers with compact sequence chunks. It does not access PostgreSQL relations, tuples, or memory contexts. The sink selects the representation declared by the destination sequence column, converts header fields to SQL values, writes sequence fragments, and creates the parent row that refers to them. It evaluates defaults and the executor row constraints available to this bulk path. This boundary keeps file processing independent of database internals while retaining transaction visibility and rollback.

The resulting contract has four main properties. Multiline FASTA and FASTQ records are framed across arbitrary refill boundaries. Sequence symbols are packed directly into the destination representation. Packed sequence delivery is bounded even for very long FASTA records. A failed load does not commit a successful prefix as if the file had been loaded completely. FASTQ quality storage has a narrower guarantee, stated in Section 12.7.

12.2. Layered input pipeline

The lowest layer follows a small byte-source contract. A source fills a caller-owned span and reports the number of bytes read, end of input, or the first error. The same parser can consequently consume a mapped plain file, a sequential file descriptor, a zlib stream, or bytes supplied by another provider. The parser is specialized for its source, so refill does not require a virtual dispatch for every buffer.

Record format and compression are selected independently. The final non-compression suffix selects the FASTA or FASTQ grammar. Common FASTA and FASTQ suffixes are accepted case-insensitively, with an optional gzip suffix. The opened file is then inspected for the gzip magic bytes defined by RFC 1952 Deutsch, GZIP File Format Specification Version 4.3.

A compressed file can therefore be decompressed even when its name omits a gzip suffix. A plain file with such a suffix is not sent through a decompressor. By contrast, the parser does not guess FASTA or FASTQ from the first record. An unrecognized or misleading format suffix produces a grammar or format error instead of a heuristic reinterpretation.

For plain input, a mapped view is preferred when it is available and suitable for the file size. Otherwise, sequential reads refill a reusable slab. The SQL loader uses zlib for gzip input. Figure 12.1 shows where the portable input work ends and database work begins.

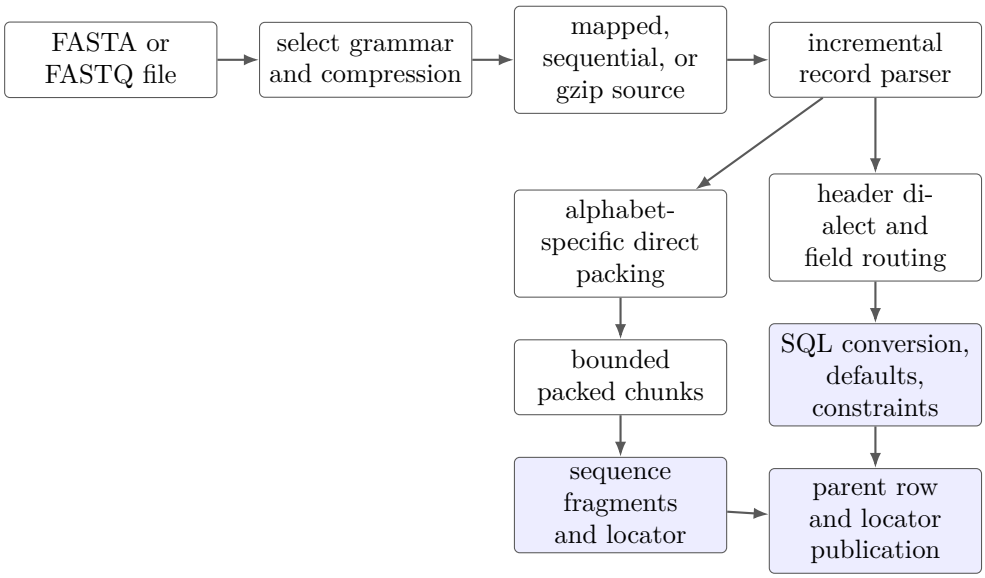


Figure 12.1: Sequence and header data follow separate paths after record framing. Packed sequence chunks are bounded. PostgreSQL operations are confined to the shaded boxes.

12.3. Incremental parsing and direct compact encoding

The parser is a finite-state machine over spans supplied by the selected source. A streamed source refills a recyclable input slab. When a record crosses a refill boundary, only its unread suffix is retained. A mapped source can expose input bytes directly. The general batch reader may enlarge its slab for an unusually large record, up to a configured hard limit. The SQL streaming path instead drains complete packed units while the body is being parsed. It does not first construct an ASCII sequence and then encode it, nor does it require the complete packed body to be contiguous in memory.

For FASTA, the states distinguish the header marker, header text, sequence body, and record end. Both LF and CRLF line endings are accepted, and the body may contain arbitrary line wrapping. On the common DNA4 path, one vectorized loop locates record boundaries and converts bases into two-bit storage words. Other registered alphabets use the same state machine with their corresponding packing operation. At record completion, the encoder flushes any partial word and records its padding. The represented base count can then exclude unused positions in the final storage unit.

FASTQ needs stricter framing because a quality line may begin with `@`, `+`, or `>`. Searching for a marker at the start of each physical line would therefore split valid records. The parser instead performs the following steps:

1. read one `@`-prefixed header

2. collect one or more sequence lines until a `+`-prefixed separator
3. skip the rest of the separator line
4. collect quality bytes until their newline-free length equals the raw sequence length

The final comparison uses the raw number of sequence bytes, not the number of symbols retained by the alphabet encoder. This matters because the current policy can discard an unknown symbol, while its corresponding quality byte is still present. A truncated quality field is a terminal input error. Excess quality bytes are rejected after a terminated line, but extra bytes in an unterminated final line are currently accepted. Multiline sequence and quality fields otherwise remain synchronized.

The destination sequence column controls representation. Before reading the first record, the loader requires exactly one parameterized sequence column and decodes its alphabet and storage-unit width. It then selects one compiled combination of alphabet, unit width, and FASTA or FASTQ grammar. Available layouts come from a compiled registry and can include DNA4, DNA5, and several storage-unit widths. Unsupported combinations fail during schema inspection rather than after a partial load. Each emitted chunk already uses the selected compact layout, so the database sink appends its bytes without a second conversion.

12.4. Bounded delivery and backpressure

The SQL path emits a begin-record event, a sequence of packed chunks, and an end-record event. Only complete storage units are emitted during a record. The last chunk carries any partial unit and the adjusted base count. A chunk view remains valid for its callback, allowing the database sink to copy it directly into fragment storage. This protocol bounds the packed body by the input slab and output window instead of the total sequence length. A multi-gigabase FASTA sequence can thus be stored without creating one multi-gigabyte database value.

PostgreSQL calls remain on the backend thread. The default path invokes the sink directly, which gives deterministic order and avoids cross-thread ownership. An optional overlap mode places owned begin, chunk, and end events in a fixed-capacity first-in, first-out queue. A producer performs only PostgreSQL-independent parsing. The backend thread consumes events and writes relations. A full queue blocks the producer, which supplies explicit backpressure. Closing the queue wakes both sides after an error, and the first consumer failure remains the reported cause. The overlap mode is disabled by default.

12.5. Header-to-column mapping

Archive headers often combine an identifier with structured metadata. UniProt, RefSeq, GenBank, NCBI, GISAID, and IMGT records use different positional fields, delimiters, and tagged values. BioSeqDB can preserve the entire header as text or extract recognized fields into SQL columns. The parser supports an explicit dialect, a plain mode, and automatic detection.

Automatic detection is bounded and occurs once per input file. The loader reads a limited decompressed prefix and samples a limited number of headers. FASTQ sampling follows logical sequence and quality lengths, so an `@` byte in quality data is not treated as a new header. Recognized dialects vote across the sample with deterministic tie handling.

If the sample is unstructured and no metadata mapping requires further parsing, the loader uses plain-header behavior. With metadata columns or user mappings, it retains automatic mode and applies generic per-record parsing.

Mapping is also resolved once. Explicit user mappings take priority. Remaining known fields are matched to normalized destination names, after which the loader chooses an identifier column and routes any generic metadata. Name normalization ignores ASCII case and common word separators. The resulting routing table maps positional or tagged header slices to destination columns. Per-record parsing then scans delimiters and passes raw byte ranges to the database boundary without knowing PostgreSQL types.

At that boundary, each destination's input conversion and typmod are cached. Unconstrained text can be constructed directly. Other scalar, range, or multirange values use PostgreSQL's normal type input rules, including session settings that affect dates and timestamps. Defaults are prepared once but evaluated for every row, so volatile defaults retain their SQL meaning. The completed tuple is checked with the executor's row-constraint step before it is written. This direct bulk path does not execute ordinary `INSERT` triggers or the secondary-index paths used for foreign-key, unique, and exclusion enforcement. Per-row memory is reset only after referenced values have been copied into database-owned storage.

The public operation can be summarized as `bioseq.load(target_table, file_path, header_format, columns)`. The optional header format defaults to automatic detection, and the optional mapping array can override inferred routes. The return value is the number of loaded records. Ambiguous mappings, unsupported sequence layouts, generated target columns, multiple sequence columns, and an unmapped required column without a default are rejected before streaming begins.

12.6. SQL publication and error behavior

Sequence data is stored outside the small parent tuple. For each record, the sink begins a sequence source, appends bounded packed chunks, and finishes the source to obtain a pending locator. It then parses the header, evaluates defaults, applies constraints, and writes the parent tuple containing that locator. Publication occurs only after the parent row has copied the tuple. This order avoids making an out-of-line sequence visible before a relational row refers to it. Publication does not commit the row. Transaction visibility still governs access to both objects.

The write is not equivalent to an ordinary `INSERT`. The loader writes through its bulk storage path after evaluating defaults and executor row constraints. It does not run normal insert triggers, maintain secondary indexes, or invoke the usual foreign-key, unique, and exclusion paths. A target relation must therefore be chosen with these limits in mind.

Input and parser failures are carried to the PostgreSQL boundary with an owned diagnostic. The boundary assigns an appropriate SQLSTATE and raises an SQL error. PostgreSQL conversion, default, and constraint failures keep their native messages. Error context adds the record number, the destination column when known, and a bounded prefix of the header. Limiting the header excerpt prevents a malformed record from creating an unbounded diagnostic.

Cleanup must preserve the earliest terminal error. Reader shutdown, queue closure, and relation cleanup cannot replace it with a later secondary failure. The target relation remains protected by a row-exclusive lock while the loader owns its executor, tuple slot, and write states. On success, both parent and fragment write states are flushed before newly loaded rows are used for optional index construction. These resources are also released on the error

```
inspect target and select format, alphabet, and unit width
detect the header dialect and resolve column routes

for each record
    begin an out-of-line sequence source
    append each bounded packed chunk
    finish the source and obtain a pending locator
    convert mapped header fields to SQL values
    evaluate unset defaults and enforce executor row constraints
    write the parent tuple with the locator
    publish the sequence source

flush parent and fragment writes
make the new rows visible to subsequent work in the transaction
optionally build configured sequence indexes
return the record count
```

Figure 12.2: Simplified loading procedure. Representation and routing decisions are made once, while row conversion and constraints are applied per record.

path. On failure, PostgreSQL aborts the enclosing transaction or subtransaction. Rows and sequence fragments written before the fault are not committed as a successful prefix. Within a larger explicit transaction, the caller can roll back to a savepoint established before the load.

This design does not introduce a second transaction protocol in the input layer. Locks, write-ahead logging, resource ownership, command visibility, and rollback remain PostgreSQL responsibilities. The reusable parser reports records and errors, while the database sink determines whether their effects become visible.

12.7. Integrity limits

The bounded claim applies to packed FASTA sequence bodies. The FASTQ adapter currently retains quality bytes until a record is complete, so its quality scratch space grows with record length. Sequence packing is still incremental, but total FASTQ memory is not independent of the largest record.

Alphabet checking is also permissive. Unknown FASTA and FASTQ symbols are currently dropped by the encoder. A strict-alphabet option exists in the reader interface but is not applied by the parser. The selected alphabet therefore determines representation, not complete validation of every input byte. Applications that require rejection of unknown symbols must validate them before relying on the current loader.

The SQL gzip path uses zlib's stream validation. A decompression error after output has begun is terminal because restarting with another decoder would duplicate the already delivered prefix.

Finally, the filename suffix remains authoritative for choosing FASTA or FASTQ. Magic-byte inspection identifies compression only. This separation keeps parser dispatch deterministic, but a mislabeled biological format is reported as an error rather than detected automatically.

12.8. Conclusion

BioSeqDB ingestion joins an incremental biological file parser to PostgreSQL without materializing a complete large sequence. The parser frames records and packs symbols into bounded native chunks. Header routing supplies typed column values, and the database sink coordinates fragment storage with parent-row publication and transaction rollback. The stated FASTQ and alphabet-validation limits define where this guarantee does not yet apply.

Chapter 13

External sequence providers

13.1. Remote databases as relations

Public sequence databases differ in identifiers, response formats, and retrieval models. NCBI separates taxonomy, assembly, gene, and sequence services National Center for Biotechnology Information, NCBI Datasets v2 REST API documentation. UCSC provides assembly and interval retrieval UCSC Genome Browser Project, UCSC Genome Browser REST API data interface. UniProt serves protein records and annotations UniProt Consortium, Programmatic access to UniProt, while EBI supports cross-domain discovery European Bioinformatics Institute, EBI Search REST API documentation.

BioSeqDB defines six remote relation shapes in the `bioseq_external` schema, but the generated table definitions currently omit the required relation-entity option. An installation must add that option before the custom scan can execute them. They cover taxonomy, provider catalogs, genes, locations, complete sequences, and bounded sequence chunks.

When configured, these relations have no local heap data. PostgreSQL replaces a physical scan with a custom scan PostgreSQL Global Development Group, PostgreSQL 17 Documentation: Writing a Custom Scan Provider. Every qualification remains a local PostgreSQL check. Supported equality and array predicates are also lowered into a provider-neutral query that constrains the remote request. The scan selects a provider adapter and emits ordinary typed tuples. A remote record can therefore participate in a projection, filter, or join without a separate download script.

Table 13.1: External relation groups and their resource contracts.

Relation group	Result	Resource behavior
Taxonomy and catalog	Normalized metadata rows	Provider response is decoded before row production
Gene and location	One row per nested item	PostgreSQL applies residual predicates after normalization
Sequences	One packed sequence value	Memory may depend on record length
Sequence chunks	Ordered packed values	Accession-based NCBI retrieval uses a bounded queue

A custom scan retains state between tuple requests, postpones network access until execution, and reports the chosen provider through `EXPLAIN`. Ordinary `EXPLAIN` performs no network request, while `EXPLAIN ANALYZE` consumes the remote stream.

13.2. Provider and execution boundaries

Provider integration combines a capability descriptor, a generated typed HTTPS client, and a handwritten adapter. The client handles validation, encoding, size limits, and transport. The adapter maps provider fields to a canonical query and message model, keeping provider-specific formats outside PostgreSQL planning.

The adapters have different capabilities. NCBI covers metadata and nucleotide retrieval, UCSC covers catalogs and intervals, UniProt supplies protein metadata, and EBI supports discovery. Protein sequence materialization is not available through the nucleotide-oriented sequence type. A request that no selected provider can serve fails explicitly. In a multi-provider scan, an adapter error can be suppressed when another adapter returns rows, so such a result may be partial.

At scan initialization, provider predicates select adapters, while supported equality and constant array predicates constrain remote requests. Array predicate expansion is currently capped at 64 equality-query combinations. Excess combinations are omitted with a warning, so callers that require complete results must remain below this limit. At execution time, the scan opens a provider session on first demand. Metadata queries can collect a bounded set of owning messages. Large NCBI nucleotide queries instead use an incremental FASTA source and a queue bounded by both bytes and item count. Only the PostgreSQL backend thread creates `bioseq.seq` values and writes tuple slots. Network workers exchange ordinary owning values and never call the PostgreSQL API.

Algorithm 16 External relation scan

Require: relation kind, predicates, requested columns
1: retain all predicates locally and lower supported predicates remotely
2: select the requested or default adapter
3: open the provider session when the first row is requested
4: while the executor requests another row do
5: obtain the next provider message and convert required sequence values
6: place typed fields in the tuple slot for residual SQL processing
7: end while
8: cancel and close the session on rescan, error, or query completion

The streaming transport verifies its HTTPS destination, restricts redirects, and applies connection and progress time limits. The archive path also applies explicit wire-size and decoded-size limits. Incremental NCBI FASTA retrieval bounds queue and carry memory but does not impose a total response-size limit. The NCBI streaming scan checks PostgreSQL cancellation between bounded waits. Retry is allowed only before the first row is emitted, which prevents duplicated visible chunks.

Whole-record retrieval still requires record-sized memory, and archive retrieval requires bounded buffering before decoding. Requests are not currently rate-coordinated across backends. Chunked retrieval is preferable for large records.

Chapter 14

Source-driven SQL code generation

14.1. One source for the PostgreSQL boundary

The preceding chapter described clients generated from provider-owned OpenAPI descriptions. The generator considered here serves a different boundary. It derives the extension's PostgreSQL interface from BioSeqDB declarations.

A PostgreSQL extension repeats the same contract in several forms. SQL needs a routine name, argument types, result type, volatility, and permissions. The shared library needs a matching C entry point. C++ code must decode the same arguments and construct the declared result. Tables, operators, casts, configuration variables, and comments introduce similar duplication. A local change can therefore leave individually valid artifacts that disagree at the boundary.

BioSeqDB makes typed C++ declarations the source of these facts. Attributes describe only information that cannot be inferred from the declaration, such as an SQL type binding or a PostgreSQL volatility category. Function names, parameter order, and C++ types remain ordinary language constructs. Repeated annotations and conflicting defaults are rejected. This keeps annotations from becoming another complete copy of the interface.

The generator performs structural compiler analysis rather than searching source text. It observes active declarations, attributes, parameter types, initializers, documentation, and module ownership under the configured build profile. These records are normalized into a closed intermediate representation. Each SQL entity receives a stable identity and explicit dependencies. Tarjan's method classifies strongly connected components, then a stable topological traversal emits the component graph in deterministic order Tarjan, "Depth-First Search and Linear Graph Algorithms."

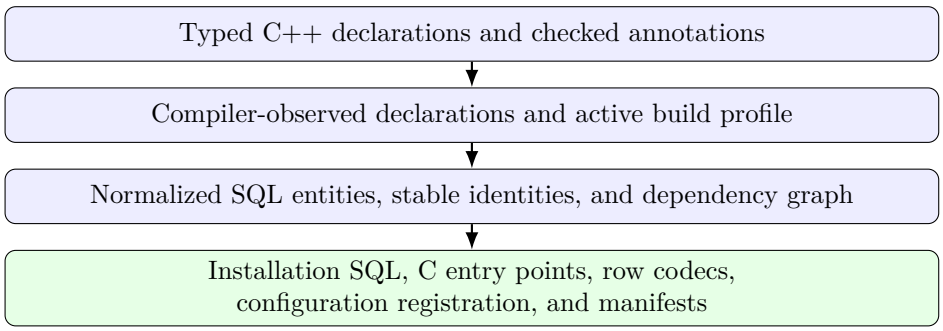


Figure 14.1: Source-driven generation of the PostgreSQL extension boundary.

14.2. Generated artifacts and safety

One declaration can produce several related artifacts. Functions normally yield installation statements, generated function-manager entry points, conversions, documentation, and manifest records. A row declaration can also yield tuple ordinals and codecs. An annotated configuration value yields registration code with a typed default, range, context, and description. A closed type registry rejects unsupported C++ and SQL combinations rather than falling back to an untyped `Datum` convention.

Dependencies determine a stable generation order. Generated outputs are written to the build tree and used directly as extension artifacts.

Error handling requires an additional boundary. PostgreSQL can report a recoverable error by a non-local jump that does not run intervening C++ destructors PostgreSQL Global Development Group, PostgreSQL 17 Documentation: Using C++ for Extensibility. Conversely, a C++ exception must not cross a PostgreSQL callback. Generated wrappers therefore translate both directions through a common boundary. Diagnostics are copied to plain state, C++ owners are destroyed, and only then is the PostgreSQL error raised. Code that calls PostgreSQL below the wrapper still needs a bridge or a region without live nontrivial C++ objects.

The model is deliberately closed. Adding a new PostgreSQL object category requires a typed representation, validation, dependency rules, and a renderer. Compiler-based collection also adds build cost. In return, SQL declarations, C symbols, converters, tuple layouts, and configuration metadata are derived from the same program rather than maintained as independent inventories.

Chapter 15

Conclusions

BioSeqDB demonstrates that sequence search can be represented as a PostgreSQL execution operation rather than as a scalar call to an external mapper. The system stores packed biological sequences, constructs persistent minimizer indexes, and runs a minimap2-derived pipeline for seeding, chaining, filtering, and final extension. Planner support and custom scans preserve the batch structure required by SIMD and GPU implementations while presenting typed SQL results that can participate in filters, projections, and joins.

The first research question concerned a shared representation for the algorithmic stages. Alphabet-generic compact ranges cover resident values, while lazy sequence views represent fragmented, sliced, and composed sources. Algorithms read the required region without first converting the complete value to a byte array. The second question concerned heterogeneous execution. Minimizer extraction and chaining provide regular work for the implemented SIMD and GPU routes. Dispatch follows explicit algorithm settings, input compatibility, index form, and resource availability. Final alignment uses banded or unbanded KSW2, adaptive Block Aligner, and GPU routes that may return specialized scripts before device dynamic programming. Optional prefilters may reject unsuitable candidates but do not replace alignment.

The third question concerned database execution. A custom scan owns the search expression, batches input rows, and substitutes match fields into planned expressions. This design also permits multiple alignments for one input row. The fourth question concerned the surrounding data system. Physical sequence storage, persistent indexes, bounded ingestion, external-provider relation definitions, cache coordination, and source-driven SQL generation give the kernels explicit storage and lifecycle contracts.

The remaining limits are operational and algorithmic. Some remote retrieval paths still require record-sized or archive-sized memory. Provider quotas are not coordinated across all PostgreSQL backends. Hardware-specific routes have batch-size thresholds and require broader evaluation on additional devices. GPU chaining may examine a bounded predecessor set, and seed and alignment backends retain route-specific contracts. The present mapper is centered on linear references rather than pangenome graphs. A relational database can nevertheless plan and execute batched sequence analysis while retaining typed data, bounded resources, persistent derived state, and explicit algorithm selection.

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